

Job Displacement, Remarriage, and Marital Sorting*

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Abstract

This paper provides new insights into marital sorting by combining quasi-experimental evidence with an equilibrium search-and-matching model of the marriage market. Leveraging Danish establishment closures, we show that job displacement leads individuals to separate from low-earning partners and rematch with higher-earning ones. These findings suggest a negative relationship between partners' incomes in how couples match, whereas spouses' incomes are positively correlated in the cross-section. We use the model to assess which mechanisms can account for this combination of patterns and draw three conclusions: (i) partners' incomes are substitutes rather than complements in the marriage market; (ii) our findings are difficult to reconcile with one-dimensional but consistent with multidimensional matching; (iii) non-uniform sorting is unlikely to be the primary explanation. To highlight the quantitative relevance of our findings, we simulate how marital sorting shapes the transmission of individual inequality to between-household inequality, finding that it is amplified in a one-dimensional calibration and dampened in a multidimensional calibration of our model.

Keywords: Marriage Market, Sorting, Search and Matching, Multidimensional Heterogeneity

JEL classification: D1, J12, C78, D83, J31

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1 Introduction

Who marries whom plays a central role in shaping inequality. This insight, dating back to [Becker \(1973\)](#), has motivated a large literature spanning macroeconomics and labor economics that studies matching patterns in the marriage market, both empirically and theoretically ([Fernández and Rogerson, 2001](#); [Greenwood, Guner, Kocharkov, and Santos, 2015](#); [Eika, Mogstad, and Zafar, 2019](#); [Almar, Friedrich, Reynoso, Schulz, and Vejlin, 2025a](#)). A well-established empirical finding in this literature is that spouses' attributes, such as income ([Chiappori, Fiorio, Galichon, and Verzillo, 2026](#)), education ([Schwartz and Mare, 2005](#)), and wealth ([Charles, Hurst, and Killewald, 2012](#); [Fagereng, Guiso, and Pistaferri, 2022](#)), are positively associated between spouses. However, the underlying mechanisms that give rise to these matching patterns remain imperfectly understood. Consider the case of income: do the observed cross-sectional associations in spouses' incomes reflect sorting on income? Are they informative about whether incomes are complements or substitutes in the marriage market? Or do they arise through other channels, for example, because individuals sort on other, potentially unobserved traits that are correlated with income?

We provide new evidence on how the association between spouses' incomes arises through marriage and divorce decisions, and interpret this evidence through the lens of an equilibrium search-and-matching model of the marriage market. Our empirical analysis exploits quasi-experimental variation from establishment closures in Danish register data, using job displacement as an exogenous shock to an individual's marriage-market value. Our empirical findings show that job displacement increases separation rates—particularly among individuals matched with lower-income partners—and raises the likelihood of rematching with a higher-income partner. These patterns suggest a negative relationship between partners' incomes in the matching process: individuals who lose income, on average, transition from lower-income to higher-income partners. In the cross-section, however, spouses' incomes are positively correlated. To understand how both patterns can arise simultaneously, we interpret the evidence through a search-and-matching framework that builds on [Shimer and Smith \(2000\)](#). The framework yields predictions about endogenous separations and rematching in response to job displacement, which we use to assess the matching mechanisms that can account for the combination of empirical patterns we document.

Our empirical analysis compares over 146,000 displaced workers to a non-displaced control group. We follow both groups over time and compare their transitions into and out of marriages and cohabiting relationships. Our research design relies on establishment closures as a source of exogenous variation, thereby circumventing the endogeneity of individual job loss and voluntary quits. We document persistent earnings losses following displacement and show that marriage market outcomes—such as separation rates, the characteristics of couples who separate, rematching rates, and the composition of new matches—differ significantly between the treatment and the control group. Specifically, displaced individuals are (i) more likely to experience a breakup, (ii) particularly likely to do so when matched with a low-earning partner, (iii) more likely to remain single after a breakup, and (iv) more likely to transition from a low-earning to a

higher-earning partner when rematching, relative to the control group. All these findings are of the same sign for men and women, though for women some of the estimates underlying (ii) and (iv) are less precise. Finding (iv) is not driven by labor supply choices: differences in partners' work hours are small. Findings (ii)–(iv) reflect a combination of treatment and selection effects. We interpret them jointly through our structural framework, which features endogenous selection into divorce and remarriage.

To interpret these findings, we begin with a one-dimensional search-and-matching environment and consider two benchmark cases of our framework: globally positive assortative matching (PAM) and globally negative assortative matching (NAM). We show formally that, under these benchmarks, it is difficult to reconcile the combination of empirical patterns we document with the model's predictions. Specifically, the rematching patterns following job displacement resemble those implied by NAM, whereas the positive cross-sectional correlation of partners' incomes points to PAM.

Intuitively, this tension arises from a tight link between the complementarity or substitutability of partners' traits and the implied cross-sectional sorting patterns. For example, complementarity in the marital match value and a positive correlation between matched partners' types can arise from complementarities in home production hours (Goussé, Jacquemet, and Robin, 2017; Chiappori, Salanié, and Weiss, 2017; Calvo, Lindenlaub, and Reynoso, 2024), education homophily (Chiappori, Costa-Dias, and Meghir, 2018; Chiappori, Iyigun, and Weiss, 2009), or the consumption of market-purchased public goods (Lam, 1988). By contrast, substitutability in the match value and a negative correlation can arise from substitutability in home production hours (leading to household specialization) or risk sharing.¹ All of these mechanisms are typically studied in one-dimensional models, and the resulting sorting patterns are usually globally positive or globally negative.²

To understand which mechanisms can reconcile the combination of patterns we document, we consider two potential explanations, each relaxing a key assumption underlying our benchmark. First, we move from one-dimensional to multidimensional heterogeneity, defining sorting dimension by dimension so that matching can be positive on some dimensions and negative on others. This allows rematching patterns following job displacement to be governed by sorting on a subset of dimensions, while cross-sectional correlations reflect sorting across all dimensions. We show formally that this extension can jointly account for both the displacement-induced rematching patterns and the positive cross-sectional correlation in partners' incomes. Compared to one-dimensional models, the link between complementarities in the match value and sorting becomes more nuanced: even if partners' incomes are substitutes, so that sorting on income itself is negative, a positive cross-sectional correlation may arise due to positive sorting on other

¹See Becker (1973) and Becker (1981) for models with substitutability in home production hours that lead to household specialization and negative sorting; risk sharing generates substitutability in spouses' types, e.g., in Chiappori et al. (2018) and Pilossoph and Wee (2021).

²While many models allow for matching on multiple characteristics, they typically assume that these characteristics are perfectly dependent (e.g., income is fully determined by education) or can be summarized by a scalar index, effectively reducing matching to one-dimensional sorting.

characteristics correlated with income.

Second, we relax the assumption of globally uniform sorting, while maintaining one-dimensional heterogeneity. This allows sorting to be negative in the part of the income distribution where displacement is concentrated while remaining positive elsewhere, so that local rematching patterns and the aggregate cross-sectional correlation need not coincide. We evaluate this possibility empirically using semiparametric estimates of local sorting patterns (following [Chiappori et al., 2026](#)). While there are regions of the bivariate income distribution where sorting is locally negative, 88% of couples in our job displacement sample are located in regions that exhibit local positive sorting. We therefore conclude that local negative sorting may contribute to the rematching patterns we document, but it is unlikely to be their primary explanation.

We discuss two broader implications of our findings for how marital sorting is interpreted. First, our multidimensional framework offers a unifying perspective: mechanisms that generate negative sorting on income (e.g., gains from household specialization; [Becker 1973](#)) and those that generate a positive cross-sectional correlation in spouses' incomes (e.g., homophily on education or other traits) need not counteract each other but may coexist and shape sorting along different dimensions. Sorting on income conditional on other characteristics can therefore be negative, consistent with Becker's specialization argument, while a positive unconditional correlation arises from sorting on correlated traits, such as through education homophily (e.g., [Chiappori et al. 2009, 2018](#)) or sorting on home productivities (e.g., [Goussé et al. 2017; Chiappori et al. 2017; Calvo et al. 2024](#)). A related implication is that unconditional correlations in spousal characteristics reflect sorting across all dimensions and therefore conflate multiple mechanisms, whereas exogenous variation in one characteristic isolates the mechanisms driving sorting along that dimension.

Second, we illustrate the quantitative relevance of our findings by calibrating one-dimensional and two-dimensional versions of our framework. In line with our theoretical results, only the two-dimensional model matches both the positive correlation in spouses' incomes and our evidence on rematching after job displacement. We then use both models to simulate how a rise in individual income inequality translates into between-household inequality. The models yield markedly different predictions: sorting on income amplifies the increase in between-household inequality in the one-dimensional model but dampens it in the two-dimensional one. Thus, how individual inequality passes through to household inequality critically depends on the marital sorting mechanisms at work.

Related Literature Our primary contribution is to the literature that studies marital sorting using equilibrium models with two-sided heterogeneity (e.g., [Becker, 1973, 1981; Wong, 2003; Choo and Siow, 2006; Greenwood, Guner, Kocharkov, and Santos, 2016; Goussé et al., 2017; Coles and Francesconi, 2019; Pilosoph and Wee, 2023; Almar, Friedrich, Reynoso, Schulz, and Vejin, 2025b](#)). We contribute by providing new empirical evidence from job displacements and assessing which mechanisms are consistent with the

patterns we document.³ Our findings point to a potential role for multidimensional matching and suggest that cross-sectional sorting patterns in income arise not only from sorting on income but also from sorting on other traits that correlate with income, as previously argued by [Becker \(1981\)](#), [Coles and Francesconi \(2011\)](#), and [Low \(2024\)](#).⁴

We also contribute to a growing literature on multidimensional marriage-market matching. Several papers study multidimensional frictionless matching (e.g., [Chiappori, Oreffice, and Quintana-Domeque, 2012](#); [Dupuy and Galichon, 2014](#); [Adda, Pinotti, and Tura, 2024](#); [Low, 2024](#)), while a smaller number of studies consider multidimensional matching with search frictions ([Coles and Francesconi, 2011, 2019](#); [Cheremukhin, Restrepo-Echavarria, and Tutino, 2024](#)). This work focuses on match formation and abstracts from divorce and remarriage.⁵ We complement this literature by extending the equilibrium search-and-matching model of [Shimer and Smith \(2000\)](#) to a multidimensional environment with recurring marriage and divorce. We also relate to recent work on non-uniform sorting, which allows the sign of sorting to vary across the type distribution (e.g., [Anderson and Smith, 2024](#); [Chiappori et al., 2026](#)). We draw on this literature to assess whether local negative sorting can account for the empirical patterns we document.

Finally, our paper relates to studies that use structural matching models to analyze how marital sorting responds to changes in the wage structure (e.g., [Fernández, Guner, and Knowles, 2005](#); [Greenwood et al., 2016](#); [Shephard, 2019](#); [Calvo et al., 2024](#)) or to policies such as taxation (e.g., [Frankel, 2014](#); [Bronson, Haanwinckel, and Mazzocco, 2024](#); [Gayle and Shephard, 2019](#)), social insurance (e.g., [Persson, 2020](#); [Low, Meghir, Pistaferri, and Voena, 2023](#); [Schulz and Siuda, 2023](#)), and divorce laws (e.g., [Fernández and Wong, 2016](#); [Reynoso, 2024](#); [Calvo, 2022](#)). Our simulations show that one-dimensional and multidimensional matching models can lead to markedly different counterfactual predictions even when both match the same cross-sectional sorting patterns.

Roadmap The paper proceeds as follows. Section 2 describes our data and empirical strategy, and Section 3 presents the empirical results. Section 4 introduces our conceptual framework, confronts its predictions with the empirical evidence, and examines the extent to which multidimensional matching or non-uniform sorting can reconcile theory and evidence. Section 5 discusses the broader implications and external validity of our findings, and Section 6 concludes.

³Our empirical analysis builds on previous work documenting that job displacement increases the risk of separation (e.g., [Charles and Stephens, 2004](#); [Eliason, 2012](#); [Huttunen and Kellokumpu, 2016](#); [Banzhaf, 2018](#)). We extend this work by studying both separation and subsequent rematching, interpreting treatment and selection effects through the lens of an equilibrium search-and-matching model of marriage and divorce.

⁴[Becker \(1981\)](#) suggests: “The positive correlation between wage rates of husbands and wives [...] may really be measuring the predicted positive correlation between a husband’s wage rate (or his non-market productivity) and his wife’s non-market productivity. Many unobserved variables, like intelligence, raise both wage rates and non-market productivity.”

⁵Closer to our approach, the labor literature has developed multidimensional search models with recurring formation and dissolution (e.g., [Lindenlaub and Postel-Vinay, 2021, 2023](#)).

2 Empirical Strategy

Our research design compares 146,523 workers who lose their jobs due to establishment closures with a control group of workers similar in observable characteristics but not affected by an establishment closure during our sample period. The following subsections describe our data, the definitions of establishment closures and job displacement, the matching procedure used to construct the control group, and the main empirical specifications we estimate.

2.1 Data

Our empirical analysis draws on Danish register data ([Statistics Denmark, 2025](#)) covering the entire Danish population between 1980 and 2007.⁶ The data are drawn from tax and social security records and include individual-level information on a wide range of demographic characteristics, employment status, labor income, occupation, work hours, the firm and establishment in which the individual is employed, marital histories, and children. In particular, the data record whether an individual is married or in a cohabiting relationship and provide an identifier for the individual's spouse or cohabiting partner. Cohabiting couples are defined as two opposite-sex individuals who share the same address, who either have at least one common child, or are less than 15 years apart in age, have no family relationship, and do not share housing with other adults apart from their partner.⁷

2.2 Establishment Closures

We define a closing establishment as one that stops operating, i.e., sheds its entire workforce within three years.⁸ The treatment year is defined as the first year in which a closing establishment sheds at least 10% of its workforce. The rationale for this definition is that layoffs that occur after this point are likely anticipated by the remaining workers. We exclude establishments with fewer than five employees in the treatment year. Our criteria yield 23,913 closing establishments. The average closing establishment employs 55 workers in the treatment year.

2.3 Treatment and Control Group

Treatment group We select the treatment group from individuals who are employed at a closing establishment in the treatment year and have at least three years of tenure. Additionally, we restrict the treatment group to individuals who are 28–48 years old in the treatment year and who were married or in a cohabiting relationship three years prior to the treatment year (i.e. in $t = -3$).⁹ Individuals who are

⁶Our sample ends in 2007 due to a change in the definition of family types in the Danish registers.

⁷This is the official definition of cohabiting couples used by Statistics Denmark. For previous studies that have relied on this definition, see (e.g., [Svarer, 2004](#); [Datta Gupta and Larsen, 2007, 2010](#); [Bruze, Svarer, and Weiss, 2015](#)). Our data do not allow us to identify same-sex couples, so we do not include same-sex couples in our analysis.

⁸By focusing on establishment closures, we minimize endogeneity concerns that arise when layoffs selectively target workers based on performance or ability (see, e.g., [Eliason and Storrie 2006](#)).

⁹Note that in our analysis, we consider an event time window ranging from five years prior to ten years after establishment closure. Within this time window the considered individuals are 23–58 years old.

employed at the same establishment as their spouse or cohabiting partner are excluded from the treatment sample. Our treatment group consists of the 146,523 individuals who meet these criteria.

Coarsened exact matching to select control group To construct a control group, we draw from individuals who, during the sample period, were never employed at an establishment within three years of its closure. We impose the same restrictions as for the treatment group on age, tenure, and relationship status, and also exclude individuals who work at the same establishment as their partner.

From this pool, we select one control individual for each treated individual using a coarsened exact matching (CEM) algorithm (Iacus, King, and Porro, 2012, 2019). The algorithm selects a control individual who, in the treatment year, provides an exact match on a set of observed characteristics. To make exact matching feasible, continuous variables are coarsened into discrete bins, and each treated individual is matched with a control whose characteristics fall into the same bin across all observables.¹⁰

Our CEM algorithm matches on gender, marital status (single, cohabiting, married, divorced), age, children (binary indicator), calendar year, occupation (6 categories), industry (9 groups), establishment size quintiles, and tenure quintiles. Matching is performed three years prior to establishment closure for all variables, except establishment size, which is matched five years prior to closure.¹¹ Our empirical analysis is based on a combined sample of 146,523 individuals in the treatment group and an equal number in the control group. For individuals in the control group, we refer to the year in which their matched treatment individual is displaced as “the year of placebo displacement”. Table B.1 reports summary statistics in the year before (actual or placebo) displacement for the treatment and control groups, separately for women and men. The table confirms that the treatment and control groups are well balanced on observables, including those that our CEM algorithm does not match on, such as own labor income and the partner’s labor income, age, and education.

2.4 Estimating Equations

We now introduce the empirical specifications used in our analysis. We begin with a standard difference-in-differences specification to estimate the impact of job displacement on labor market outcomes and relationship status. To study how job displacement affects rematching patterns, we use a second specification that compares the partners individuals match with before and after displacement. Since this comparison is only meaningful for individuals who change partners, we estimate this specification on the subsample who re-partner within the considered event-time window.¹² Unless otherwise noted, all

¹⁰Coarsened exact matching has favorable statistical properties in finite samples compared to methods such as propensity score matching (see Iacus et al., 2012) and the appeal of being straightforward to interpret. See, e.g., Azoulay, Graff Zivin, and Wang (2010) and Jäger and Heining (2022) for previous applications.

¹¹We match on establishment size five years before closure because differences between the treatment and control groups begin to emerge in this variable three years prior to closure.

¹²Conditioning on re-partnering aligns our empirical estimates with the model analog to which we compare them later on. Both the empirical estimate and the model analog capture the selection effect, whereby job displacement changes the composition of singles who rematch, as well as the treatment effect on the type of partner a given individual matches with.

empirical specifications are estimated separately by gender, i.e., on separate samples of displaced men and displaced women and their respective control groups.

Unconditional difference-in-differences specification We estimate the effect of job displacement on labor market and marriage market outcomes using the following difference-in-differences specification:

$$Y_{it} = \alpha + \sum_{\tau=-5}^{10} \psi_{\tau} \mathbb{1}\{t = \tau\} + \sum_{\tau=-5}^{10} \phi_{\tau} D_i \mathbb{1}\{t = \tau\} + \phi_D D_i + e_{it}, \quad (1)$$

where Y_{it} denotes the outcome for individual i in year t , measured relative to the year of actual or placebo displacement. D_i is an indicator for whether individual i is in the treatment group, and e_{it} is the residual error term. The coefficients ψ_{τ} capture event-time effects common to both groups, while the coefficients of interest, ϕ_{τ} , capture differential changes in outcomes over event-time in the treatment group relative to the control group. We normalize $\psi_{-3} = \phi_{-3} = 0$. The key identifying assumption is that absent job displacement, outcomes in the treatment and control group would have followed parallel trends.

Difference-in-differences specification conditional on rematching To estimate how job displacement affects rematching patterns, we use the following difference-in-differences specification, conditional on rematching. We consider the sample of all partners who are matched with a treatment or control individual between $t = -5$ and $t = 10$. Let j denote the partner for whom we observe the outcome Y_{jt} , and let i denote the corresponding treatment or control group individual matched with j in period t .¹³ We define an indicator variable $D_{i(j)}$, which equals 1 if the individual i matched with j belongs to the treatment group. We also define an indicator variable $Post_{ij}$, which equals 1 if the match between i and j was formed after i 's (actual or placebo) displacement. We then estimate the following regression:

$$Y_{jt} = \alpha + \psi Post_{ij} + \gamma D_{i(j)} Post_{ij} + \phi_D D_{i(j)} + e_{jt}. \quad (2)$$

The coefficient ϕ_D captures differences between partners of treatment and control group individuals in matches formed prior to (actual or placebo) displacement. The coefficient ψ captures general differences between partners matched before and after displacement. The key coefficient of interest, γ , captures whether the difference in partner characteristics between the treatment and the control group changes as we move from pre- to post-displacement matches. This specification identifies γ under the assumption that, absent job displacement, matching patterns would have evolved similarly across the two groups.

¹³We exclude partners who are matched with one treatment or control group individual at one point in time and with a different treatment or control group individual at another point in time (0.05% of our sample) because in that case the function $i(j)$ is not well defined.

3 Empirical Results

This section presents our empirical results. We document four main findings: (i) job displacement increases the risk of relationship dissolution; (ii) job displacement induces relationship dissolution particularly among individuals matched with lower-earning partners; (iii) displaced individuals are more likely to remain single after a breakup, relative to non-displaced individuals; (iv) displaced individuals are more likely to transition to higher-earning partners following a breakup, relative to non-displaced individuals. These findings hold for both displaced men and women, although some estimates for women are less precise and not all are statistically significant.¹⁴ Section 3.1 documents the effect of job displacement on employment and earnings. Sections 3.2–3.4 present the effects on relationship status, separations, and rematching patterns. Section 3.5 provides several robustness checks.

3.1 Labor Market Outcomes

We begin by documenting the effects of job displacement on employment and earnings. Following displacement, labor incomes drop sharply and remain persistently low for both displaced men and women. Figure 1A presents estimates of ϕ_τ from equation (1), for displaced men and women, respectively. The trend in labor income prior to displacement is flat for both genders, followed by a pronounced drop post-displacement, reaching a decline of 4.4% in $t = 5$ for men and 3.8% for women relative to pre-displacement earnings in $t = -3$. Labor incomes remain depressed, with women's incomes recovering only by $t = 10$ and men's incomes still significantly reduced after ten years.¹⁵ Figures 1B and B.1A and B show that the long-run decline in labor incomes is primarily driven by transitions to jobs with lower hourly wages, and to a lesser extent by reductions in work hours and employment. The gender difference in the income effect operates mainly through hourly wages, with employment responses very similar, and work hours responses only deviating marginally between men and women.

3.2 Relationship Status

Next, we examine how job displacement affects relationship status. Figures 2A-C show the dynamic effects of job displacement on the probability of being separated (Figure 2A), being single (Figure 2B) or being matched with a new partner (Figure 2C).

Figure 2A reports the effect on separations, using an indicator for not living with the same partner as in $t = -3$ as the outcome variable.¹⁶ We find a statistically significant increase in separations following

¹⁴For women, the point estimates underlying findings (i)-(iv) are of the same sign as for men, and most are similar in magnitude. However, some do not reach statistical significance, reflecting smaller samples underlying these estimates and, consequently, lower power.

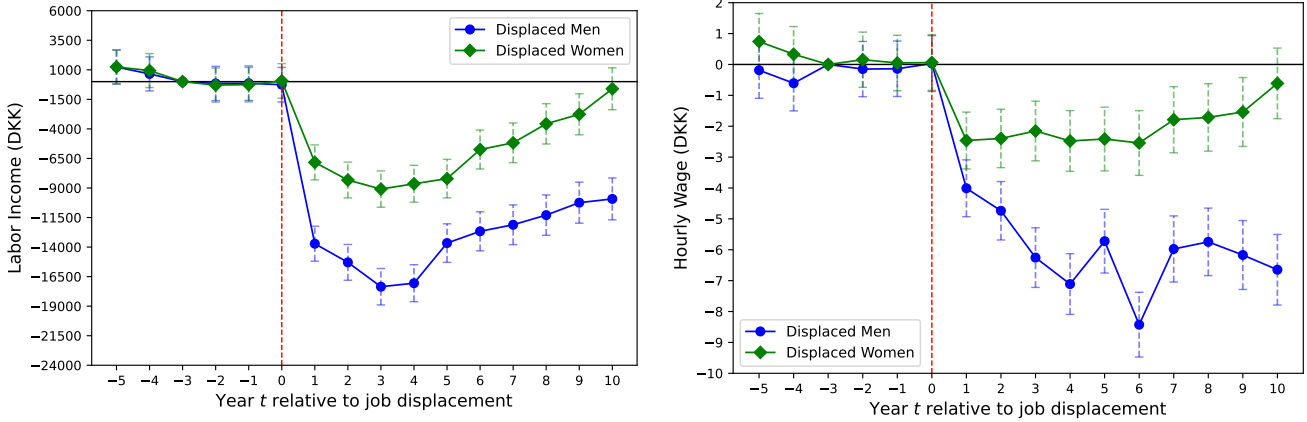
¹⁵A possible explanation for the gender difference is that women tend to have flatter job ladders and wage profiles, which may allow for a faster recovery following job displacement (see, e.g., Bronson and Thoursie, 2021; Goldin, 2021; Card, Devicienti, Rossi, and Weber, 2025).

¹⁶This includes both not cohabiting with any partner and living with a new partner different from the $t = -3$ partner. Since the Statistics Denmark definition of cohabitation requires partners to reside at the same address, cohabiting couples who move into

Figure 1: Labor Market Outcomes of Displaced Individuals:
Labor Income and Hourly Wages

(A) Labor Income (in DKK)

(B) Hourly Wage (in DKK)



Notes: The figure shows the impact of job displacement on annual labor income (Panel A) and hourly wages (Panel B). Labor income includes zeros for nonemployed individuals, while hourly wages are conditional on employment. Both outcomes are measured in DKK (CPI 2004). Dashed vertical lines around each point estimate represent 95% confidence intervals. The estimates correspond to ϕ_τ from Equation (1). All results are based on samples of men and women displaced due to establishment closures between 1980 and 2007, and matched control groups selected via coarsened exact matching. Sample selection criteria and the matching procedure are described in Subsection 2.3.

job displacement, reaching 0.01 for men and 0.02 for women by $t = 10$, corresponding to 6% and 11% relative to control-group separation rates of 0.18 and 0.19.¹⁷ Figure 2B and C decompose the effect of job displacement on separation into being single and living with a new partner. We use indicators for not living with a partner (Panel B) and for living with a partner different from the one in $t = -3$ (Panel C) as outcome variables.¹⁸ By construction, the estimates sum to the overall effects on separations. For men, approximately two-thirds of the increase is driven by individuals who are single post-breakup, and one-third by individuals living with new partners. For women, most of the increase in separations is driven by individuals who remain single rather than by individuals who find a new partner.

3.3 Post-Displacement Separation Patterns

Next, we examine how job displacement affects selection into relationship dissolution. To avoid capturing direct effects of displacement, we measure earnings prior to displacement in $t \in \{-5, \dots, -3\}$. We then compare the pre-displacement earnings of displaced individuals' partners in couples that dissolve within

separate households but continue their relationship are recorded as separating. Financial hardship following job displacement may induce residential relocation (as our robustness check in Section 3.5 confirms). However, it is not obvious why financial hardship would induce partners to move into separate households rather than relocating jointly, especially given the loss of economies of scale associated with maintaining two households. If such cases nevertheless occur, they would bias our estimated separation effects upward. They are less likely to affect the subsequent rematching patterns discussed below. In particular, this would require a couple to move apart due to financial hardship, remain in a committed relationship, while at the same time having one partner form a new household with another opposite-sex individual, with no other adults residing in the household.

¹⁷Huttunen and Kellokumpu (2016) and Eliason (2012) report comparable findings for Finland and Sweden, respectively.

¹⁸Over our ten-year observation window, around 5% of separating individuals reunite with their pre-displacement partner, whereas rematching with a different partner is 4.1 times more common. Since we define rematching as marriage or cohabitation with a partner distinct from the pre-displacement partner, our results are unaffected by couples who separate and subsequently reconcile.

ten years of actual or placebo displacement, across the treatment and control group. Figure 2D plots the average labor income of displaced individuals' partners in couples that experience a breakup, separately for the treatment and control group, relative to the respective group mean. First, the figure shows that the partners of displaced individuals who experience a breakup tend to have below-average earnings, regardless of gender or treatment status. Importantly, among couples who separate, women matched with displaced men have significantly lower earnings than those matched with control group men. The difference is statistically significant at -3063 DKK, 23% of the average income loss of a displaced man. This finding indicates that job displacement induces separations particularly among men matched with low-earning women. By contrast, for women the corresponding difference is smaller at -540 DKK, 9.2% of the average income loss of a displaced woman and not statistically significant, though of the same sign as the effect for men. We repeat the analysis for outcomes other than labor income. Figure B.2 shows no statistically significant differences in the age or number of children of separating partners, and only minimal differences (less than 0.04 years) in years of schooling.

3.4 Rematching Patterns of Displaced Men and Women

We now examine how job displacement affects the types of partners displaced individuals rematch with after a breakup. We begin by comparing the propensity to transition to a higher-, similarly-, or lower-earning new partner relative to the pre-displacement partner in the treatment and the control group. To this end, we estimate Equation (1) using three indicator variables as outcomes:

$$\begin{aligned} Y_{it}^+ &= \mathbf{1} \{ Y_{it} \geq (1 + \rho) Y_{it}^{\text{pre}} \} \cdot D_{R,it}, \\ Y_{it}^0 &= \mathbf{1} \{ (1 - \rho) Y_{it}^{\text{pre}} < Y_{it} < (1 + \rho) Y_{it}^{\text{pre}} \} \cdot D_{R,it}, \\ Y_{it}^- &= \mathbf{1} \{ Y_{it} \leq (1 - \rho) Y_{it}^{\text{pre}} \} \cdot D_{R,it}, \end{aligned}$$

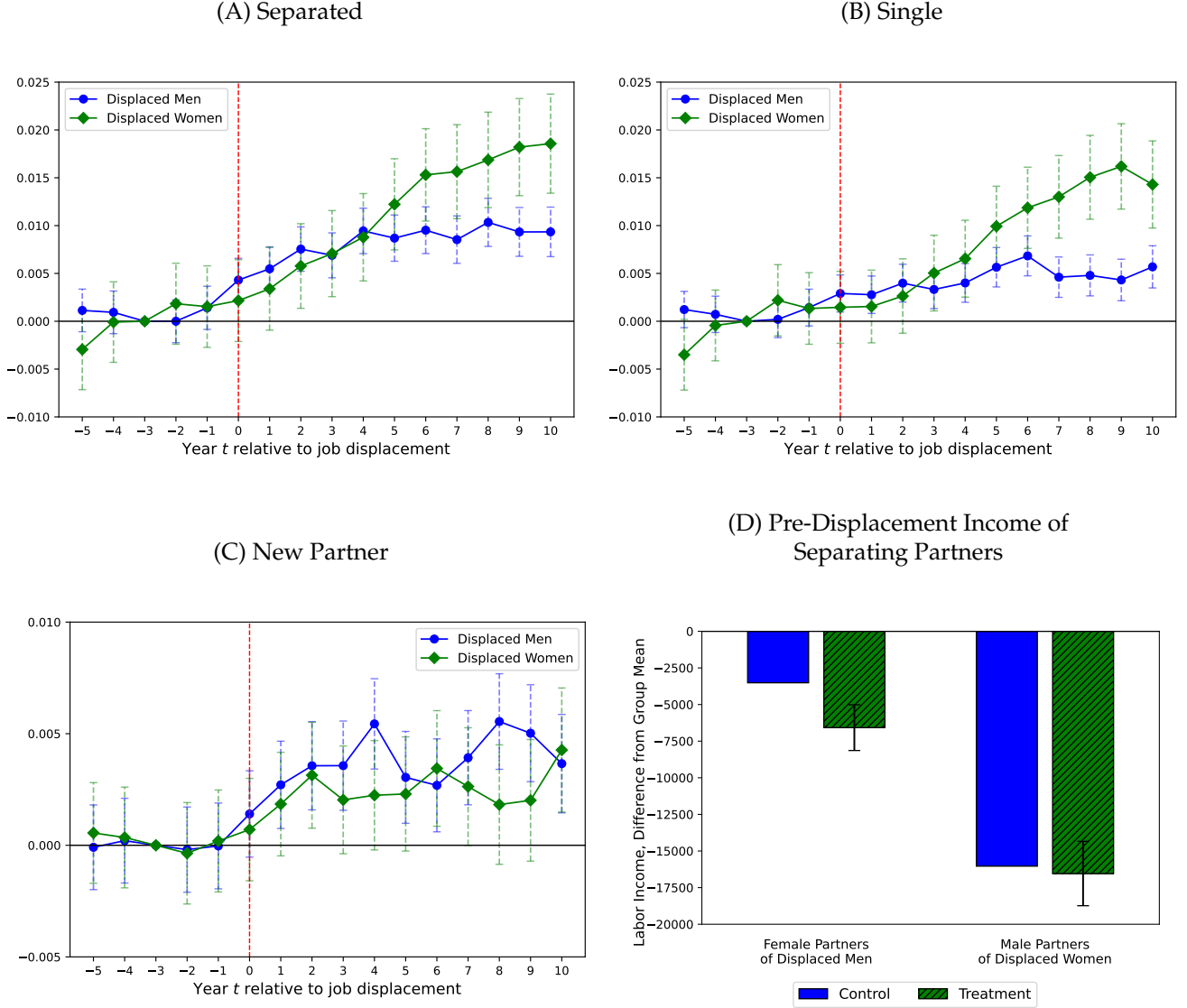
where Y_{it} denotes the earnings of the partner with whom individual i is matched in period t , Y_{it}^{pre} denotes the earnings of their pre-displacement partner in $t = -3$, and Y_{it}^+ , Y_{it}^0 , and Y_{it}^- indicate transitions to a higher-, similarly-, or lower-earning new partner, defined relative to the pre-displacement partner's earnings with threshold ρ .¹⁹ $D_{R,it}$ indicates whether individual i is living with a partner different from the one in $t = -3$. For our main analysis, we set $\rho = 0.05$ and use annual labor income as the measure of earnings.²⁰ We also report average effects of job displacement on these outcomes from simple difference-in-differences specifications that collapse the event time into a pre-displacement ($t \leq 0$) and a post-displacement period ($t > 0$).

Figures 3A–C present estimates of ϕ_τ from Equation (1), using Y_{it}^+ , Y_{it}^0 , and Y_{it}^- as outcome variables. For men, the results show statistically significant increases in the likelihood of transitioning to higher-

¹⁹We compare the earnings of pre- and post-displacement partners in the same time period t , so our estimates are not affected by time trends that affect both partners.

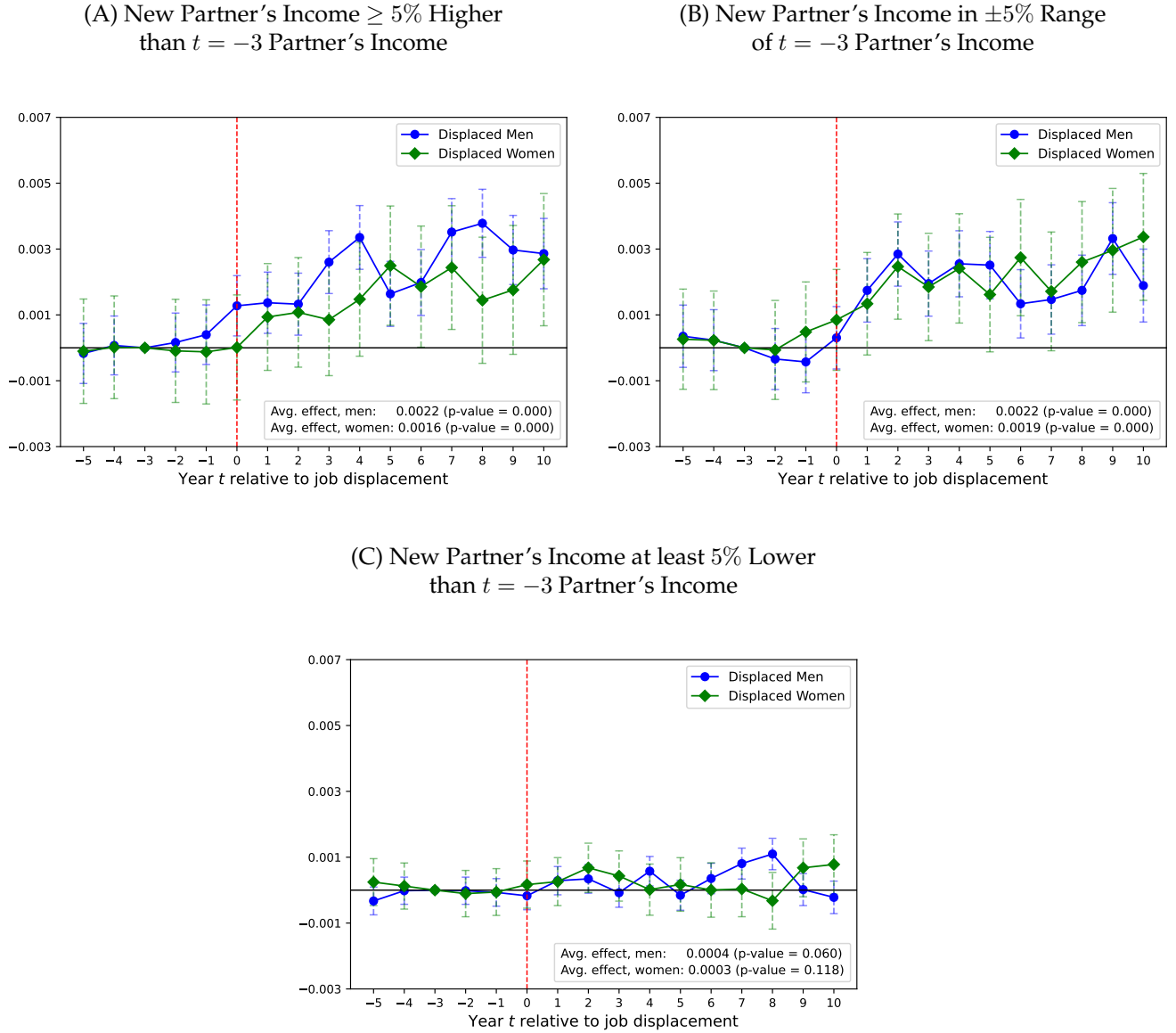
²⁰Varying the value of ρ or using hourly wages instead of annual labor income yields qualitatively similar results. Figures B.3 and B.4 report results for $\rho = 0.1$ and for using hourly wages.

Figure 2: Relationship Status of Displaced Individuals and Pre-Displacement Income of Separating Partners



Notes: Panels A–C show the effects of job displacement on different measures of relationship status. Panel A displays the effect on the probability of being separated from the pre-displacement partner, defined as the partner at $t = -3$. Panel B shows the effect on the probability of being single (i.e., unmarried and not cohabiting), and Panel C shows the effect on the probability of being matched (married or cohabiting) with a new partner who is distinct from the pre-displacement partner. Dashed vertical lines around point estimates indicate 95% confidence intervals. The values in Panels A–C correspond to coefficient estimates of ϕ_τ from Equation (1). Panel D displays the pre-displacement labor income of the partners of displaced individuals in couples that subsequently experience a breakup. Each bar represents average labor income in $t \in \{-5, \dots, -3\}$ (before the displaced individual's actual or placebo displacement) for partners of displaced individuals who separate between $t = 0$ and $t = 10$. All values in Panel D are differences from the respective (treatment or control) group mean. The underlying sample for all panels consists of men and women displaced as part of establishment closures between 1980–2007, and matched control groups selected by coarsened exact matching. The specific sample selection criteria and matching procedure are described in Subsection 2.3.

Figure 3: Rematching Patterns of Displaced Individuals:
New Partners' vs. Old Partners' Income



Notes: The figure shows the effect of job displacement on rematching patterns following a breakup. Panel A displays the effect on the probability of matching with a new partner with at least 5% higher labor income than that of the pre-displacement partner (defined as the partner at $t = -3$). Panel B shows the effect on the probability of matching with a new partner with labor income between -5% and +5% of the pre-displacement partner's labor income. Panel C shows the effect on the probability of matching with a new partner with at least 5% lower labor income than that of the pre-displacement partner. The estimates correspond to ϕ_τ from Equation (1). Dashed vertical lines around point estimates indicate 95% confidence intervals. Average effects are estimated from regressions of each outcome on a treatment indicator, a post-displacement ($t > 0$) indicator, and their interaction. All estimates are based on samples of men and women displaced due to establishment closures between 1980 and 2007 and matched control groups selected by coarsened exact matching. Sample selection criteria and the matching procedure are described in Subsection 2.3.

and similarly-earning partners following job displacement, and no effect on transitions to lower-earning partners. For women, the point estimates exhibit the same qualitative pattern and are of similar magnitude, although they are estimated less precisely and are therefore not always statistically significant; as for men, we find no effect on transitions to lower-earning partners. Taken together, these results suggest that job displacement shifts rematching patterns toward higher-earning partners for both men and women.

Table 1: Rematching Patterns of Displaced Men and Women:
New Partners' Income, Wage, and Work Hours

Partners' Outcomes	Labor Income	Hourly Wage	Work Hours
<i>Panel (a): Displaced men</i>			
Treated $\times$ Post-Displacement, γ	3278.61** (1614.71)	3.98*** (1.46)	-0.12 (0.13)
No. of observations	108,982	60,362	53,704
<i>Panel (b): Displaced women</i>			
Treated $\times$ Post-Displacement, γ	2282.65 (2342.12)	0.72 (1.52)	0.09 (0.17)
No. of observations	68,908	46,628	39,664

Notes: The table shows the effect of job displacement on the types of partners displaced individuals transition to, measured by partners' labor income, hourly wage (conditional on employment), and weekly work hours (including zeros for non-employed individuals). Panel (a) reports the effect of men's job displacement on the characteristics of new female partners. Panel (b) reports the effect of women's job displacement on the characteristics of new male partners. The table reports coefficient estimates of γ from Equation (2). Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To provide a continuous measure of the change in rematching patterns following job displacement, we estimate specification (2). Table 1 reports estimates of γ , using annual labor income, hourly wages, and work hours as outcome variables.²¹ The estimate in Column (1) shows that displaced individuals who transition to a new partner experience an increase in partner earnings relative to individuals in the control group. The point estimates are positive for both men and women and are of similar magnitude, although only the estimate for men is statistically significant, reflecting lower power due to the smaller sample of women who rematch. Scaling the estimates by the average income loss from job displacement (-13,332 DKK for men, -5,897 DKK for women), yields a ratio of -0.25 for men and -0.39 for women. This implies that a one-unit loss in own income is associated with matching with a 0.25 unit higher-earning partner for men and 0.39 units for women. Columns (2) and (3) of Table 1 present estimates using hourly wages and work hours as outcome variables, indicating that the estimated increase in partner income is driven by differences in partners' hourly wages rather than their labor supply.

To assess whether job displacement affects rematching patterns along other dimensions, we repeat the analysis using partner age, education (years of schooling), and the number of children from previous relationships as outcome variables. Table B.2 shows that displaced men tend to rematch with slightly younger partners who have more children, though the effects are modest in size (-0.27 years for partner age and 0.04 additional children). The effect on partner education is small and statistically insignificant.

²¹While labor income is observed for our full sample, wages and work hours are only recorded for subsamples of the population. In particular, we observe hourly wages for jobs in the public sector and in private firms with at least 10 employees, whereas work hours are available only for the individual's main job in November (defined as the job with the highest income or hours that month), provided that the job was subject to pension contributions (ATP) (see Lund and Vejlin, 2016, for details).

For displaced women, all corresponding estimates are statistically insignificant.

3.5 Robustness and Additional Results

This subsection presents robustness checks and additional results supporting our main findings. We show that our results are not driven by endogenous labor supply adjustments, residential moves to marriage markets with a more favorable composition, or particularly strong labor market recoveries for some displaced individuals. We also provide back-of-the-envelope calculations suggesting that establishment closures are unlikely to generate substantial equilibrium effects, examine to what extent the separation patterns we document are driven by women's financial independence, and present additional evidence on separation patterns and the rematching behavior of the partners of displaced individuals.

Endogenous Labor Supply A potential concern is that labor income is not an exogenous trait, but a consequence of endogenous labor supply decisions. This raises the question to what extent our empirical findings are driven by partners who endogenously adjust their work hours. However, our empirical results for labor income, work hours, and hourly wages show that endogenous adjustments of labor supply do not appear to be the primary driver of our findings. In particular, Figure B.4 shows that the main patterns hold when we consider hourly wages as the outcome, for which this concern is less relevant. Moreover, Table 1 shows that our empirical finding (iv) is mainly driven by displaced individuals matching with new partners who have higher hourly wages, while differences in partners' work hours choices are small (0.09 weekly hours for displaced women and -0.12 for displaced men) and insignificant.

Moves Across Municipalities and Local Marriage Market Conditions We examine whether job displacement affects residential mobility and, if so, whether such moves are directed toward municipalities with more favorable marriage market conditions. Figure B.5A presents estimates of Equation (1), using as the outcome an indicator for whether the individual resides in a municipality different from that in $t = -3$ for at least three consecutive years. Displacement leads to a statistically significant increase in the likelihood of having moved, reaching approximately 1.5 percentage points for both displaced men and women. To assess whether these moves are directed toward municipalities with more favorable marriage market conditions, we consider several additional outcomes: the average earnings of single women and single men, and the local sex ratio (defined as the number of single women divided by the number of single men), each measured as the difference between the individual's municipality of residence in year t and in $t = -3$. Figures B.5B and C show no statistically significant effect of displacement on any of these outcomes, except for small effects (of about -0.001) on the female sex ratio in event-study periods $t = 9$ and $t = 10$. Taken together, while displaced individuals are more likely to move across municipalities, we find no evidence that these moves systematically improve local marriage market conditions.

Labor Market Recovery Among Rematching Individuals Another potential concern is that the upward rematching patterns we document arise because a subset of displaced individuals experiences particularly strong labor-market recoveries and ultimately earns more than before displacement. This subgroup might be more likely to rematch with a high-earning new partner and thereby drive our results. Figure B.6 addresses this concern by comparing the labor-market trajectories of displaced individuals who do and do not find a new partner. The figure shows that displaced individuals who rematch experience substantial and persistent declines in labor income and hourly wages that are broadly comparable to those of displaced individuals who do not rematch. Additionally, we reestimate the displacement effects on rematching patterns after excluding displaced individuals with exceptionally strong post-displacement income growth. Tables B.3 and B.4 show that doing so leaves the sign and statistical significance of our main estimates unchanged. If anything, the effects on partner income become slightly larger. Together, these findings suggest that the rematching patterns we document are unlikely to be driven by a subgroup of displaced individuals who experience strong labor-market recoveries.

Labor Market and Marriage Market Equilibrium Effects of Establishment Closures We assess the likelihood that establishment closures exert notable labor-market or marriage-market equilibrium effects by using a back-of-the-envelope calculation. The average closing establishment in our sample employs 55 workers, 0.6% of the average local labor force in municipalities with an establishment closure. The rate at which displaced workers separate from their partners within the 10 years following establishment closure is 0.2 in our sample. The average inflow of singles into the marriage market over 10 years due to an establishment closure is thus approximately $0.2 \times 55 = 11$. This amounts to an influx of 1.5% relative to the average local population of singles who are 28-48 years old in municipalities that contain a closing establishment. We conclude that the inflows of displaced workers into the labor market and of newly single individuals into the marriage market are relatively small and, therefore, unlikely to generate general equilibrium effects.

Separation Patterns and Women's Financial Independence We also examine whether the displacement-induced increase in separations is stronger among couples in which the female partner is employed or has high earnings before her partner's displacement, as would be expected if more financially independent women are more likely to separate upon job displacement of their partner. To this end, we estimate specifications that allow the treatment effect to vary with the female partner's pre-displacement employment status and labor income, controlling for the male partner's income throughout (Appendix Table B.5, Columns 2-4). Across all specifications, the interaction terms are negative, indicating that displacement-induced separations are concentrated among couples in which the female partner is non-employed or has low pre-displacement earnings. This finding mirrors the descriptive evidence in Figure 2D, and is inconsistent with the hypothesis that more financially independent women are more likely to separate following their partner's displacement. Instead, the pattern we find is consistent with the male partner's displacement

generating particularly severe distress in couples with little or no secondary income to buffer the shock from displacement, making rematching with a higher-earning partner more attractive for both spouses.

Rematching Patterns of Displaced Individuals' Partners We also examine the rematching patterns of the partners of displaced individuals. We reestimate empirical specifications (1) and (2) but follow the outcomes of displaced individuals' partners rather than those of the displaced individuals themselves.²² Figure B.7 shows that the labor incomes of displaced individuals' partners remain unchanged around displacement. As expected, the partners experience an increase in separations that mirrors the corresponding increase among displaced individuals, driven both by an increase in the share remaining single and by an increase in the share forming a new union (see Figure B.8). Appendix Table B.6 shows that former partners of displaced individuals who rematch with a new partner do not experience statistically significant upward or downward rematching. On average, they rematch with new partners whose earnings are similar to those of the displaced individual prior to displacement.

Our interpretation of these findings (which we formalize in Section 4) is that job displacement changes the displaced individual's type while leaving the type of the partner unaffected. Consequently, we expect displaced individuals to rematch with systematically different partners, whereas their former partners are expected to rematch with individuals whose type is similar to that of the displaced worker prior to displacement. This is exactly what we see in the data. By contrast, if job displacement did not change the affected partner's type but was merely a temporary adverse shock to relationship quality that causes the couple to separate, both the displaced individual and the former partner would be expected to rematch with similar partners. The data do not support this alternative hypothesis about the link between job displacement, selection into divorce, and remarriage.

4 Confronting Theory and Evidence

We interpret our empirical findings within a general search-and-matching framework of the marriage market. We adopt a frictional setting because it naturally captures endogenous divorce and rematching decisions, which are central to our analysis.²³ Importantly, the framework accounts for both endogenous selection into divorce and remarriage and changes in search behavior following displacement, allowing us to interpret the combination of treatment and selection effects that we find empirically.

As a starting point, we focus on either globally positive assortative matching (PAM) or globally negative assortative matching (NAM) equilibria in a one-dimensional search-and-matching environment. These benchmark cases are central to much of the marriage-market matching literature. We show formally that

²²Specification (2) estimates rematching patterns by comparing pre- and post-displacement partners. Since the pre-displacement partner in this comparison is the displaced individual, we define the outcome variables for the rematching analysis using pre-displacement characteristics (measured at $t = -3$) to avoid mechanically capturing the direct effects of displacement.

²³Frictional search models have been widely used to study marriage markets in settings where divorce and rematching play an important role (e.g., Goussé et al. 2017; Coles and Francesconi 2019; Chiappori 2017; Shephard 2019).

under this benchmark it is difficult to reconcile the model's predictions with the specific combination of empirical patterns we find. We then explore the extent to which multidimensional matching or non-uniform sorting can account for these patterns.

4.1 Conceptual Framework

This section introduces our conceptual framework. We build on the frictional extension of Becker's (1973) assignment model developed by Shimer and Smith (2000), which features two-sided heterogeneity and transferable utility. The framework is deliberately general, placing minimal structure on functional forms and the distributions of agent types.

4.1.1 Setup

We consider a two-sided matching environment populated by women and men, denoted by f and m , respectively. Time is continuous and discounted at rate r . Women and men are characterized by their types, $q_f \in Q_f$ and $q_m \in Q_m$. We begin with the benchmark case of one-dimensional heterogeneity, where $q_f \in [\underline{q}, \bar{q}]$ and $q_m \in [\underline{q}, \bar{q}]$. Later, we discuss the extension to multidimensional heterogeneity, where types are vectors $q_f, q_m \in \prod_{k=1}^K [\underline{q}_k, \bar{q}_k]$, and each dimension represents a distinct attribute. Search is random. Denote by G_f and G_m the cumulative distribution functions (CDFs) of single women's and men's types, respectively.²⁴ At rate λ_f , a single woman meets a single man drawn from G_m . Conversely, at rate λ_m , a single man meets a single woman drawn from G_f . Following Shimer and Smith (2000), we assume that meeting rates for men and women are proportional to the mass of singles of the opposite gender: $\lambda_f = \bar{\lambda} \int dG_m(q_m)$ and $\lambda_m = \bar{\lambda} \int dG_f(q_f)$, where $\bar{\lambda}$ is a scaling factor that governs how meeting rates depend on the mass of singles on the other side of the marriage market. Upon meeting, agents observe each other's types and jointly decide whether to form a match or continue searching.

4.1.2 Flow Utilities

The flow utility of a single agent of gender $g \in \{f, m\}$ and type q_g is given by $u_g^0(q_g)$, and is weakly increasing in q_g .²⁵ When matched, women and men receive flow utilities $u_f^1(q_f, q_m)$ and $u_m^1(q_f, q_m)$, which are weakly increasing in both own and partner's type. The flow value of a match, $f(q_f, q_m)$, is the sum of the partners' individual flow utilities,

$$f(q_f, q_m) = u_f^1(q_f, q_m) + u_m^1(q_f, q_m). \quad (3)$$

We do not impose functional form assumptions on $u_g^1(q_f, q_m)$ or $u_g^0(q_g)$. The framework is therefore flexible, nesting a wide range of models of intra-period household decision-making, as these flow utilities can be

²⁴Note that G_f and G_m are endogenous equilibrium objects.

²⁵Throughout, model objects with superscript 0 refer to singles, whereas those with superscript 1 refer to matched agents.

interpreted as indirect utilities derived from an underlying model of household behavior.²⁶

4.1.3 Bellman Equations, Matching, and Surplus Division

A model agent's decision problem is characterized by two Bellman equations. The value of being a single man of type q_m is

$$rV_m^0(q_m) = u_m^0(q_m) + \lambda_m \int \max\{V_m^1(q_f, q_m) - V_m^0(q_m), 0\} dG_f(q_f), \quad (4)$$

where $V_m^1(q_f, q_m)$ denotes the value of matching with a woman of type q_f . This value satisfies

$$rV_m^1(q_f, q_m) = u_m^1(q_f, q_m) + t_m + \delta(V_m^0(q_m) - V_m^1(q_f, q_m)), \quad (5)$$

where t_m denotes the intra-household utility transfer.²⁷ δ denotes the exogenous separation rate, capturing match dissolution unrelated to job displacement.²⁸ We additionally model endogenous separations triggered by unexpected job displacement, as described in Sections 4.1.5 and 4.1.6.²⁹

The marital surplus is defined as

$$S(q_f, q_m) = V_m^1(q_f, q_m) + V_f^1(q_f, q_m) - V_m^0(q_m) - V_f^0(q_f). \quad (6)$$

Under transferable utility, the marital surplus can be freely redistributed between spouses. Accordingly, a couple chooses to match if and only if the surplus is weakly positive, $S(q_f, q_m) \geq 0$. We close the model by assuming that the surplus is divided according to Nash bargaining (see Appendix E.1 for details).

4.1.4 Equilibrium and Sorting

For the one-dimensional case, $K = 1$, Shimer and Smith (2000) prove the existence of a steady-state equilibrium and characterize equilibrium sorting by generalizing Becker (1973)'s concepts of PAM and NAM to a frictional matching environment.³⁰ Denote by $\mathcal{M}(q_m)$ the matching set for a man of type q_m , i.e., the set of female types with whom he matches upon meeting. Shimer and Smith (2000) show that matching sets take the form of closed intervals, $[a(q_m), b(q_m)]$, in equilibrium. Under PAM, the interval bounds $a(q_m)$ and $b(q_m)$ are weakly increasing in q_m , while under NAM, they are weakly decreasing.³¹

²⁶For an example of this approach, see Goussé et al. (2017) who embed a model of intra-period household time and resource allocation into the Shimer and Smith (2000) search and matching framework.

²⁷The value functions for single women and for type- q_f women matched with type- q_m men are defined analogously to (4) and (5). Transfers are restricted to be net-zero, i.e., $t_m = -t_f$.

²⁸In contrast to most labor markets, marriage markets are characterized by much lower turnover. In our calibration in Section 5.2, which targets moments from the Danish register data, the annual separation rate is $\delta = 0.06$, implying an average relationship duration of 16.7 years, and single individuals encounter a potential partner only once every four years on average.

²⁹Since job displacement is unexpected, it does not occur at a Poisson rate in the Bellman equations.

³⁰We state the formal definition of PAM and NAM by Shimer and Smith (2000) in Appendix A.

³¹See Shimer and Smith (2000), Propositions 2 and 3. Analogous results hold for the matching sets of women.

These relationships allow us to analyze how an unexpected shock to an agent's type (e.g., job displacement) affects their matching set and, in turn, which couples dissolve and which new couples form in response.

By these relationships, PAM and NAM also imply the following cross-sectional correlations:

$$\text{PAM} \Rightarrow \text{Corr}(q_f, q_m) \geq 0, \quad (7)$$

$$\text{NAM} \Rightarrow \text{Corr}(q_f, q_m) \leq 0. \quad (8)$$

These relationships form the basis for interpreting positive correlations in spouses' attributes, such as earnings and educational attainment, as evidence in favor of PAM and against NAM, assuming that agent types map into these attributes.

4.1.5 Job Displacement and Marriage Market Transitions

We connect this conceptual framework to our empirical setting by modeling job displacement as a permanent, unexpected reduction in an agent's type.³² We assume that agent types map into earnings potential, which we measure empirically using labor income and hourly wages.³³

For expositional convenience, we formulate the analysis in terms of male job displacement, although analogous results hold for female job displacement. Formally, a displaced man of type q_m experiences a permanent type reduction to $q_m - d$, where $d > 0$.

Job displacement triggers an endogenous separation whenever the marital surplus becomes negative following the type reduction, $S(q_f, q_m - d) < 0$. Following separation, the displaced man remarries if he meets a single woman of type q_f , such that $S(q_f, q_m - d) \geq 0$. We derive predictions that characterize which couple types (q_f, q_m) separate endogenously following male job displacement, and which new matches form.

To connect the model to our empirical design, consider a treatment group that experiences job displacement and a control group that does not. We denote by D an indicator equal to one for displaced individuals and zero otherwise. Let t_0 denote the last period before displacement and $\tau > t_0$ a later period after displacement. Assume that all men in both groups are initially matched with a female partner in period t_0 . Further, let D_B indicate whether an individual separates from their pre-displacement partner and D_R whether they rematch with a new partner.

³²The assumption of a permanent type reduction can be relaxed without affecting our qualitative conclusions. If displaced individuals recover to their pre-displacement type at rate $\mu > 0$, they are at any point in time a mixture of recovered and unrecovered workers. Since the treatment effect is zero for recovered workers and has the same sign as under permanent type reduction for unrecovered workers, the expected treatment effect retains the same sign as in the benchmark model.

³³Formally, we assume strictly increasing mappings from agent types to earnings. Throughout, we report results for both labor income and hourly wages, and all findings hold for each measure. The assumption that type reductions are persistent is consistent with the extensive literature documenting long-run earnings losses following job displacement (e.g., [Jacobson, LaLonde, and Sullivan, 1993](#); [Sullivan and von Wachter, 2009](#)). Beyond reductions in earnings potential, job displacement may have other consequences, such as mental health effects. To the extent that these effects reduce an individual's attractiveness as a partner, they would be expected to attenuate the upward rematching on income that we document, implying that our estimates are conservative.

We derive predictions about the following treatment and selection effects.

1. Treatment effect on breakup risk:

$$\gamma_B = P(D_B = 1|D = 1) - P(D_B = 1|D = 0),$$

which captures the effect of job displacement on breakup risk.

2. Selection effect on the types of men and women who break up:

$$\begin{aligned}\gamma_{q_m|B} &= \mathbb{E}[q_m(t_0)|D_B = 1, D = 1] - \mathbb{E}[q_m(t_0)|D_B = 1, D = 0], \\ \gamma_{q_f|B} &= \mathbb{E}[q_f(t_0)|D_B = 1, D = 1] - \mathbb{E}[q_f(t_0)|D_B = 1, D = 0],\end{aligned}$$

which capture how job displacement affects the composition of dissolving relationships.

3. Combined selection and treatment effect on the probability of remaining single after a breakup:

$$\gamma_{R=0|B} = P(D_R = 0|D_B = 1, D = 1) - P(D_R = 0|D_B = 1, D = 0),$$

which reflects both changes in the composition of singles and in a given man's likelihood of rematching, as a result of job displacement.

4. Combined selection and treatment effect on rematching patterns:

$$\gamma_{\Delta q_f|R} = \mathbb{E}[q_f(\tau) - q_f(t_0)|D_R = 1, D_B = 1, D = 1] - \mathbb{E}[q_f(\tau) - q_f(t_0)|D_R = 1, D_B = 1, D = 0],$$

which reflects changes in the composition of singles who rematch, as well as in the average type of partner a given man is expected to match with following job displacement.

The following proposition summarizes the model's predictions regarding the treatment and selection effects of job displacement under PAM and NAM.³⁴ Throughout, we assume that the treatment group is small (i.e., of measure zero), so that job displacements affect the displaced agents but do not induce a transition to a different steady-state equilibrium.³⁵ Effects 3 and 4 reflect a combination of selection and treatment effects. The proof shows that under both PAM and NAM, both treatment and selection effects point in the same direction. As a result, the corresponding model predictions provide a natural benchmark for our empirical estimates, which likewise capture both treatment and selection effects.

Proposition 1. *Consider the one-dimensional case, $K = 1$, of the described matching environment in steady-state equilibrium.*

³⁴For proofs and derivations, see Appendix A.

³⁵Section 3.5 provides back-of-the-envelope calculations suggesting that establishment closures are unlikely to generate substantial equilibrium effects in the labor or marriage market, supporting this assumption.

Under either PAM or NAM:

1. Job displacement increases the breakup risk: $\gamma_B \geq 0$.
2. Job displacement may increase or decrease the probability of remaining single after a break up:
 $\gamma_{R=0|B}$ may be positive or negative.

Under PAM:

- 3.-a Job displacement leads men to rematch with women of lower type: $\gamma_{\Delta q_f|R} \leq 0$.
- 4.-a The association between job displacement and partner type is bounded above: $\gamma_{\Delta q_f|R} \leq \bar{\gamma}_{\Delta q_f|R}$.
The upper bound is given by

$$\bar{\gamma}_{\Delta q_f|R} = - \int \int_0^d \frac{\partial \mathbb{E}[q_f|q_m]}{\partial q_m} \Big|_{q_m=q-x} dx dF(q|D_R = 1, D_B = 1, D = 1) \leq 0.$$

- 5.-a If $F(q_m|D_B = 1, D = 1) \leq F(q_m|D_B = 1, D = 0)$ holds additionally, then, on average, women from whom displaced men separate are of higher type than women from whom non-displaced men separate: $\gamma_{q_f|B} \geq 0$.

Under NAM:

- 3.-b Job displacement leads men to rematch with women of higher type: $\gamma_{\Delta q_f|R} \geq 0$.
- 4.-b The association between job displacement and partner type is bounded below: $\gamma_{\Delta q_f|R} \geq \underline{\gamma}_{\Delta q_f|R}$.
The lower bound is given by

$$\underline{\gamma}_{\Delta q_f|R} = - \int \int_0^d \frac{\partial \mathbb{E}[q_f|q_m]}{\partial q_m} \Big|_{q_m=q-x} dx dF(q|D_R = 1, D_B = 1, D = 1) \geq 0.$$

- 5.-b If $F(q_m|D_B = 1, D = 1) \leq F(q_m|D_B = 1, D = 0)$ holds additionally, then, on average, women from whom displaced men separate are of lower type than women from whom non-displaced men separate: $\gamma_{q_f|B} \leq 0$.

Relationships (3.-a) and (5.-a) imply that under PAM, displaced men tend to separate from higher-type partners and rematch with lower-type partners. By contrast, relationships (3.-b) and (5.-b) imply that under NAM, displaced men tend to separate from lower-type partners and rematch with higher-type partners. Relationships (4.-a) and (4.-b) further imply that these rematching responses are bounded away from zero. Because the bounds depend on the slope of $\mathbb{E}[q_f|q_m]$ in q_m , they generate additional quantitative restrictions that are empirically testable.

We note that the intuition underlying Proposition 1 extends beyond our specific structural framework. The key property underlying the proof is that the expected partner type a person matches with, $E[q_f|q_m]$, is weakly increasing in own type, q_m , under global PAM and weakly decreasing under global NAM. This

property follows directly from the definition of PAM and NAM by [Shimer and Smith \(2000\)](#) and does not depend on assumptions regarding search frictions, utility functions, or the equilibrium concept in our framework. The intuition underlying our results therefore extends to any model that shares this property, which typically follows directly from the definition of global PAM and NAM in one-dimensional models.³⁶

4.1.6 Mechanisms: Endogenous Separations and Rematching

To provide the intuition behind Proposition 1, we now describe the underlying model mechanisms. When a matched male agent loses his job, his type q_m decreases to $q_m - d$, reducing both partners' flow utilities from marriage. The female partner's outside option remains unchanged, while the male partner's outside option declines.³⁷ Transfers adjust to compensate the partner who experienced the larger decline in marital surplus share. If the surplus turns negative, there is no level of transfers such that both partners prefer staying together over breaking up, and the couple separates. This provides the intuition for part 1. of Proposition 1.³⁸

Which couples are most likely to separate depends on whether partners' types are complements or substitutes in the marital surplus. Intuitively, under PAM, high-type men and high-type women are complements. Consequently, a decline in male type reduces marital surplus most strongly when the female partner has a high type, making these couples more likely to separate following job loss. Under NAM, by contrast, types are substitutes: a high-type woman can partly compensate for a decline in her partner's type, making the marital surplus more resilient to male job loss. Consequently, a decline in male type reduces marital surplus most strongly among couples with low-type women, and separations are concentrated there. This provides the intuition for parts 5.-a and 5.-b of Proposition 1.

The same logic determines rematching patterns. Under PAM, high-type men and high-type women are complements: a decline in male type particularly reduces the gains from matching with a higher-type woman, while increasing the transfer she requires. Both forces push toward downward rematching, consistent with part 3.-a. Under NAM, by contrast, types are substitutes: a higher-type woman can partly compensate for the decline in male type, making such matches relatively more attractive after job loss. This leads displaced men to rematch with higher-type partners, consistent with part 3.-b.

Finally, even when displaced men rematch with higher-type women under NAM, this does not imply that

³⁶While we are not aware of frictionless extensions of [Becker \(1973\)](#) or [Choo and Siow \(2006\)](#) that incorporate divorce and remarriage, our results would naturally extend to such frictionless models provided that $E[q_f | q_m]$ remains weakly monotone in own type, q_m .

³⁷This logic is closely related to limited commitment models, where bargaining power adjusts when individual participation constraints are violated (e.g., [Mazzocco 2007](#); [Voena 2015](#); [Foerster 2024](#); [Verriest 2024](#)). Our model differs in that, under transferable utility, redistribution occurs through transfers rather than bargaining power, and transfers adjust immediately following job loss (as would be the case under no commitment).

³⁸In limited commitment models of marriage and divorce, it is common to interpret a partner whose participation constraint is violated as initiating the divorce (this may be either partner or both simultaneously). We can adopt a similar interpretation and define a partner for whom $u_g^1(q_f, q_m) < rV_g^0(q_g)$ as initiating the divorce. Following male job loss, either partner may initiate the separation. Which partner does so depends on whose participation constraint was closer to binding prior to male job loss, on the respective magnitude of the decline in marital utilities, and the reduction in the male partner's outside option.

job loss increases welfare. Although such men match with higher-type partners, they need to compensate these partners through higher transfers. As a result, displaced men receive a smaller share of marital surplus and, on average, remain worse off than before job loss.³⁹

4.2 Taking Stock

We now compare the predictions of our conceptual framework to our empirical findings. Table 2 summarizes the main empirical findings from Sections 3.2–3.4 alongside the implications of Proposition 1. The comparison shows that our evidence from job displacements is consistent with NAM, while difficult to reconcile with PAM.⁴⁰ At the same time, the cross-sectional income correlation among matched partners in our sample is positive, which is in line with PAM but inconsistent with NAM.

In addition to the sign restrictions summarized in Table 2, Proposition 1 implies quantitative bounds on the magnitude of rematching. Approximating $\frac{\partial \mathbb{E}[q_f | q_m]}{\partial q_m}$ by the slope coefficient from a regression of wives' income on husbands' income yields a lower bound of 0.15 for $\gamma_{\Delta q_f | R} / \Delta q_m$.⁴¹ By contrast, our estimate of $\gamma_{\Delta q_f | R} / \Delta q_m$ is -0.25 , indicating a substantial discrepancy with the model implied lower bound. For displaced women, the corresponding lower bound is 0.25, while our estimate of $\gamma_{\Delta q_m | R} / \Delta q_f$ is -0.39 , indicating a similarly large discrepancy.

Taken together, the different pieces of evidence point to opposite sorting patterns, with rematching patterns following job displacement resembling those implied by NAM, whereas the cross-sectional pattern resembles that implied by PAM. Reconciling both patterns requires relaxing the link between complementarities in match formation and cross-sectional sorting patterns relative to our benchmark model. We consider relaxing two key assumptions that shape this link: first, we move from one-dimensional to multidimensional heterogeneity. This allows rematching patterns following job displacement to be governed by sorting on a subset of dimensions, while cross-sectional correlations reflect sorting across all dimensions. Second, we relax the assumption of globally uniform sorting. This allows sorting to be negative in the part of the income distribution where displacement shocks are concentrated, while remaining positive elsewhere, so that local rematching patterns and the aggregate cross-sectional correlation need not coincide. In the following subsections, we explore both extensions and discuss the extent to which each can reconcile the empirical patterns we document.

³⁹Formally, the expected marital surplus of a man of type q_m , averaged over the types of partners with whom he matches, is weakly increasing in q_m (see Lemma 1 in Shimer and Smith, 2000). Consequently, the expected marital surplus of a displaced man of type $q_m - d$ is lower than that of an otherwise identical man of type q_m , even under NAM where lower-type men rematch with higher-type women.

⁴⁰Note that the theoretical predictions regarding partner types experiencing separations (Proposition 1, 5.a and 5.b) additionally rely on the condition $F(q_m | D_B=1, D=1) \leq F(q_m | D_B=1, D=0)$. Appendix Figure B.9 assesses this condition empirically. For men, the empirical CDFs of pre-displacement labor income for displaced and placebo-displaced individuals who subsequently separate are nearly identical, and a two-sided Kolmogorov–Smirnov test fails to reject equality (p-value: 0.539). For women, a one-sided Kolmogorov–Smirnov test supports the required ordering, rejecting the null hypothesis $F(q_m | D_B=1, D=1) > F(q_m | D_B=1, D=0)$ (p-value: 0.001). The empirical evidence is therefore consistent with the condition for both displaced men and women.

⁴¹Using $\frac{\partial \mathbb{E}[q_f | q_m]}{\partial q_m} \approx \beta$, prediction 4.-a implies $\gamma_{\Delta q_f | R} / \Delta q_m \geq \beta$. In our estimation sample, $\beta = 0.15$.

Table 2: Contrasting Theory and Evidence

Impact of Job Displacement on		Data		Model	
		Displaced Men	Displaced Women	NAM	PAM
Breakup risk	γ_B	> 0	> 0	≥ 0	≥ 0
Risk of remaining single post breakup	$\gamma_{R=0 B}$	> 0	> 0	unrestricted	unrestricted
Partner types experiencing breakup	$\gamma_{q_f B}$	< 0	$< 0^a$	≤ 0	≥ 0
Partner types displaced individuals rematch with	$\gamma_{\Delta q_f R}$	> 0	$> 0^b$	≥ 0	≤ 0
Cross-sectional income correlation		> 0		≤ 0	≥ 0

Notes: The table summarizes the predictions implied by our conceptual framework under NAM and PAM from Proposition 1 (in Section 4.1) and contrasts them with our empirical findings reported in Sections 3.2–3.4. ^a The supporting estimate is negative but small in magnitude and not statistically significant. ^b The supporting estimates are positive throughout, but not all are statistically significant.

4.3 Reconciling Theory and Evidence I: Multidimensional Matching

This section discusses whether multidimensional matching can reconcile the positive cross-sectional correlation between spouses' earnings with the rematching patterns revealed by job displacement. The key intuition is that job displacement may affect only certain dimensions of an individual's type (such as earnings potential or health), while leaving others unchanged (such as age or height). In this case, rematching patterns following job displacement reveal sorting along the affected dimensions, whereas cross-sectional correlations reflect sorting across all dimensions. We formalize this idea and show that multidimensional matching can simultaneously account for both empirical patterns.

4.3.1 Multidimensional Sorting

We now extend our conceptual framework to the multidimensional case, $K > 1$. Agent types are vectors $q_f, q_m \in \prod_{k=1}^K [\underline{q}_k, \bar{q}_k]$, but flow utilities, value functions, and marital surplus remain as defined in the one-dimensional framework in Subsection 4.1.1.

We extend the notions of positive and negative assortative matching to multidimensional environments by defining sorting dimension by dimension (as in [Lindenlaub and Postel-Vinay, 2023](#)). Let $\mathcal{M}(q_m) = \{q_f \in Q_f : S(q_f, q_m) \geq 0\}$ denote the multidimensional matching set for a man of type q_m . At times, it will be convenient to write $q_f = (q_{fi}, q_f^{-i})$, where q_{fi} denotes the i -th component of the type vector and q_f^{-i} the remaining components.

We define positive and negative assortative mating in dimension i , denoted by $\text{PAM}(i)$ and $\text{NAM}(i)$, as follows:

Definition 1. Consider $q'_{fi} < q''_{fi}$, $q'_{mi} < q''_{mi}$.

There is $\text{PAM}(i)$ if for all q_f^{-i}, q_m^{-i} : $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mi}, q_m^{-i})$ and $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$

$\Rightarrow (q'_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mi}, q_m^{-i})$ and $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$.

There is $\text{NAM}(i)$ if for all q_f^{-i}, q_m^{-i} : $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mi}, q_m^{-i})$ and $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$

$\Rightarrow (q''_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mi}, q_m^{-i})$ and $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$.

To characterize multidimensional matching sets, we impose the following regularity assumption, which ensures that sets of the form $\{q_{fi} : (q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)\}$ are nonempty.

A-1. For any given q_m and q_f^{-i} there exists a q_{fi} , such that $(q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)$. For any given q_f and q_m^{-i} there exists a q_{mi} , such that $(q_{mi}, q_m^{-i}) \in \mathcal{M}(q_f)$.

Intuitively, Assumption A-1 requires that for any man of type q_m and any woman with characteristics q_f^{-i} , there exists a sufficiently favorable value of q_{fi} such that the two would agree to match. We show that, under this assumption, multidimensional matching sets inherit the same monotonicity properties in each dimension as in the one-dimensional model (see Appendix A, Lemma A.1).

4.3.2 Job Loss and Multidimensional Marriage Market Dynamics

We now link the multidimensional framework to our empirical setting by modeling job displacement as a permanent, unexpected reduction in the i -th component of an individual's type. Formally, a displaced man of type q_m experiences a permanent reduction in q_{mi} to $q_{mi} - d < q_{mi}$, while all remaining components of q_m remain unchanged. As in the one-dimensional framework, we interpret q_{mi} as earnings potential.⁴²

We consider the same treatment-control comparison as in Section 4.1.5. The definitions of the treatment effects on breakup risk and the probability of remaining single, γ_B and $\gamma_{R=0|B}$, carry over directly. Analogously, we define

$$\gamma_{q_{mi}|B} = \mathbb{E}[q_{mi}(t_0)|D_B = 1, D = 1] - \mathbb{E}[q_{mi}(t_0)|D_B = 1, D = 0],$$

$$\gamma_{q_{fi}|B} = \mathbb{E}[q_{fi}(t_0)|D_B = 1, D = 1] - \mathbb{E}[q_{fi}(t_0)|D_B = 1, D = 0],$$

$$\gamma_{\Delta q_{fi}|R} = \mathbb{E}[q_{fi}(\tau) - q_{fi}(t_0)|D_R = 1, D_B = 1, D = 1] - \mathbb{E}[q_{fi}(\tau) - q_{fi}(t_0)|D_R = 1, D_B = 1, D = 0],$$

⁴²Other characteristics, in addition to earnings potential, may also depend on q_{mi} and therefore be affected by job displacement. The distinguishing feature of the multidimensional framework is that additional dimensions $j \neq i$ remain unchanged. The idea is that some characteristics, such as earnings potential or health, are permanently affected by job displacement (see, e.g., Eliason and Storrie 2006; Browning, Moller Dano, and Heinesen 2006; Sullivan and von Wachter 2009), whereas others, such as age or height, are not.

which measure the treatment and selection effects of job displacement.

We show that the qualitative predictions of Proposition 1 extend to the multidimensional setting (see Appendix A, Proposition A.1). In particular, breakup risk continues to increase following job displacement, and the direction of rematching and partner selection in the dimension affected by job displacement depends on whether sorting is positive or negative in that dimension. Under PAM(i), displaced individuals separate disproportionately from partners with high values of the affected characteristic and rematch with partners having lower values of that characteristic. Under NAM(i), the opposite holds: displaced individuals separate disproportionately from partners with low values of the affected characteristic and rematch with partners with high values of it.

Multidimensional matching thus preserves the key implications of Proposition 1. If earnings potential is a dimension along which sorting is negative, job displacement induces upward rematching in earnings potential. The remaining question is whether this prediction can be reconciled with the positive cross-sectional correlation in spouses' earnings observed in the data. In the next subsection, we show that multidimensional matching can indeed generate both patterns simultaneously.

4.3.3 How Multidimensional Sorting Shapes Cross-Sectional Correlations

We now analyze how multidimensional matching shapes the cross-sectional correlation between partners' incomes. We show that a positive cross-sectional correlation in dimension i can arise even when sorting is negative in that dimension, provided indirect effects from sorting on other dimensions are sufficiently strong.⁴³ Thus, multidimensional matching can reconcile upward rematching following job displacement with a positive cross-sectional correlation between spouses' earnings.

To characterize when this occurs, we impose the following additional assumption on the orientation of matching sets.

A-2. For any given dimensions i and j , and any $q'_{fi} < q''_{fi}$, $q'_{fj} < q''_{fj}$, $q_f^{-(i,j)}$, and q_m it holds that:

$$\begin{aligned} & (q'_{fi}, q'_{fj}, q_f^{-(i,j)}) \in \mathcal{M}(q_m) \text{ and } (q''_{fi}, q''_{fj}, q_f^{-(i,j)}) \in \mathcal{M}(q_m) \\ \Rightarrow & (q'_{fi}, q''_{fj}, q_f^{-(i,j)}) \in \mathcal{M}(q_m) \text{ and } (q''_{fi}, q'_{fj}, q_f^{-(i,j)}) \in \mathcal{M}(q_m). \end{aligned}$$

Intuitively, Assumption A-2 requires that there is a trade-off between dimensions i and j : for a man of type q_m , matches with women who have a high value in one dimension and a low value in the other are more feasible than matches with women who have high values in both dimensions or low values in both dimensions.

⁴³In particular, we use monotonicity of the conditional expectation $\mathbb{E}[q_{fi} \mid q_{mi}]$ as a sufficient condition for the sign of the cross-sectional correlation. Specifically, a weakly increasing (decreasing) conditional expectation $\mathbb{E}[q_{fi} \mid q_{mi}]$ implies $\text{Corr}(q_{fi}, q_{mi}) \geq 0$ (≤ 0).

For expositional simplicity, we state the result for the two-dimensional case ($K = 2$). The corresponding result for the general multidimensional case ($K > 1$) is provided in Proposition 3 in Appendix A.⁴⁴

Proposition 2. *Consider the bidimensional case, $K = 2$, of the described search and matching environment in steady-state equilibrium, and suppose Assumptions A-1 and A-2 hold.*

Consider the following decomposition for $q''_{mi} \geq q'_{mi}$

$$\begin{aligned} \mathbb{E}[q_{fi}|q''_{mi}] - \mathbb{E}[q_{fi}|q'_{mi}] &= \underbrace{\int \mathbb{E}[q_{fi}|q''_{mi}, q_{mj}] - \mathbb{E}[q_{fi}|q'_{mi}, q_{mj}] dG(q_{mj}|q''_{mi})}_{:= \text{DE (Direct effect)}} \\ &+ \underbrace{\int \mathbb{E}[q_{fi}|q'_{mi}, q_{mj}] dG(q_{mj}|q''_{mi}) - \int \mathbb{E}[q_{fi}|q'_{mi}, q_{mj}] dG(q_{mj}|q'_{mi})}_{:= \text{IE (Indirect effect)}}. \end{aligned}$$

In a bidimensional steady-state matching equilibrium, the following implications hold:

$$\begin{aligned} PAM(i) &\Rightarrow DE \geq 0, \\ NAM(i) &\Rightarrow DE \leq 0. \end{aligned}$$

Given PAM (i) or NAM (i), the following additional implications hold:

$$\begin{aligned} PAM(j) \text{ and } G(q_{mj}|q_{mi}) \text{ is weakly decreasing in } q_{mi} &\Rightarrow IE \geq 0, \\ NAM(j) \text{ and } G(q_{mj}|q_{mi}) \text{ is weakly decreasing in } q_{mi} &\Rightarrow IE \leq 0, \\ PAM(j) \text{ and } G(q_{mj}|q_{mi}) \text{ is weakly increasing in } q_{mi} &\Rightarrow IE \leq 0, \\ NAM(j) \text{ and } G(q_{mj}|q_{mi}) \text{ is weakly increasing in } q_{mi} &\Rightarrow IE \geq 0. \end{aligned}$$

Proposition 2 shows that the cross-sectional association between spouses' characteristics can be decomposed into two components. The direct effect (DE) captures how an increase in q_{mi} , holding the man's remaining characteristics fixed, affects the expected type of his partner in dimension i . The indirect effect (IE) captures the association between q_{fi} and q_{mi} that arises from sorting on the remaining attributes q_m^{-i} .

The proposition provides sufficient conditions for the sign of each component. Under $NAM(i)$, the direct effect is negative: holding all other characteristics fixed, higher earnings potential is associated with lower partner earnings. However, if higher earnings potential is positively associated with other characteristics that are themselves positively sorted, the indirect effect is positive. When sufficiently strong, this indirect effect outweighs the negative direct effect, generating a positive cross-sectional correlation between spouses' earnings despite negative sorting on earnings potential.

This result provides a potential explanation for the empirical patterns documented above: job displacement reveals sorting along the dimension affected by the shock, whereas cross-sectional correlations reflect

⁴⁴The condition that $G(q_{mj}|q_{mi})$ is weakly increasing in q_{mi} (often referred to as *positive regression dependence*) implies $Corr(q_{mj}, q_{mi}) \geq 0$ (see, e.g., Lehmann, 1966).

the combined influence of sorting across all dimensions. Consequently, multidimensional matching can simultaneously generate upward rematching following job displacement and a positive cross-sectional correlation between spouses' earnings.

4.4 Reconciling Theory and Evidence II: Local Sorting

This section assesses empirically to what extent non-uniform sorting patterns may explain the empirical patterns we document within a one-dimensional matching framework. In particular, local sorting may be negative in the part of the income distribution where most job displacements occur, giving rise to the upward rematching patterns we document, while positive assortative matching elsewhere gives rise to an overall positive cross-sectional income correlation. We now investigate to what extent local sorting can explain our empirical findings.

To do so, we estimate local sorting patterns and examine whether couples in our estimation sample are located in regions characterized by local positive or local negative sorting. Specifically, we follow [Chiappori et al. \(2026\)](#) and estimate local log-odds ratios across the bivariate distribution of female and male partners' labor income. Intuitively, the local log-odds ratio measures whether, in a given region of the type space, similar types are more likely to match with each other than with dissimilar types. Following the specification of [Chiappori et al. \(2026\)](#), we flexibly estimate the log-odds ratio as the cross-partial derivative of a higher-order polynomial and assess its sign over the support of the joint income distribution.⁴⁵ The estimation is based on all couples in the Danish administrative data observed in the same calendar period and age range covered by our analysis of job displacements. Appendix D provides further details on the sample construction and estimation procedure.

Figure 4A displays the estimated local sorting patterns across the bivariate distribution of women's and men's labor income. Most of the income distribution is characterized by local positive sorting. Local negative sorting is confined to three regions: the very top of the joint income distribution and two off-diagonal regions where one spouse has substantially higher labor income than the other.⁴⁶

As a next step, we assess which local sorting patterns are relevant for the sample of couples used in our job displacement analysis. To this end, Figure 4B plots the bivariate income distribution of these couples, using each partner's pre-displacement income (in $t = -3$). The figure shows that the majority of observations are concentrated in the part of the distribution, where local positive sorting prevails. More precisely, 88% of the observations are located in regions exhibiting local positive sorting, with the remaining 12% falling into regions where local sorting is negative.

⁴⁵Note that the local log-odds ratio equals twice the second cross-difference of the surplus in a [Choo and Siow \(2006\)](#) model. However, it also has a model free interpretation as an empirical measure of local sorting (see, e.g., [Chiappori, Dias, Meghir, and Zhang, 2025](#)).

⁴⁶This pattern is qualitatively similar to that reported by [Chiappori et al. \(2026\)](#) based on their main specification (see their Figure 3). They use data from a different country and time period (the Netherlands, 2013–2019).

partners' incomes are substitutes and therefore predicts negative sorting on wages as this maximizes the gains from household specialization (Becker, 1973, 1981). Various arguments have been put forward to resolve this apparent discrepancy. Becker (1981) argues that missing wage data for non-working women may bias the observed correlation toward positive values. Lam (1988) shows that joint consumption of a market-purchased good that is public within the household can give rise to positive sorting (see also Low 2024). Recently, complementarities in spouses' housework hours (e.g., Gayle and Shephard 2019; Calvo et al. 2024) and homophily (e.g., Goussé et al. 2017; Gayle and Shephard 2019; Adda et al. 2024) have often been used as mechanisms to generate earnings-based positive sorting.

Our findings offer an alternative perspective: within our multidimensional framework, they can be rationalized by negative sorting on income, conditional on other dimensions of agent type. This is consistent with the prediction of Becker's theory, in which partners' incomes are substitutes and couples match to maximize gains from specialization. At the same time, a positive cross-sectional correlation in partners' income may arise through indirect effects of sorting on other characteristics, for example driven by homophily in education or other traits (see, e.g., Chiappori et al. 2009, 2018). In our multidimensional framework, forces toward negative sorting on income and positive sorting on other characteristics do not counteract each other but coexist and jointly shape matching patterns. As a result, unconditional correlations of spousal characteristics reflect sorting across all dimensions. In contrast, leveraging exogenous variation in one characteristic, while holding all others fixed, allows inference about sorting along that specific dimension.⁴⁷

5.2 Counterfactual Simulations: Marital Sorting and Income Inequality

This section explores quantitative implications of our findings for how marital sorting shapes the relationship between individual- and household-level income inequality. We simulate a rise in individual-level income inequality in calibrated one-dimensional and two-dimensional versions of our framework and show that both versions have different implications for the link between marital sorting and household-level inequality.

Model Specification To calibrate our framework, we impose additional functional form and distributional assumptions. We also extend the framework to include a match-specific "love shock", z , which captures non-economic motives for marriage.⁴⁸ In the 1D specification, individuals match on incomes, q_f and q_m . In the 2D specification, they match on income, q_{f1} and q_{m1} , and on a residual characteristic, q_{f2} and q_{m2} , which captures all remaining traits, including unobserved characteristics. We denote the correlation

⁴⁷This argument also has implications for interpreting cross-sectional regressions of female partners' incomes on male partners' incomes. Appendix C combines our theoretical framework with our empirical findings to show that such regressions provide a lower bound on the contribution of unobserved characteristics to the observed cross-sectional association between partners' incomes.

⁴⁸The love shock is drawn from a $N(\mu_z, \sigma_z)$ distribution upon meeting a potential partner, is equal for both partners, and remains unchanged for the duration of a match. It allows the model to generate empirically plausible matching patterns. Without it, income would fully determine matching in the 1D model. See Jacquemet and Robin (2013), Goussé et al. (2017), and Borovicková and Shimer (2024).

coefficient between income and the residual characteristic by ρ . We specify the flow utilities of married and single individuals as follows:

$$\begin{aligned}
&\text{1D model:} && \text{Couples: } u_g^1(q_f, q_m) &= \kappa_1 \frac{(q_f + q_m)^{1-\eta}}{1-\eta} - \kappa_2 (q_f - q_m)^2 + z \\
&&& \text{Singles: } u_g^0(q_g) &= \kappa_1 \frac{q_g^{1-\eta}}{1-\eta}, \quad g \in \{f, m\} \\
&\text{2D model:} && \text{Couples: } u_g^1(q_f, q_m) &= \omega_1 \frac{(q_{f1} + q_{m1})^{1-\eta}}{1-\eta} - \omega_2 (q_{f2} - q_{m2})^2 + z \\
&&& \text{Singles: } u_g^0(q_g) &= \omega_1 \frac{q_{g1}^{1-\eta}}{1-\eta} \quad g \in \{f, m\},
\end{aligned} \tag{9}$$

$$\tag{10}$$

In both models, the first term in $u_g^1(q_f, q_m)$ induces negative sorting when $\eta > 0$.⁴⁹ Intuitively, the marginal utility of partner income declines with own income, which generates incentives for NAM. The second term in $u_g^1(q_f, q_m)$ captures homophily by penalizing unequal matches, thereby creating incentives for PAM.⁵⁰ The relative magnitudes of κ_2 and κ_1 in the 1D model, and of ω_2 and ω_1 in the 2D model, determine which of these incentives dominates. The key distinction between the model versions is that the homophily term penalizes income differences in the 1D model, but differences in residual characteristics in the 2D model. This creates incentives for PAM on income in the 1D model, and on residual traits in the 2D model. Finally, the flow utility of singles only depends on own income in both models. Further details about the model specification (including type spaces, matching technology, household bargaining, equilibrium characterization, and numerical solution procedure) are provided in Appendix E.

Calibration We calibrate both the 1D and the 2D specifications of our framework. A subset of parameters is fixed at standard values: we set the annual discount rate to 0.05 and the utility curvature parameter, η , to 1.5.⁵¹ We assume equal Nash-bargaining power, $\mu_m = 1 - \mu_f = 0.5$, and normalize the meeting rates λ_f and λ_m by fixing the scaling factor to $\bar{\lambda} = 1$.⁵² We fix the annual relationship dissolution rate at $\delta = 0.06$ to match its empirical counterpart in the data. The remaining parameters are calibrated by minimizing the relative distance between theoretical and empirical moments. In the 1D model, we calibrate $\{\kappa_1, \kappa_2, \mu_z, \sigma_z\}$ to match four moments: the share of married individuals, the marriage rate (i.e., the flow into marriage), the cross-sectional correlation in matched partner's incomes, and the variance of log-household income.⁵³ In the 2D model, we calibrate $\{\omega_1, \omega_2, \mu_z, \sigma_z, \rho\}$, additionally targeting the effect of job displacement on rematching patterns for men, $\frac{\gamma \Delta q_f |^R}{\Delta q_m} = -0.25$, as estimated in Section 3.4. Since the displacement effect on

⁴⁹To see this, note that the cross-partial derivative of the term is negative for $\eta > 0$, which implies a negative cross-partial derivative of the match flow value, $f(q_f, q_m) = u_f^1(q_f, q_m) + u_m^1(q_f, q_m)$. This is part of the sufficient conditions for negative assortative matching (NAM) derived in [Shimer and Smith \(2000\)](#).

⁵⁰See, e.g., [Gihleb and Lang \(2016\)](#), who use a similar penalty term to capture homophily in marriage markets. [Marimon and Zilibotti \(2001\)](#) use a comparable notion to model suitability of workers for jobs.

⁵¹See, e.g., [Attanasio, Low, and Sánchez-Marcos \(2008\)](#).

⁵²Recall that we assume a quadratic matching technology, implying that the meeting rates for men and women are $\lambda_f = \lambda \int dG_m(q_m)$ and $\lambda_m = \lambda \int dG_f(q_f)$. Fixing the common scaling factor at $\lambda = 1$ implies the rate at which women and men meet potential partners is equal to the mass of singles of the opposite gender.

⁵³A standard measure of income inequality, see, e.g., [Blundell, Pistaferri, and Preston 2008](#).

Table 3: Model Fit

Moment	Value 1D Model	Value 2D Model	Empirical Value
Population share of married individuals	0.75	0.70	0.76
Income correlation, $\text{Corr}(\text{income}_f, \text{income}_m)$	0.22	0.22	0.22
Income inequality, $\text{Var}(\log(\text{income}_f + \text{income}_m))$	0.14	0.14	0.15
Marriage rate	0.04	0.04	0.05
Displacement effect, $\frac{\gamma_{\Delta q_f R}}{\Delta q_m}$	0.17	-0.19	-0.25

Notes: The table shows the fit of the calibrated 1D and 2D version of our quantitative framework compared to the data. Each row corresponds to one of the 5 moments that we target in the calibration. The data moments are computed based on our main estimation sample described in Subsection 2.3.

partner income upon rematching for women, $\frac{\gamma_{\Delta q_m | R}}{\Delta q_f} = -0.39$, is estimated noisily, we leave it untargeted. To compute the model analogues of both estimates, we simulate an exogenous income shock and measure the average change in partner income when individuals subsequently transition into a new match. All theoretical moments are computed in steady state. Empirical moments are computed in the estimation sample described in Section 2. Table B.7 summarizes all parameter values.

Model Fit Table 3 reports the model fit. Both the 1D and 2D versions of our framework match the targeted empirical moments closely. The key difference is that the 1D model fails to capture the observed rematching patterns following job displacement: it predicts a positive value for $\frac{\gamma_{\Delta q_f | R}}{\Delta q_m}$, contrary to the empirical evidence. The 1D model’s inability to reconcile post-displacement rematching patterns with the cross-sectional income correlation is in line with the theoretical results developed in Section 4.1. In contrast, the 2D model reproduces the negative sign of the displacement effect for both men (-0.19) and women (-0.23).

Simulation Results We use the calibrated models to simulate an increase in individual income inequality by applying the following transformation to individual income, separately for men and women.⁵⁴

$$\tilde{q} = \max \left(c \cdot (q - \mu_{q,g}) + \mu_{q,g}, q_{\min} \right). \quad (11)$$

This transformation scales deviations from the gender-specific mean $\mu_{q,g}$, where c governs the degree of dispersion and $q_{\min}$ ensures that incomes remain strictly positive. In our simulations, we set $c = 1.15$ and $q_{\min} = 5000$ DKK, resulting in a 23% increase in the variance of income for men and a 29% increase for women. At the household level, the variance of log household income, increases by 62% to 67% due to the transformation in the 1D and 2D model, respectively (holding the marriage market fixed). This corresponds to an increase of approximately 2 Gini points, which is roughly a third to a half of the overall Gini increase in Denmark between 1985 and 2013.⁵⁵

⁵⁴In the 1D model, the transformation is applied to the scalar agent type, q_g , in the 2D model it is applied to the first component of type, q_{1g} , which captures income.

⁵⁵4-6 points according to the OECD (Causa, Hermansen, Ruiz, Klein, and Smidova, 2016). Note that we compute the Gini based on gross household labor income, whereas the referenced number is based on equivalized disposable income for the whole

Table 4: Simulation Results: Income Inequality and Marital Sorting

	Var(log(income _f + income _m))	
	1D model	2D model
(1) Baseline	0.144	0.143
(2) Counterfactual, marital sorting fixed	0.233	0.239
(3) Counterfactual	0.250	0.174

Notes: The table shows household income inequality, measured by the variance of log household income, in the baseline scenario (row 1) and in the counterfactual experiment that increases individual income inequality (row 3). Row 2 shows how inequality increases under the counterfactual if the sorting of individuals into couples is kept constant at the baseline distribution.

Table 4 reports between-household inequality under three scenarios: (1) the calibrated baseline, (2) the counterfactual with increased individual income inequality holding the sorting of individuals into couples fixed, and (3) the counterfactual allowing sorting to adjust. Comparing scenarios (1) and (3), the 1D model predicts a substantially larger increase in between-household inequality than the 2D model. The comparison between scenarios (2) and (3) shows that this difference is driven by marriage market sorting. The 1D model predicts that positive sorting on income amplifies the rise in inequality. This amplification of 0.017 in the variance of log household income corresponds to 0.3 Gini points, which is quantitatively comparable to the effect of major Danish tax reforms.⁵⁶ In contrast, the 2D model predicts that negative sorting on income dampens the rise in inequality. Quantitatively, under the counterfactual, the 2D model predicts a 30% lower variance of log household income (0.174) than the 1D model (0.250). Overall, the simulations illustrate important differences between the 1D and the 2D model for the link between marriage market sorting and between-household inequality.⁵⁷

5.3 External Validity and Cross-Country Differences

The relative strength of the mechanisms underlying our findings may depend on the economic environment and the institutional setting. In the following, we briefly discuss how Denmark compares to other countries along these dimensions.

In terms of marital sorting, the best cross-country evidence is available for sorting based on educational attainment. Here, Denmark exhibits levels of positive sorting broadly comparable to those observed in Norway and the United Kingdom, and somewhat lower than in the United States (see Eika et al., 2019). At the same time, relative to many OECD countries, Denmark is characterized by high female labor force

population. Thus, the levels are not directly comparable but the comparison of changes in the Gini coefficient is still meaningful.

⁵⁶The Danish Ministry of Economic Affairs and the Interior estimates that the 2009 tax reform (*Forårspakke 2.0*) increased the Gini coefficient of equivalized disposable income by 0.5 Gini points (*Økonomi-og Indenrigsministeriet*, 2013).

⁵⁷Note that our simulation results do not contradict an overall positive contribution of marital sorting to between-household inequality: positive sorting on non-income traits increases between-household inequality, although negative sorting on income dampens this effect to some extent. For example, in the 2D model, a rise in sorting on education over time can be captured by an increase in ω_2 , which strengthens homophily on non-income traits. Since these traits are positively correlated with income in our calibration, this leads to a rising correlation between partners' incomes over time (as documented by, e.g., Lise and Seitz 2011).

participation, while part-time employment among women is comparatively low (see, e.g., [OECD, 2018, 2020](#); [Lind and Rasmussen, 2008](#)). Similarly, the gender earnings gap has historically been relatively small in Denmark, although other OECD countries have recently converged toward the Danish level (see, e.g., [Kleven, Landais, and Sogaard, 2019](#)). These features tend to reduce the gains from household specialization and may therefore weaken forces contributing to negative sorting on income, arguably working against the upward rematching mechanism following job displacement that we find.

Moreover, Denmark is characterized by low income inequality (see, e.g., [Kreiner and Svarer, 2022](#)), which, in the presence of matching frictions, reduces incentives for sorting on income. Denmark's low poverty risk and generous social safety net further attenuate the economic consequences of displacement relative to many other countries (see [Bertheau, Acabbi, Barceló, Gulyas, Lombardi, and Saggio, 2023](#)). In the context of our model, this corresponds to a smaller reduction in type following displacement and may therefore reduce the magnitude of the marriage market responses we estimate.

Finally, Denmark's divorce rate has been somewhat higher than that of Germany and the United Kingdom, and markedly lower than that of the United States (see [OECD, 2019](#)). These differences indicate where Denmark lies relative to these countries in terms of relationship stability, although we note that the relationship between aggregate separation rates and separation following job loss is not straightforward.

6 Conclusion

Understanding marital sorting is central to explaining household inequality and assessing how marriage markets respond to changes in the wage structure or policy. In this paper, we use exogenous variation from establishment closures to provide new evidence on how sorting arises through marriage and divorce decisions. Our evidence shows that displaced workers are more likely to separate from low-income partners and to rematch with higher-income partners. Interpreting these findings through the lens of an equilibrium search-and-matching framework suggests that the positive cross-sectional correlation in partners' incomes reflects not only sorting on income itself but also sorting on other traits correlated with income, while non-uniform sorting appears to play at most a secondary role.

Our findings point to several directions for future research. One is to exploit exogenous variation in characteristics other than income, such as health or education, to better understand the mechanisms governing sorting along these dimensions. Another is to study how observed sorting patterns reflect not only complementarities in the marital match value but also heterogeneity in meeting opportunities (e.g., due to geographic segregation; [Alonzo, Guner, and Luccioletti, 2023](#)) or the contexts in which couples meet (e.g., workplaces or universities; [Kirkeboen, Leuven, Mogstad, and Mountjoy, 2025](#)). More broadly, our results highlight the value of combining structural modeling with quasi-experimental variation to uncover the mechanisms that give rise to marital sorting.

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Supplemental Appendix

A Proofs and Derivations

Definition of PAM and NAM: Shimer and Smith (2000) characterize sorting by defining the following notions of PAM and NAM, which generalize the corresponding definition for the frictionless case by Becker (1973).⁵⁸

Definition A.1. Consider $q'_f < q''_f, q'_m < q''_m$.

There is PAM if: $q''_f \in \mathcal{M}(q'_m)$ and $q'_f \in \mathcal{M}(q''_m) \Rightarrow q'_f \in \mathcal{M}(q'_m)$ and $q''_f \in \mathcal{M}(q''_m)$

There is NAM if: $q'_f \in \mathcal{M}(q'_m)$ and $q''_f \in \mathcal{M}(q''_m) \Rightarrow q''_f \in \mathcal{M}(q'_m)$ and $q'_f \in \mathcal{M}(q''_m)$.

Intuitively, under PAM, whenever two couples, (q'_f, q'_m) and (q''_f, q''_m) , can form more positively sorted matches by trading partners, they are willing to do so.

Proof of proposition 1: We start by proving that under PAM or NAM, $\gamma_B \geq 0$:

As men in the control group by definition are not displaced between period t_0 and τ , their types are unchanged between these points in time, i.e., $q_m(\tau) = q_m(t_0)$. A control group couple that was matched in period τ , therefore continues to have the identical (non-negative) marital surplus it had in t_0 .

It follows that no endogenous breakups occur in the control group. Exogenous breakups, by assumption, occur at rate δ . The overall probability that a man in the control group experiences a breakup from his t_0 -partner between t_0 and τ is thus given by:

$$P(D_B = 1 | D = 0) = 1 - e^{-\delta(\tau - t_0)} \quad (\text{A.1})$$

Note that this holds under PAM as well as under NAM.

In the treatment group, by contrast, men's types change between t_0 and τ due to job displacement. Specifically, $q_m(\tau) = q_m(t_0) - d < q_m(t_0)$.

For a given man with pre-displacement type $q_m(t_0)$, job displacement will lead to a breakup if it changes the couples' marital surplus from weakly positive to negative, or equivalently if $q_f(t_0) \in \mathcal{M}(q_m(t_0))$ but $q_f(t_0) \notin \mathcal{M}(q_m(t_0) - d)$.

Shimer and Smith (2000) show that under NAM or PAM matching sets are closed intervals, $\mathcal{M}(q_m) = [a(q_m), b(q_m)]$, with interval bounds, $a(q_m), b(q_m)$, that are weakly increasing in q_m and PAM and that are

⁵⁸Note that as matching is symmetric, $q_f \in \mathcal{M}(q_m)$ is equivalent to $q_m \in \mathcal{M}(q_f)$. The definitions of PAM and NAM thus imply that the respective relationships with q_m and q_f interchanged also hold.

weakly decreasing under NAM. It follows under PAM that job displacement leads to a breakup for a man of pre-displacement type q_m if and only if he is matched with a woman of type $q_f \in (\max\{b(q_m - d), a(q_m)\}, b(q_m)]$.

Similarly, it follows under NAM that job displacement will lead to a breakup for a man with pre-displacement type q_m if and only if he is matched with a woman of type $q_f \in [a(q_m), \min\{a(q_m - d), b(q_m)\})$.

Additionally, breakups occur exogenously at rate δ under PAM as well as under NAM.

It follows that under PAM the overall probability that a man in the treatment group experiences a breakup from his t_0 -partner between t_0 and τ is given by:

$$P(D_B = 1|D = 1) = \underbrace{1 - e^{-\delta(\tau - t_0)}}_{\text{prob. of exogenous breakups}} + \underbrace{\int G_f(b(q_m(t_0))) - G_f((\max\{b(q_m(t_0) - d), a(q_m(t_0))\})) dF(q_m(t_0)|D = 1)}_{\text{prob. of endogenous breakups}}. \quad (\text{A.2})$$

Note that $G_f(b(q_m(t_0))) - G_f((\max\{b(q_m(t_0) - d), a(q_m(t_0))\}))$ is the mass of men of type $q_m(t_0)$ matched with a woman of type $q_f \in (\max\{b(q_m - d), a(q_m)\}, b(q_m)]$, i.e., the mass of $q_m(t_0)$ -type men who experience an endogenous breakup after displacement.

Similarly, under NAM, the overall probability that a man in the treatment group experiences a breakup from his t_0 -partner between t_0 and τ is:

$$P(D_B = 1|D = 1) = \underbrace{1 - e^{-\delta(\tau - t_0)}}_{\text{prob. of exogenous breakup}} + \underbrace{\int G_f(\min\{a(q_m(t_0) - d), b(q_m(t_0))\}) - G_f(a(q_m(t_0))) dF(q_m(t_0)|D = 1)}_{\text{prob. of endogenous breakup}}, \quad (\text{A.3})$$

From (A.1), (A.2), and (A.3) it follows that under PAM as well as under NAM

$\gamma_B = P(D_B = 1|D = 1) - P(D_B = 1|D = 0) \geq 0$. This concludes the proof of statement 1.

To see that the sign of the impact of job displacement on the probability of staying single post-breakup is undetermined, note that for a given man of type q_m

$$P(D_R = 0|q_m) = \exp\left(-(\tau - t_0)\lambda_m(G_f(b(q_m)) - G_f(a(q_m)))\right).$$

It follows that $P(D_R = 0|q_m - d) \geq P(D_R = 0|q_m)$ if and only if

$$G_f(b(q_m - d)) - G_f(b(q_m)) \geq G_f(a(q_m - d)) - G_f(a(q_m)). \quad (\text{A.4})$$

Under PAM a, b are weakly increasing in q_m , implying that $G_f(b(q_m - d)) - G_f(b(q_m))$ is weakly negative. However, as the same is implied for $G_f(b(q_m - d)) - G_f(b(q_m))$, (A.4) may or may not hold. By similar arguments it follows that the sign of $P(D_R = 0|q_m - d) \geq P(D_R = 0|q_m)$ is also undetermined under NAM.

Note that the above arguments show that for a given type q_m the sign of $P(D_R = 0|q_m - d) - P(D_R = 0|q_m)$ is undetermined, i.e., even if the compared groups of men were to overlap perfectly (i.e., if $F(q_m|D_B = 1, D = 1) = F(q_m|D_B = 1, D = 0)$) the sign of $\gamma_{R=0|B}$ is undetermined.⁵⁹ In the general case, $F(q_m|D_B = 1, D = 1) \neq F(q_m|D_B = 1, D = 0)$ is a further reason why the sign of $\gamma_{R=0|B}$ may be weakly positive or negative. These arguments confirm statement 2.

Next, we turn to proving that under PAM the impact of job displacement on partner type, $\gamma_{\Delta q_f|R}$, is weakly negative and bounded above by

$$\bar{\gamma}_{\Delta q_f|R} = - \int_0^d \int_0^d \frac{\partial \mathbb{E}[q_f|q_m]}{\partial q_m} \Big|_{q_m=q-x} dx dF(q|D_R = 1, D_B = 1, D = 1) \leq 0.$$

Denote by D_δ an indicator that equals 1 for men who experience an exogenous breakup between t_0 and τ , and 0 for all other men. Consider men in the treatment group of pre-displacement type q_m who separate from their t_0 -partner and rematch with a new partner between t_0 and τ . The average female type this group of men is matched with in t_0 can be written as weighted average:

$$\begin{aligned} & \mathbb{E}[q_f(t_0)|D_R = 1, D_B = 1, D = 1, q_m(t_0) = q_m] = \\ & \mathbb{E}[q_f(t_0)|D_R = 1, D_B = 1, D = 1, q_m, D_\delta = 1] P(D_\delta = 1|D_R = 1, D_B = 1, D = 1, q_m) \\ & + \mathbb{E}[q_f(t_0)|D_R = 1, D_B = 1, D = 1, q_m, D_\delta = 0] P(D_\delta = 0|D_R = 1, D_B = 1, D = 1, q_m) \\ & = \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \cdot \frac{1}{G_f(b(q_m)) - G_f(a(q_m))} \int_{a(q_m)}^{b(q_m)} q_f dG_f(q_f) \\ & + \frac{G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \\ & \cdot \frac{1}{G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \int_{\max\{b(q_m - d), a(q_m)\}}^{b(q_m)} q_f dG_f(q_f) \\ & = \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] \\ & + \frac{G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \mathbb{E}[q_f|\max\{b(q_m - d), a(q_m)\} < q_f < b(q_m)] \end{aligned}$$

⁵⁹The fact that even for a given individual the sign of $P(D_R = 0|q_m - d) - P(D_R = 0|q_m)$ is undetermined implies that under additional assumptions on the stochastic ordering of $F(q_m|D_B = 1, D = 1)$ and $F(q_m|D_B = 1, D = 0)$, NAM and PAM still do not determine the sign of $\gamma_{R=0|B}$.

(A.5)

Next we turn to computing the corresponding average for period τ , taking into account that men in the treatment group are displaced in t_0 . Their type when rematching with a new partner in $(t_0, \tau]$ is therefore $q_m - d$, and the average female type they are matched with in τ is:

$$\begin{aligned}\mathbb{E}[q_f(\tau)|D_R = 1, D_B = 1, D = 1, q_m(t_0) = q_m] &= \frac{1}{G_f(b(q_m - d)) - G_f(a(q_m - d))} \int_{a(q_m - d)}^{b(q_m - d)} q_f dG_f(q_f) \\ &= \mathbb{E}[q_f|a(q_m - d) < q_f < b(q_m - d)].\end{aligned}\quad (\text{A.6})$$

For the control group, by contrast, as men's types are unchanged between t_0 and τ , the corresponding expressions are given by:

$$\begin{aligned}\mathbb{E}[q_f(t_0)|D_R = 1, D_B = 1, D = 0, q_m(t_0) = q_m] &= \mathbb{E}[q_f(\tau)|D_R = 1, D_B = 1, D = 0, q_m(t_0) = q_m] \\ &= \frac{1}{G_f(b(q_m)) - G_f(a(q_m))} \int_{a(q_m)}^{b(q_m)} q_f dG_f(q_f) \\ &= \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)].\end{aligned}\quad (\text{A.7})$$

Using (A.5), (A.6), and (A.7) it follows for $\gamma_{\Delta q_f|R}$ that

$$\begin{aligned}\gamma_{\Delta q_f|R} &= \int \mathbb{E}[q_f(\tau) - q_f(t_0)|D_R = 1, D_B = 1, D = 1, q_m] dF(q_m|D_R = 1, D_B = 1, D = 1) \\ &\quad - \int \mathbb{E}[q_f(\tau) - q_f(t_0)|D_R = 1, D_B = 1, D = 0, q_m] dF(q_m|D_R = 1, D_B = 1, D = 0) \\ &= \int \frac{1 - e^{-\delta(\tau - t_0)}}{1 - e^{-\delta(\tau - t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \\ &\quad \cdot \left(\mathbb{E}[q_f|a(q_m - d) < q_f < b(q_m - d)] - \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] \right) \\ &\quad dF(q_m|D_R = 1, D_B = 1, D = 1) \\ &\quad + \int \frac{G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})}{1 - e^{-\delta(\tau - t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m - d), a(q_m)\})} \\ &\quad \cdot \left(\mathbb{E}[q_f|a(q_m - d) < q_f < b(q_m - d)] - \mathbb{E}[q_f|\max\{b(q_m - d), a(q_m)\} < q_f < b(q_m)] \right) \\ &\quad dF(q_m|D_R = 1, D_B = 1, D = 1) \\ &\leq \int \mathbb{E}[q_f|a(q_m - d) < q_f < b(q_m - d)] \\ &\quad - \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] dF(q_m|D_R = 1, D_B = 1, D = 1) \\ &= \int \mathbb{E}[q_f|q_m - d] - \mathbb{E}[q_f|q_m] dF(q_m|D_R = 1, D_B = 1, D = 1)\end{aligned}$$

$$\begin{aligned}
&= - \int_0^d \int \frac{\partial \mathbb{E}[q_f|q_m]}{\partial q_m} \Big|_{q_m=q-x} dx dF(q|D_R=1, D_B=1, D=1) \\
&= \bar{\gamma}_{\Delta q_f|R},
\end{aligned}$$

where the weak inequality follows as⁶⁰

$$\mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] \leq \mathbb{E}[q_f|\max\{b(q_m-d), a(q_m)\} < q_f < b(q_m)]. \quad (\text{A.8})$$

As shown by [Shimer and Smith \(2000\)](#), $\mathbb{E}[q_f|q_m]$ is weakly increasing in q_m under PAM, from which $\bar{\gamma}_{\Delta q_f|R} \leq 0$ follows. This concludes the proof of statements 3.-a and 3.-b.

By analogous steps it can be shown that under NAM $\gamma_{\Delta q_f|R}$ is weakly positive and bounded below by $\underline{\gamma}_{\Delta q_f|R} \geq 0$ (statements 4.-a and 4.-b).

Finally, we prove that under PAM, if $F(q_m|D_B=1, D=1) \leq F(q_m|D_B=1, D=0)$, then $\gamma_{q_f|B} \geq 0$. As noted above, under PAM $\mathcal{M}(q_m) = [a(q_m), b(q_m)]$ with interval bounds that are weakly increasing in q_m (see [Shimer and Smith \(2000\)](#)). By implication, under PAM $\mathbb{E}[q_f|a(q_m) < q_f < b(q_m)]$ is weakly increasing in q_m . From $F(q_m|D_B=1, D=1) \leq F(q_m|D_B=1, D=0)$ it follows that⁶¹

$$\begin{aligned}
&\int \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] dF(q_m|D_B=1, D=1) \\
&\geq \int \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] dF(q_m|D_B=1, D=0).
\end{aligned} \quad (\text{A.9})$$

Using (A.5) and (A.7) it follows for $\gamma_{q_f|B}$ that

$$\begin{aligned}
\gamma_{q_f|B} &= \int \mathbb{E}[q_m(t_0)|D_B=1, D=1, q_m] dF(q_m|D_B=1, D=1) \\
&\quad - \int \mathbb{E}[q_m(t_0)|D_B=1, D=0] dF(q_m|D_B=1, D=0) \\
&= \int \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m-d), a(q_m)\})} \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] \\
&\quad + \frac{G_f(b(q_m)) - G_f(\max\{b(q_m-d), a(q_m)\})}{1 - e^{-\delta(\tau-t_0)} + G_f(b(q_m)) - G_f(\max\{b(q_m-d), a(q_m)\})} \\
&\quad \mathbb{E}[q_f|\max\{b(q_m-d), a(q_m)\} < q_f < b(q_m)] dF(q_m|D_B=1, D=1) \\
&\quad - \int \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] dF(q_m|D_B=1, D=0) \\
&\geq \int \mathbb{E}[q_f|a(q_m) < q_f < b(q_m)] dF(q_m|D_B=1, D=1)
\end{aligned}$$

⁶⁰Note that in general for any random variable X , and $a \leq a'$ it holds that $\mathbb{E}[X|a \leq X \leq b] \leq \mathbb{E}[X|a' \leq X \leq b]$.

⁶¹Note that in general, if $F_1(x) \geq F_2(x)$ for all x , then $\int h(x)dF_2(x) \geq \int h(x)dF_1(x)$ for any weakly increasing measurable function $h(x)$.

$$\begin{aligned}
& - \int \mathbb{E} [q_f | a(q_m) < q_f < b(q_m)] dF(q_m | D_B = 1, D = 0) \\
& \geq 0,
\end{aligned}$$

where the first weak inequality follows by (A.8) and the second follows by (A.9). This concludes the proof of statement 5-a. Statement 5-b can be proved by analogous steps. $\square$

Lemma A.1. *Under assumption A-1, and given either PAM(i) or NAM(i), multidimensional matching sets, $\mathcal{M}(q_m)$, are characterized by one-dimensional sets*

$$\{q_{fi} : (q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)\} = [a_i(q_{mi}, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})],$$

where

$$q_f \in \mathcal{M}(q_m) \Leftrightarrow q_{fi} \in [a_i(q_{mi}, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})]$$

and a_i, b_i are

- (i) weakly increasing in q_{mi} under PAM(i),
- (ii) weakly decreasing in q_{mi} under NAM(i).

Proof of Lemma A.1: Define $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i}) := \{q_{fi} : (q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)\}$. We proceed by first proving that any set $\mathcal{M}_i$ is a convex set and then show that its bounds are weakly increasing under PAM (i).⁶²

1. $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i})$ is convex:

Consider $q'_{fi} < q''_{fi} < q'''_{fi}$, with q'_{fi} and q'''_{fi} in $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i})$, i.e.,

$$(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q_m), \tag{A.10}$$

$$(q'''_{fi}, q_f^{-i}) \in \mathcal{M}(q_m). \tag{A.11}$$

Now consider $\mathcal{M}(q_f)$. By A-1 there exists a $\hat{q}_{mi}$ such that $(\hat{q}_{mi}, q_m^{-i}) \in \mathcal{M}(q''_{fi}, q_f^{-i})$. As matching is symmetric, equivalently:

$$(q''_{fi}, q_f^{-i}) \in \mathcal{M}(\hat{q}_{mi}, q_m^{-i}). \tag{A.12}$$

In case $\hat{q}_{mi} = q_{mi}$, (A.12) yields $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q_{mi}, q_m^{-i})$ and we have shown convexity of $\mathcal{M}_i$. Now suppose $\hat{q}_{mi} < q_{mi}$ then PAM (i) together with (A.10) and (A.12) implies $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)$. If $\hat{q}_{mi} > q_{mi}$ the same follows from PAM (i), together with (A.11) and (A.12). In each case we have shown that $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q_{mi}, q_m^{-i})$, and thus that $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i})$ is convex. $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i})$ is thus an interval

⁶²Note that $\mathcal{M}_i$ is bounded, as it is a subset of $[\underline{q}_i, \bar{q}_i]$ by assumption.

described by bounds $a_i(q_{mi}, q_m^{-i}, q_f^{-i})$, $b_i(q_{mi}, q_m^{-i}, q_f^{-i})$.

2. $a_i(q_{mi}, q_m^{-i}, q_f^{-i})$ and $b_i(q_{mi}, q_m^{-i}, q_f^{-i})$ are weakly increasing in q_{mi} under $PAM(i)$:

b_i is weakly increasing in q_{mi} : Suppose not, then $b_i(q'_{mi}, q_m^{-i}, q_f^{-i}) > b_i(q''_{mi}, q_m^{-i}, q_f^{-i})$ for some $q'_{mi} < q''_{mi}$. Note that as $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i}) = [a_i(q_{mi}, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})]$ it follows that $b_i(q'_{mi}, q_m^{-i}, q_f^{-i}) \in \mathcal{M}_i(q'_{mi}, q_m^{-i}, q_f^{-i})$ and $b_i(q''_{mi}, q_m^{-i}, q_f^{-i}) \in \mathcal{M}_i(q''_{mi}, q_m^{-i}, q_f^{-i})$. Equivalently $(b_i(q'_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}((q'_{mi}, q_m^{-i}))$ and $(b_i(q''_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$. By $PAM(i)$ this constellation implies $(b_i(q'_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}(q''_{mi}, q_m^{-i})$. Equivalently, $b_i(q'_{mi}, q_m^{-i}, q_f^{-i}) \in \mathcal{M}_i(q''_{mi}, q_m^{-i}, q_f^{-i})$, in contradiction to $b_i(q''_{mi}, q_m^{-i}, q_f^{-i})$ being the upper bound of $\mathcal{M}_i(q''_{mi}, q_m^{-i}, q_f^{-i})$.

That a_i is weakly increasing in q_{mi} follows by similar steps that yield, $a_i(q''_{mi}, q_m^{-i}, q_f^{-i}) \in \mathcal{M}_i(q'_{mi}, q_m^{-i}, q_f^{-i})$, in contradiction to $a_i(q'_{mi}, q_m^{-i}, q_f^{-i})$ being the lower bound of $\mathcal{M}_i(q'_{mi}, q_m^{-i}, q_f^{-i})$.

The proof that $a_i(q_{mi}, q_m^{-i}, q_f^{-i})$ and $b_i(q_{mi}, q_m^{-i}, q_f^{-i})$ are weakly decreasing in q_{mi} under $NAM(i)$ proceeds analogously. $\square$

Proposition A.1. Consider the multidimensional case, $K > 1$, of the described search and matching environment in steady-state equilibrium, and suppose Assumption A-1 holds.

Under either $PAM(i)$ or $NAM(i)$:

1. Job displacement increases the separation risk: $\gamma_B \geq 0$.
2. Job displacement may increase or decrease the probability of staying single:
 $\gamma_{R=0|B}$ may be positive or negative.

Under $PAM(i)$:

- 3.-a Job displacement leads men to rematch with women of lower type: $\gamma_{\Delta q_{fi}|R} \leq 0$.
- 4.-a The association between job displacement and partner type is bounded above: $\gamma_{\Delta q_{fi}|R} \leq \bar{\gamma}_{\Delta q_{fi}|R}$.
The upper bound is given by

$$\bar{\gamma}_{\Delta q_{fi}|R} = - \int \int \int_0^d \frac{\partial \mathbb{E} [q_{fi} | q_{mi}, q_m^{-i}, q_f^{-i}]}{\partial q_{mi}} \bigg|_{q_{mi}=q-x} dx dG_f^{-i}(q_f^{-i}) dF(q | D_R = 1, D_B = 1, D = 1) \leq 0.$$

- 5.-a If $F(q_{mi} | D_B = 1, D = 1) \leq F(q_{mi} | D_B = 1, D = 0)$ holds additionally, then, on average, women from whom displaced men separate are of higher type than women from whom non-displaced men separate: $\gamma_{q_{fi}|B} \geq 0$.

Under $NAM(i)$:

- 3.-b Job displacement leads men to rematch with women of higher type: $\gamma_{\Delta q_{fi}|R} \geq 0$.

4.-b The association between job displacement and partner type is bounded below: $\gamma_{\Delta q_{fi}|R} \geq \underline{\gamma}_{\Delta q_{fi}|R}$.

The lower bound is given by

$$\underline{\gamma}_{\Delta q_{fi}|R} = - \int \int \int_0^d \frac{\partial \mathbb{E} [q_{fi}|q_{mi}, q_m^{-i}, q_f^{-i}]}{\partial q_{mi}} \bigg|_{q_{mi}=q-x} dx dG_f^{-i}(q_f^{-i}) dF(q|D_R = 1, D_B = 1, D = 1) \geq 0.$$

5.-b If $F(q_{mi}|D_B = 1, D = 1) \geq F(q_{mi}|D_B = 1, D = 0)$ holds additionally, then, on average, women from whom displaced men separate are of lower type than women from whom non-displaced men separate: $\gamma_{q_{fi}|B} \leq 0$.

Proof of Proposition A.1: We first prove that under PAM(i) or NAM(i), $\gamma_B \geq 0$.

As men in the control group are not displaced, their types are unchanged between t_0 and τ , i.e., $q_m(\tau) = q_m(t_0)$. It follows that no endogenous breakups occur in the control group, while exogenous breakups occur at rate δ . Like in the one-dimensional case, the probability that control group couples break up between t_0 and τ is thus given by

$$P(D_B = 1|D = 0) = 1 - e^{-\delta(\tau-t_0)} \quad (\text{A.13})$$

under PAM as well as under NAM.

In the treatment group, the i -th dimension of men's type changes between t_0 and τ due to job displacement. Specifically, $q_{mi}(\tau) = q_{mi}(t_0) - d < q_{mi}(\tau)$. For a given man, with pre-displacement type $q_m(t_0)$, job displacement leads to a breakup if and only if $q_f(t_0) \in \mathcal{M}((q_m^i, q_m^{-i}))$ and $q_f(t_0) \notin \mathcal{M}((q_m^i - d, q_m^{-i}))$. By Lemma A.1, equivalently $q_{fi} \in [a_i(q_{mi}, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})]$ and $q_{fi} \notin [a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), b_i(q_{mi} - d, q_m^{-i}, q_f^{-i})]$. Further, $a_i(q_{mi}, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})$ are weakly increasing in q_{mi} under PAM(i) and weakly decreasing in q_{mi} under NAM(i).

It follows under PAM(i) that job displacement leads to a breakup for a man of pre-displacement type q_m if and only if he is matched with a q_f -type woman, such that

$$q_{fi} \in \left(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}, b_i(q_{mi}, q_m^{-i}, q_f^{-i}) \right].$$

Similarly, under NAM job displacement leads to breakup for a man of pre-displacement type q_m if and only if he is matched with a woman of type q_f , such that

$$q_{fi} \in \left[a_i(q_{mi}, q_m^{-i}, q_f^{-i}), \max\{a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), b_i(q_{mi}, q_m^{-i}, q_f^{-i})\} \right).$$

Additionally, breakups occur exogenously at rate δ under PAM(i) as well as under NAM(i).

Denote by $G_{fi}(q_{fi})$ the marginal CDF of q_{fi} , by $G_f^-(q_f^-)$ the joint CDF of q_f^- , and by $G_{fi}(q_{fi}|q_f^-)$ the marginal CDF of q_{fi} conditional on q_f^- .

Under PAM(i) the overall probability that a man in the treatment group experiences a breakup between t_0 and τ is:

$$\begin{aligned} P(D_B = 1|D = 1) &= 1 - e^{-\delta(\tau-t_0)} \\ &+ \int \int G_{fi} \left(b_i(q_{mi}, q_m^-, q_f^-) | q_f^- \right) \\ &- G_{fi} \left(\max\{b_i(q_{mi} - d, q_m^-, q_f^-), a_i(q_{mi}, q_m^-, q_f^-)\} | q_f^- \right) dG_f^-(q_f^-) dF(q_m) \end{aligned} \quad (\text{A.14})$$

Similarly, under NAM(i) the overall probability that a man in the treatment group experiences a breakup between t_0 and τ is:

$$\begin{aligned} P(D_B = 1|D = 1) &= 1 - e^{-\delta(\tau-t_0)} \\ &+ \int \int G_{fi} \left(\min\{a_i(q_{mi} - d, q_m^-, q_f^-), b_i(q_{mi}, q_m^-, q_f^-)\} | q_f^- \right) \\ &- G_{fi} \left(a_i(q_{mi} - d, q_m^-, q_f^-) | q_f^- \right) dG_f^-(q_f^-) dF(q_m). \end{aligned} \quad (\text{A.15})$$

From (A.13), (A.14), and (A.15) it follows that under PAM(i) as well as under NAM(i)

$\gamma_B = P(D_B = 1|D = 1) - P(D_B = 1|D = 0) \geq 0$, concluding the proof of statement 1.

Next, we turn to proving that under PAM(i), $\gamma_{\Delta q_{fi}} \leq 0$.

Denote by D_δ an indicator that equals 1 for men who experience an exogenous breakup between t_0 and τ , and 0 for all other men. Consider men in the treatment group of pre-displacement type q_m , who separate from their t_0 -partner and rematch with a new partner between t_0 and τ . Moreover, condition on the t_0 -partner's type in all but the i -th dimension, $q_f^-(t_0) = q_f^-$. The conditional mean of the t_0 -partner's type in the i -th dimension can be written as weighted average:

$$\begin{aligned} &\mathbb{E} \left[q_{fi}(t_0) | D_R = 1, D_B = 1, D = 1, q_m(t_0) = q_m, q_f^-(t_0) = q_f^- \right] = \\ &\mathbb{E} \left[q_{fi}(t_0) | D_R = 1, D_B = 1, D = 1, q_m, q_f^-, D_\delta = 1 \right] P(D_\delta = 1 | D_R = 1, D_B = 1, D = 1, q_m, q_f^-) \\ &+ \mathbb{E} \left[q_{fi}(t_0) | D_R = 1, D_B = 1, D = 1, q_m, q_f^-, D_\delta = 0 \right] P(D_\delta = 0 | D_R = 1, D_B = 1, D = 1, q_m, q_f^-) \\ &= \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^-, q_f^-) | q_f^-) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^-, q_f^-), a_i(q_{mi}, q_m^-, q_f^-)\} | q_f^-)} \end{aligned}$$

$$\begin{aligned}
& \frac{1}{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(a_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i})} \int_{a_i(q_{mi}, q_m^{-i}, q_f^{-i})}^{b_i(q_{mi}, q_m^{-i}, q_f^{-i})} q_{fi} dG_{fi}(q_{fi}|q_f^{-i}) \\
& + \frac{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})} \\
& \cdot \frac{1}{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})} \\
& \int_{\max\{b_i(q_{mi}-d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}}^{b_i(q_{mi}, q_m^{-i}, q_f^{-i})} q_{fi} dG_{fi}(q_{fi}|q_f^{-i}) \\
& = \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})} \\
& \cdot \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \\
& + \frac{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})|q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\}|q_f^{-i})} \\
& \cdot \mathbb{E} \left[q_{fi} | \max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \tag{A.16}
\end{aligned}$$

Taking into account that treatment group men are displaced in period t_0 , the corresponding average for period τ is:

$$\begin{aligned}
& \mathbb{E} \left[q_{fi}(\tau) | D_R = 1, D_B = 1, D = 1, q_m, q_f^{-i} \right] \\
& = \frac{1}{G_{fi}(b_i(q_{mi} - d, q_m^{-i}, q_f^{-i})) - G_{fi}(a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}))} \int_{a_i(q_{mi}-d, q_m^{-i}, q_f^{-i})}^{b_i(q_{mi}-d, q_m^{-i}, q_f^{-i})} q_{fi} dG_{fi}(q_{fi}|q_f^{-i}) \\
& = \mathbb{E} \left[q_{fi} | a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), q_f^{-i} \right]. \tag{A.17}
\end{aligned}$$

For the control group, by contrast, men's types are unchanged between t_0 and τ . The corresponding expressions therefore are:

$$\begin{aligned}
& \mathbb{E} \left[q_{fi}(t_0) | D_R = 1, D_B = 1, D = 0, q_m, q_f^{-i} \right] \\
& = \mathbb{E} \left[q_{fi}(\tau) | D_R = 1, D_B = 1, D = 0, q_m, q_f^{-i} \right] \\
& = \frac{1}{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})) - G_{fi}(a_i(q_{mi}, q_m^{-i}, q_f^{-i}))} \int_{a_i(q_{mi}, q_m^{-i}, q_f^{-i})}^{b_i(q_{mi}, q_m^{-i}, q_f^{-i})} q_{fi} dG_{fi}(q_{fi}|q_f^{-i})
\end{aligned}$$

$$= \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \quad (\text{A.18})$$

Using (A.16), (A.23), and (A.18) it follows for $\gamma_{\Delta q_{fi}}$ that

$$\begin{aligned} \gamma_{\Delta q_{fi}|R} &= \int \mathbb{E} \left[q_{fi}(\tau) - q_{fi}(t_0) | D_R = 1, D_B = 1, D = 1, q_m, q_f^{-i} \right] dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 1) \\ &\quad - \int \mathbb{E} \left[q_{fi}(\tau) - q_{fi}(t_0) | D_R = 1, D_B = 1, D = 0, q_m, q_f^{-i} \right] dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 0) \\ &= \int \int \frac{1 - e^{-\delta(\tau-t_0)}}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i}) | q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} | q_f^{-i})} \\ &\quad \left(\mathbb{E} \left[q_{fi} | a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] - \right. \\ &\quad \left. \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \right) \\ &\quad + \frac{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i}) | q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} | q_f^{-i})}{1 - e^{-\delta(\tau-t_0)} + G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i}) | q_f^{-i}) - G_{fi}(\max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} | q_f^{-i})} \\ &\quad \left(\mathbb{E} \left[q_{fi} | a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] - \right. \\ &\quad \left. \mathbb{E} \left[q_{fi} | \max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \right) \\ &\quad dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 1) \\ &\leq \int \int \left(\mathbb{E} \left[q_{fi} | a_i(q_{mi} - d, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \right. \\ &\quad \left. - \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \right) dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 1) \\ &= \int \int \mathbb{E} \left[q_{fi} | q_{mi} - d, q_m^{-i}, q_f^{-i} \right] - \mathbb{E} \left[q_{fi} | q_{mi}, q_m^{-i}, q_f^{-i} \right] dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 1) \\ &= \int \int \mathbb{E} \left[q_{fi} | q_{mi} - d, q_m^{-i}, q_f^{-i} \right] - \mathbb{E} \left[q_{fi} | q_{mi}, q_m^{-i}, q_f^{-i} \right] dG_f^{-i}(q_f^{-i}) dF(q_m | D_R = 1, D_B = 1, D = 1) \\ &= - \int \int \int_0^d \frac{\partial \mathbb{E} \left[q_{fi} | q_{mi}, q_m^{-i}, q_f^{-i} \right]}{\partial q_{mi}} \Big|_{q_{mi}=q-x} dx dG_f^{-i}(q_f^{-i}) dF(q | D_R = 1, D_B = 1, D = 1) \\ &= \bar{\gamma}_{\Delta q_{fi}|R}, \end{aligned}$$

where the weak inequality follows as

$$\begin{aligned} &\mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] \\ &\leq \mathbb{E} \left[q_{fi} | \max\{b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), a_i(q_{mi}, q_m^{-i}, q_f^{-i})\} < q_{fi} < b_i(q_{mi} - d, q_m^{-i}, q_f^{-i}), q_f^{-i} \right]. \quad (\text{A.19}) \end{aligned}$$

By Lemma A.1 under PAM(i)

$$\mathbb{E} \left[q_{fi} | q_{mi}, q_m^{-i}, q_f^{-i} \right] = \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right]$$

is weakly increasing in q_{mi} , from which $\bar{\gamma}_{\Delta q_{fi}|R} \leq 0$ follows. This concludes the proof of statements 3.-a and 3.-b.

By analogous steps it can be shown that under NAM(i) $\gamma_{\Delta q_{fi}|R}$ is weakly positive and bounded below by $\bar{\gamma}_{\Delta q_{fi}|R} \geq 0$ (statements 4.-a and 4.-b). $\square$

Lemma 1. *Given the assumptions of Lemma A.1 and A-2, under PAM(j) or NAM(j)*

$$\{q_{fi} : (q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)\} = [a_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i}), b_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i})],$$

where a_i, b_i are

(i) increasing in q_{mj} under PAM(j),

(ii) decreasing in q_{mj} under NAM(j).

Proof of Lemma 1: We start by proving that for any $q'_{fi} < q''_{fi}$, $q'_{mj} < q''_{mj}$, q_f^{-i} , and q_m^{-j} :

$$\begin{aligned} (q'_{fi}, q_f^{-i}) &\in \mathcal{M}(q''_{mj}, q_m^{-j}) \text{ and } (q''_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mj}, q_m^{-j}) \\ \Rightarrow (q'_{fi}, q_f^{-i}) &\in \mathcal{M}(q'_{mj}, q_m^{-j}) \text{ and } (q''_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mj}, q_m^{-j}). \end{aligned} \quad (\text{A.20})$$

Under PAM(j) it follows by Lemma A.1 that:

$$(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mj}, q_m^{-j}) \Leftrightarrow q_{fj} \in [a_j(q''_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j}), b_j(q''_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j})],$$

with a_j, b_j weakly increasing in q_{mj} . It follows that $a_j(q'_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j}) \leq a_j(q''_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j})$. In the special case $q_{fj} = a_j(q'_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j})$, it follows trivially that $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mj}, q_m^{-j})$. Outside this special case, it holds that $a_j(q'_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j}) < q_{fj}$. It follows that there exists a $\check{q}_{fj} < q_{fj}$ such that $\check{q}_{fj} \in [a_j(q'_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j}), b_j(q'_{mj}, q_m^{-j}, q'_{fi}, q_f^{-i,j})]$, or equivalently $(q'_{fi}, \check{q}_{fj}, q_f^{-i,j}) \in \mathcal{M}(q'_{mj}, q_m^{-j})$. Together with $(q''_{fi}, q_{fj}, q_f^{-i,j}) \in \mathcal{M}(q'_{mj}, q_m^{-j})$ by A-2, $(q'_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mj}, q_m^{-j})$ is implied (the first part of the right hand side of implication A.20).

By analogous steps, using PAM(j) together with Lemma A.1 and A-2, it can be shown that $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q'_{mj}, q_m^{-j})$ implies $(q''_{fi}, q_f^{-i}) \in \mathcal{M}(q''_{mj}, q_m^{-j})$, proving the second part of implication A.20.

By Lemma A.1 we have

$$\mathcal{M}_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i}) = [a_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i}), b_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i})]$$

for $\mathcal{M}_i(q_{mi}, q_m^{-i}, q_f^{-i}) := \{q_{fi} : (q_{fi}, q_f^{-i}) \in \mathcal{M}(q_m)\}$. Next, we use A.20 to show that under PAM(j) b_i is weakly increasing in q_{mj} : Suppose not, then $b_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i}) > b_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i})$ for some $q'_{mj} < q''_{mj}$.

From $\mathcal{M}_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i}) = [a_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i}), b_i(q_{mi}, q_{mj}, q_m^{-i,j}, q_f^{-i})]$ it follows that $b_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i}) \in \mathcal{M}_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i})$ and $b_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i}) \in \mathcal{M}_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i})$. Equivalently, $(b_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}((q_{mi}, q'_{mj}, q_m^{-i,j}))$ and $(b_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}((q_{mi}, q''_{mj}, q_m^{-i,j}))$.

By A.20 this constellation implies $(b_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i}), q_f^{-i}) \in \mathcal{M}((q_{mi}, q''_{mj}, q_m^{-i,j}))$, or equivalently, $b_i(q_{mi}, q'_{mj}, q_m^{-i,j}, q_f^{-i}) \in \mathcal{M}_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i})$, in contradiction to $b_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i})$ being the upper bound of $\mathcal{M}_i(q_{mi}, q''_{mj}, q_m^{-i,j}, q_f^{-i})$.

By similar steps it can be shown that a_i is weakly increasing in q_{mj} .

The proof that a_i, b_i are weakly decreasing in q_{mj} under NAM(j) proceeds analogously. $\square$

Proof of Proposition 2: We first show that PAM(i) implies $DE \geq 0$. By Lemma A.1

$$\begin{aligned} \mathbb{E}[q_{fi}|q_{mi}, q_{mj}, q_{fj}] &= \frac{1}{G_{fi}(b_i(q_{mi}, q_{mj}, q_{fj})) - G_{fi}(a_i(q_{mi}, q_{mj}, q_{fj}))} \int_{a_i(q_{mi}, q_{mj}, q_{fj})}^{b_i(q_{mi}, q_{mj}, q_{fj})} q_{fi} dG_{fi}(q_{fi}|q_{fj}) \\ &= \mathbb{E}[q_{fi}|a_i(q_{mi}, q_{mj}, q_{fj}) < q_{fi} < b_i(q_{mi}, q_{mj}, q_{fj}), q_{fj}]. \end{aligned}$$

where, under PAM(i), $a_i(q_{mi}, q_{mj}, q_{fj})$ and $b_i(q_{mi}, q_{mj}, q_{fj})$ are weakly increasing in q_{mi} implying the same for $\mathbb{E}[q_{fi}|q_{mi}, q_{mj}, q_{fj}]$. It follows that

$$E[q_{fi}|q_{mi}, q_{mj}] = \int \mathbb{E}[q_{fi}|a_i(q_{mi}, q_{mj}, q_{fj}) < q_{fi} < b_i(q_{mi}, q_{mj}, q_{fj}), q_{fj}] dG_{fj}(q_{fj})$$

is also weakly increasing in q_{mi} , and

$$DE = \int E[q_{fi}|q''_{mi}, q_{mj}] - E[q_{fi}|q'_{mi}, q_{mj}] dG_{mj}(q_{mj}|q''_{mi}) \geq 0.$$

By analogous steps it follows that NAM(i) implies $DE \leq 0$.

Next, we establish that under PAM(j) if $G_{mj}(q_{mj}|q_{mi})$ is weakly decreasing in q_{mi} , $IE \geq 0$ follows. By

Lemma A.1

$$\mathbb{E}[q_{fi}|q_{mi}, q_{mj}, q_{fj}] = \mathbb{E}[q_{fi}|a_i(q_{mi}, q_{mj}, q_{fj}) < q_{fi} < b_i(q_{mi}, q_{mj}, q_{fj}), q_{fj}].$$

By Lemma 1 $a_i(q_{mi}, q_{mj}, q_{fj})$ and $b_i(q_{mi}, q_{mj}, q_{fj})$ are weakly increasing in q_{mj} under PAM (j), implying the same for $\mathbb{E}[q_{fi}|q_{mi}, q_{mj}, q_{fj}]$. It follows that

$$E[q_{fi}|q_{mi}, q_{mj}] = \int \mathbb{E}[q_{fi}|a_i(q_{mi}, q_{mj}, q_{fj}) < q_{fi} < b_i(q_{mi}, q_{mj}, q_{fj}), q_{fj}] dG_{fj}(q_{fj})$$

weakly increasing in q_{mj} . As $G(q_{mj}|q''_{mi})$ first order stochastically dominates $G(q_{mj}|q'_{mi})$ this implies

$$IE = \int \mathbb{E}[q_{fi}|q'_{mi}, q_{mj}] dG(q_{mj}|q''_{mi}) - \int \mathbb{E}[q_{fi}|q'_{mi}, q_{mj}] dG(q_{mj}|q'_{mi}) \geq 0.$$

The remaining implications for IE follow analogously. □

Proposition 3. Consider the described matching environment in the multidimensional case, $K > 1$ and suppose that A-1 and A-2 hold. Consider the following decomposition for $q''_{mi} \geq q'_{mi}$

$$\begin{aligned} \mathbb{E}[q_{fi}|q''_{mi}] &- \mathbb{E}[q_{fi}|q'_{mi}] \\ &= \underbrace{\int \mathbb{E}[q_{fi}|q''_{mi}, q_m^{-i}] - \mathbb{E}[q_{fi}|q'_{mi}, q_m^{-i}] dG(q_m^{-i}|q''_{mi})}_{=DE \text{ (Direct effect)}} \\ &+ \sum_{k \neq i} \left(\underbrace{\int \int \int \mathbb{E}[q_{fi}|q'_{mi}, q_m^{-i}] dG(q_{m,1:k-1} \setminus \{i\} | q_{m,k:K} \setminus \{i\}, q'_{mi}) dG(q_{mk} | q_{m,k+1:K} \setminus \{i\}, q''_{mi})}_{\vdots = IE_k \text{ (Indirect effect from k-th dimension)}} \right. \\ &\quad \left. - \underbrace{\int \int \mathbb{E}[q_{fi}|q'_{mi}, q_m^{-i}] dG(q_{m,1:k-1} \setminus \{i\} | q_{m,k:K} \setminus \{i\}, q'_{mi}) dG(q_{mk} | q_{m,k+1:K} \setminus \{i\}, q'_{mi})}_{\vdots = IE_k \text{ (Indirect effect from k-th dimension)}} \right. \\ &\quad \left. dG(q_{m,k+1:K} \setminus \{i\} | q''_{mi}) \right). \end{aligned}$$

In a multi-dimensional steady state matching equilibrium the following implications hold:

$$PAM(i) \Rightarrow DE \geq 0,$$

$$NAM(i) \Rightarrow DE \leq 0$$

Given $PAM(j)$ for $j \in A_{PAM}$ and $NAM(j)$ for $j \in A_{NAM}$, where $A_{PAM} \cup A_{NAM} = \{1, \dots, K\}$, the following additional implications hold.⁶³

⁶³Note that analogous sufficient conditions for $IE_k \geq 0$ and $IE_k \leq 0$ can be proved assuming in the antecedent that $G(q_{mk}|q_{m,k+1:K} \setminus \{i\}, q_{mi})$ is weakly increasing in q_{mi} . We omit these additional implications for brevity.

$$\left. \begin{aligned}
& (i) \text{ PAM}(k) \text{ and } G(q_{mk}|q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly decreasing in } q_{mi}. \\
& (ii) G(q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}|q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}, q_{mk}, q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly decreasing in } q_{mk}, \\
& \quad \text{and weakly increasing in } q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}. \\
& (iii) G(q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}|q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}, q_{mk}, q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly increasing in } q_{mk}, \\
& \quad \text{and weakly increasing in } q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}.
\end{aligned} \right\} \Rightarrow IE_k \geq 0, \quad (\text{A.21})$$

$$\left. \begin{aligned}
& (i) \text{ NAM}(k) \text{ and } G(q_{mk}|q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly decreasing in } q_{mi}. \\
& (ii) G(q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}|q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}, q_{mk}, q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly increasing in } q_{mk}, \\
& \quad \text{and weakly increasing in } q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}. \\
& (iii) G(q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}|q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}, q_{mk}, q_{m,k+1:K}\setminus\{i\}, q_{mi}) \text{ is weakly decreasing in } q_{mk}, \\
& \quad \text{and weakly increasing in } q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}.
\end{aligned} \right\} \Rightarrow IE_k \leq 0, \quad (\text{A.22})$$

Proof of Proposition 3: We first show that PAM(i) implies $DE \geq 0$. By Lemma A.1

$$\begin{aligned}
\mathbb{E} [q_{fi}|q_{mi}, q_m^{-i}, q_f^{-i}] &= \frac{1}{G_{fi}(b_i(q_{mi}, q_m^{-i}, q_f^{-i})) - G_{fi}(a_i(q_{mi}, q_m^{-i}, q_f^{-i}))} \int_{a_i(q_{mi}, q_m^{-i}, q_f^{-i})}^{b_i(q_{mi}, q_m^{-i}, q_f^{-i})} q_{fi} dG_{fi}(q_{fi}|q_f^{-i}) \\
&= \mathbb{E} [q_{fi}|a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i}].
\end{aligned}$$

where, under PAM(i), $a_i(q_{mi}, q_m^{-i}, q_f^{-i})$ and $b_i(q_{mi}, q_m^{-i}, q_f^{-i})$ are weakly increasing in q_{mi} implying the same for $\mathbb{E} [q_{fi}|q_{mi}, q_m^{-i}, q_f^{-i}]$. It follows that

$$E[q_{fi}|q_{mi}, q_m^{-i}] = \int \mathbb{E} [q_{fi}|a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i}] dG_{fj}(q_f^{-i})$$

is also weakly increasing in q_{mi} , and

$$DE = \int E[q_{fi}|q_{mi}'', q_m^{-i}] - E[q_{fi}|q_{mi}', q_m^{-i}] dG_{mj}(q_m^{-i}|q_{mi}'') \geq 0.$$

By analogous steps it follows that NAM(i) implies $DE \leq 0$.

Next, we assume that $\text{PAM}(j)$ for $j \in A_{\text{PAM}}$ and $\text{NAM}(j)$ for $j \in A_{\text{NAM}}$, where $A_{\text{PAM}} \cup A_{\text{NAM}} = \{1, \dots, K\}$, and establish that $IE_k \geq 0$ follows from premise (i) - (iii) of implication (A.21).

Note that IE_k can be expressed as

$$IE_k = \int \int \int \mathbb{E} [q_{fi}|q_{mi}', q_m^{-i}] dG(q_{m,A_{\text{PAM}}\setminus\{i,k:K\}}|q_{m,A_{\text{NAM}}\setminus\{i,k:K\}}, q_{mk}, q_{m,k+1:K}\setminus\{i\}, q_{mi}') dG_{mj}(q_m^{-i}|q_{mi}'')$$

$$\begin{aligned}
& dG(q_{m,A_{\text{NAM}} \setminus \{i,k:K\}} | q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) dG(q_{mk} | q_{m,k+1:K \setminus \{i\}}, q''_{mi}) \\
& - \int \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) \\
& dG(q_{m,A_{\text{NAM}} \setminus \{i,k:K\}} | q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) dG(q_{mk} | q_{m,k+1:K \setminus \{i\}}, q'_{mi}) dG(q_{m,k+1:K \setminus \{i\}} | q''_{mi})
\end{aligned}$$

By Lemma A.1

$$\mathbb{E}[q_{fi} | q_{mi}, q_m^{-i}] = \int \mathbb{E} \left[q_{fi} | a_i(q_{mi}, q_m^{-i}, q_f^{-i}) < q_{fi} < b_i(q_{mi}, q_m^{-i}, q_f^{-i}), q_f^{-i} \right] dG_{fj}(q_f^{-i}).$$

By Lemma 1 $a_i(q_{mi}, q_m^{-i}, q_f^{-i})$ and $b_i(q_{mi}, q_m^{-i}, q_f^{-i})$ are weakly increasing in q_{mj} for all $j \in A_{\text{PAM}}$, and weakly decreasing in q_{mj} for all $j \in A_{\text{NAM}}$, implying the same for $\mathbb{E}[q_{fi} | q_{mi}, q_m^{-i}]$.

By the premise, $G(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q_{mi})$ is weakly decreasing in q_{mk} and weakly increasing in $q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}$. It follows that for any $q'_{mk} \leq q''_{mk}$ and $q'_{m,A_{\text{NAM}} \setminus \{i,k:K\}} \leq q''_{m,A_{\text{NAM}} \setminus \{i,k:K\}}$

$$\begin{aligned}
& \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q''_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q'_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) \\
& \leq \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q'_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q''_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}),
\end{aligned}$$

by $\mathbb{E}[q_{fi} | q_{mi}, q_m^{-i}]$ being weakly increasing in $q_{m,A_{\text{PAM}} \setminus \{i,k:K\}}$, weakly decreasing in $q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}$, and by first order stochastic dominance of $G(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q'_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q''_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi})$ over $G(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q''_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q'_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi})$, implying that

$$\int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) \quad (\text{A.23})$$

is weakly increasing in q_{mk} , and weakly decreasing in $q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}$.

By analogous arguments it follows from $G(q_{m,A_{\text{NAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{PAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q_{mi})$ being weakly increasing in q_{mk} , and by (A.23) being weakly decreasing in q_{mk} , and weakly decreasing in $q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}$ that

$$\begin{aligned}
& \int \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) \\
& dG(q_{m,A_{\text{NAM}} \setminus \{i,k:K\}} | q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi})
\end{aligned} \quad (\text{A.24})$$

is weakly increasing in q_{mk} .

By the premise, $G(q_{mk} | q_{m,k+1:K \setminus \{i\}}, q''_{mi})$ first order stochastically dominates $G(q_{mk} | q_{m,k+1:K \setminus \{i\}}, q'_{mi})$, implying together with (A.24) being weakly increasing in q_{mk} that

$$\begin{aligned}
& \int \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m,A_{\text{PAM}} \setminus \{i,k:K\}} | q_{m,A_{\text{NAM}} \setminus \{i,k:K\}}, q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) \\
& dG(q_{m,A_{\text{NAM}} \setminus \{i,k:K\}} | q_{mk}, q_{m,k+1:K \setminus \{i\}}, q'_{mi}) dG(q_{mk} | q_{m,k+1:K \setminus \{i\}}, q''_{mi})
\end{aligned}$$

$$\begin{aligned}
& - \int \int \mathbb{E}[q_{fi} | q'_{mi}, q_m^{-i}] dG(q_{m, A_{\text{PAM}} \setminus \{i, k: K\}} | q_{m, A_{\text{NAM}} \setminus \{i, k: K\}}, q_{mk}, q_{m, k+1: K \setminus \{i\}}, q'_{mi}) \\
& dG(q_{m, A_{\text{NAM}} \setminus \{i, k: K\}} | q_{mk}, q_{m, k+1: K \setminus \{i\}}, q'_{mi}) dG(q_{mk} | q_{m, k+1: K \setminus \{i\}}, q'_{mi}) \geq 0
\end{aligned} \tag{A.25}$$

As (A.25) is satisfied for any $q_{m, k+1: K \setminus \{i\}}$, integrating over $G(q_{m, k+1: K \setminus \{i\}} | q''_{mi})$ preserves the weak inequality, implying $IE_k \geq 0$.

Implication (A.22) can be proved by analogous steps. □

B Additional Empirical Results

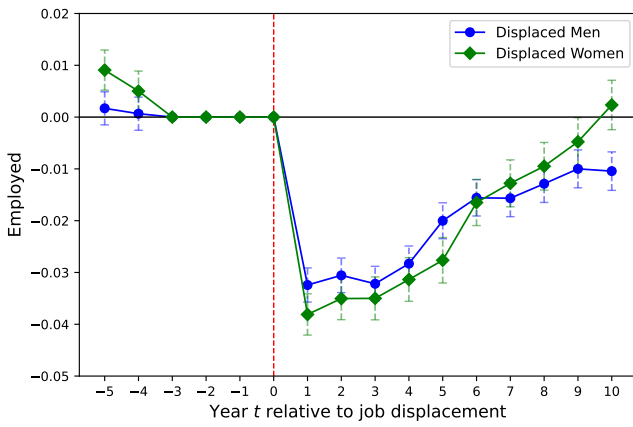
Table B.1: Pre-displacement Summary Statistics, Treatment and Control Group

	Displaced men		Displaced women	
	Treatment	Control	Treatment	Control
Age	38.1 (5.6)	38.1 (5.6)	38.0 (5.7)	37.9 (5.7)
Partner's age	36.2 (6.2)	36.2 (6.2)	40.6 (6.7)	40.7 (6.8)
Years of education	12.5 (2.4)	12.6 (2.4)	12.5 (2.4)	12.5 (2.4)
Partner's years of education	12.2 (2.5)	12.2 (2.5)	12.6 (2.4)	12.7 (2.4)
Job tenure	6.3 (4.0)	6.3 (3.9)	6.1 (4.0)	6.0 (3.9)
No. of children	1.5 (1.0)	1.5 (1.0)	1.4 (0.9)	1.4 (0.9)
Labor income (in DKK)	324,508 (96,438)	323,097 (96,127)	228,746 (74,875)	226,222 (75,179)
Partner's labor income (in DKK)	175,460 (105,535)	176,478 (105,619)	284,461 (140,888)	285,708 (142,544)
Corr(age _f , age _m)	0.83	0.83	0.83	0.83
Corr(education _f , education _m)	0.38	0.39	0.37	0.37
Corr(income _f , income _m)	0.15	0.15	0.14	0.15
<i>N</i>	72,667	72,667	73,856	73,856

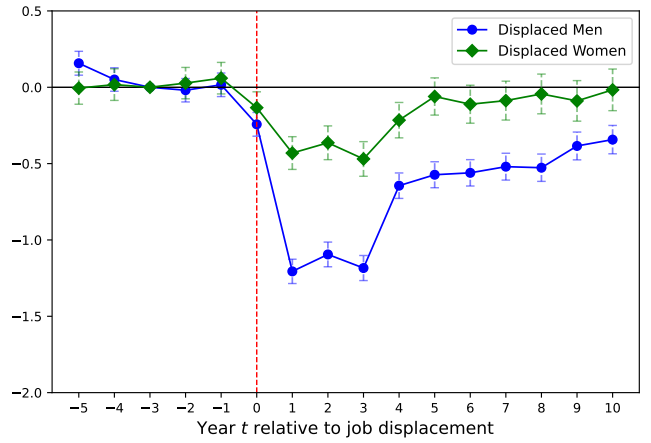
Notes: This table shows summary statistics for actual and placebo displaced men and women in the treatment and control groups. Standard deviations are reported in parentheses. All variables are measured in $t = -1$, i.e., one year before actual or placebo displacement. Years of education are calculated as follows: 9 years for individuals with compulsory education, 12 years for individuals with a high school degree ("Gymnasium"), 13 years for individuals with a vocational degree, 13.5 years for individuals with a degree from professional schools or technical colleges ("Professionsbachelor"), 15 years for individuals with a Bachelor's degree, and 18.5 years for individuals with a Master's or Doctoral degree. Tenure measures the years of employment at the establishment. Labor incomes are real annual labor earnings in DKK (2004 CPI).

Figure B.1: Labor Market Outcomes of Displaced Individuals:
Employment and Weekly Work Hours

(A) Employment

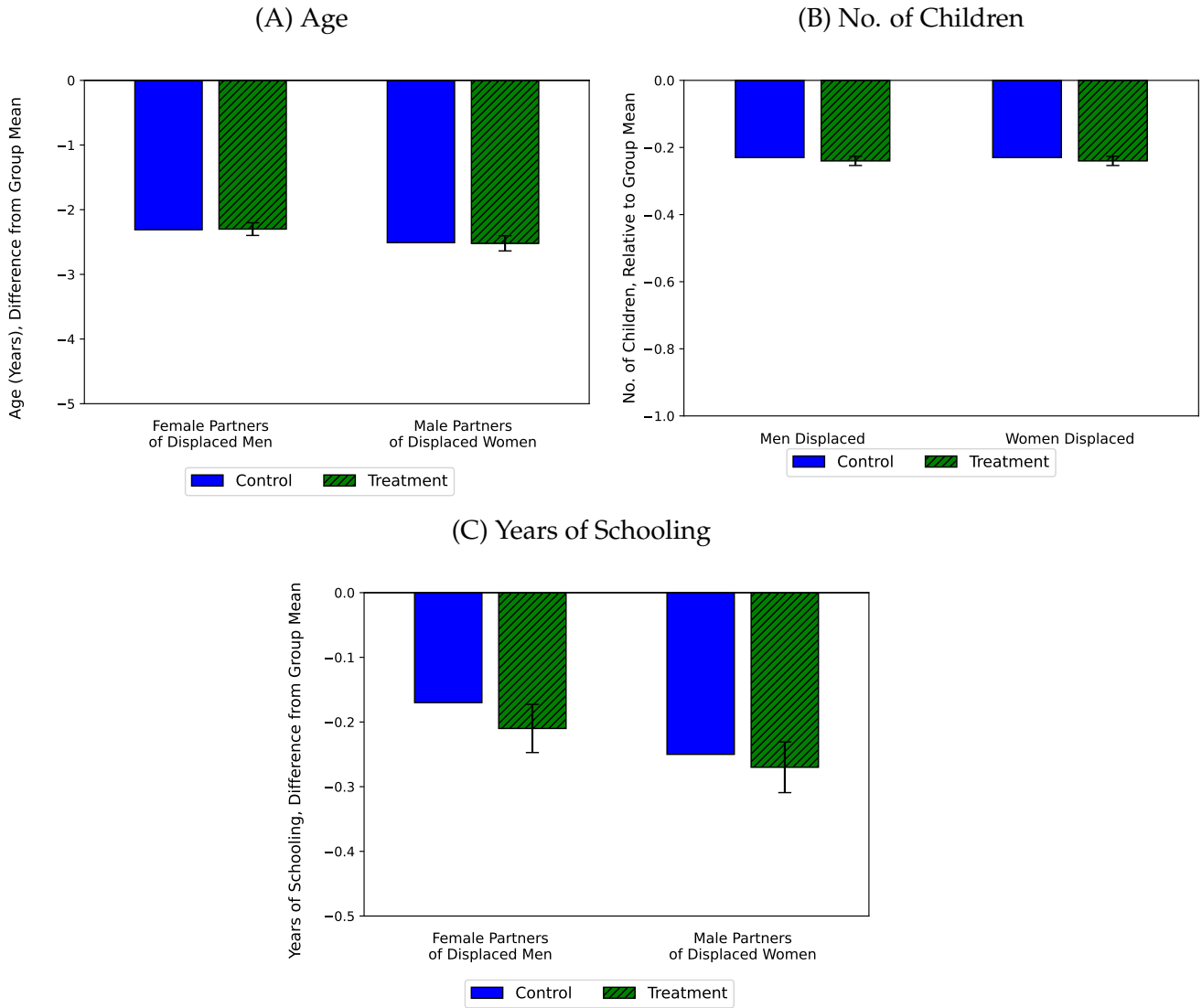


(B) Weekly Work Hours



Notes: The figure shows the impact of job displacement on employment (Panel A) and weekly work hours (Panel B). Dashed vertical lines around each point estimate represent 95% confidence intervals. The estimates correspond to ϕ_τ from Equation (1). All results are based on samples of men and women displaced due to establishment closures between 1980 and 2007, and matched control groups selected via coarsened exact matching. Sample selection criteria and the matching procedure are described in Subsection 2.3.

Figure B.2: Pre-Displacement Age, Education and No. of Children of Separating Partners



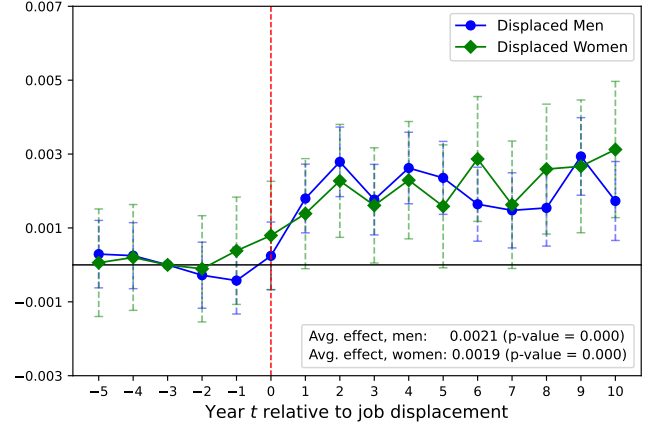
Notes: The figure displays the pre-displacement characteristics of the partners of displaced individuals in couples that subsequently experience a breakup. Panel A shows average age, Panel B shows the average number of children, and Panel C shows average years of schooling in $t \in \{-5, \dots, -3\}$ (before the male partner's actual or placebo displacement) for individuals who separate between $t = 0$ and $t = 10$ (after the male partner's actual or placebo displacement). All values are differences from the respective (treatment or control) group mean. The underlying sample includes men who were displaced as part of an establishment closure between 1980 and 2007, and an equally sized control group selected by coarsened exact matching. The specific sample selection criteria and matching procedure are described in Subsection 2.3.

Figure B.3: Rematching Patterns of Displaced Men:
New Partners' vs. Old Partners' Income

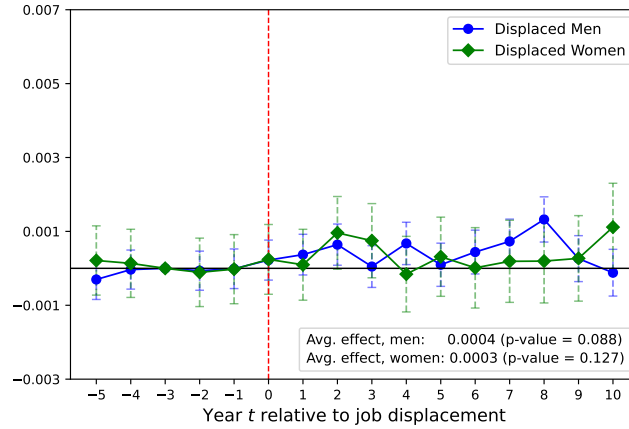
(A) New Partner's Income $\geq 10\%$ Higher
than $t = -3$ Partner's Income



(B) New Partner's Income in $\pm 10\%$ Range
of $t = -3$ Partner's Income



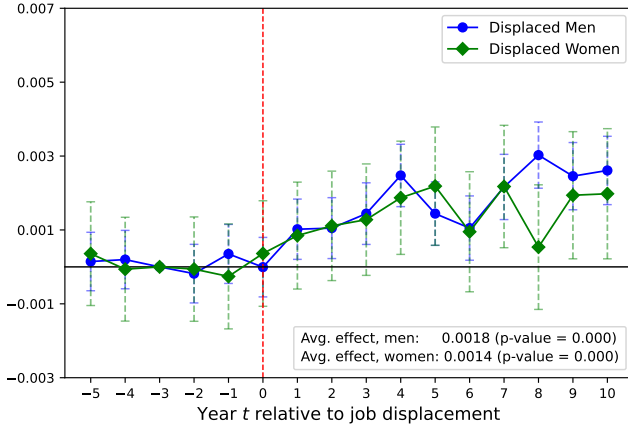
(C) New Partner's Income at least 10% Lower
than $t = -3$ Partner's Income



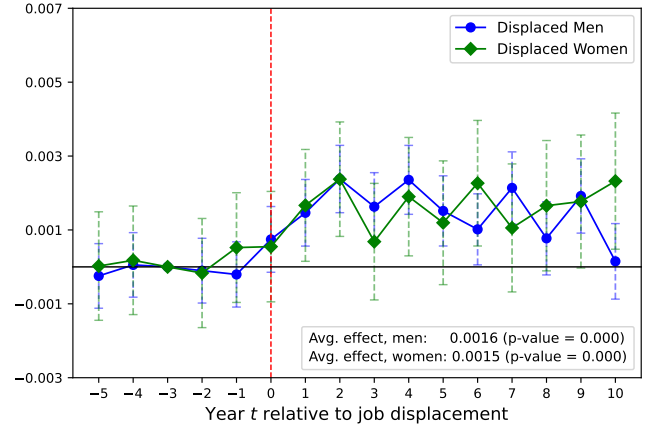
Notes: The figure shows the effect of job displacement on the rematching patterns of men after a breakup. Panel A displays the effect on the probability of matching with a new partner with at least 10% higher labor income than that of the pre-displacement partner (defined as the partner at $t = -3$). Panel B shows the effect on the probability of matching with a new partner with labor income between -10% and $+10\%$ of the pre-displacement partner's labor income. Panel C shows the effect on the probability of matching with a new partner with at least 10% lower labor income than that of the pre-displacement partner. The estimates correspond to ϕ_τ from Equation (1). Dashed vertical lines around point estimates indicate 95% confidence intervals. All estimates are based on a sample of men displaced due to establishment closures between 1980 and 2007 and a control group selected by coarsened exact matching. Sample selection and matching procedures are detailed in Subsection 2.3.

Figure B.4: Rematching Patterns of Displaced Men:
New Partners' vs. Old Partners' Hourly Wage

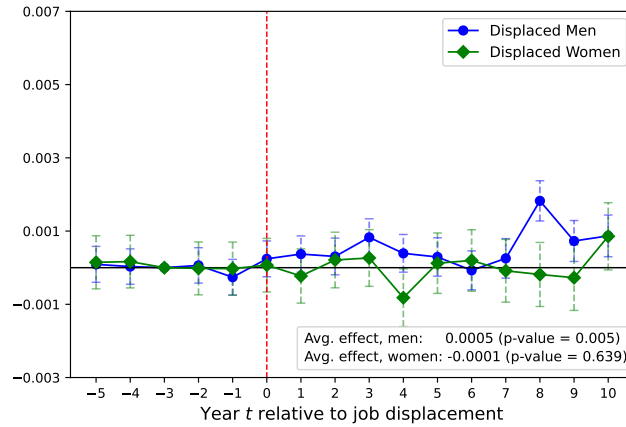
(A) New Partner's Hourly Wage $\geq 5\%$ Higher
than $t = -3$ Partner's Hourly Wage



(B) New Partner's Hourly Wage in $\pm 5\%$ Range
of $t = -3$ Partner's Hourly Wage



(C) New Partner's Hourly Wage at least 5% Lower
than $t = -3$ Partner's Hourly Wage



Notes: The figure shows the effect of job displacement on the rematching patterns of men after a breakup. Panel A displays the effect on the probability of matching with a new partner with at least 5% higher hourly wage than that of the pre-displacement partner (defined as the partner at $t = -3$). Panel B shows the effect on the probability of matching with a new partner with hourly wage between -5% and $+5\%$ of the pre-displacement partner's hourly wage. Panel C shows the effect on the probability of matching with a new partner with at least 5% lower hourly wage than that of the pre-displacement partner. The estimates correspond to ϕ_τ from Equation (1). Dashed vertical lines around point estimates indicate 95% confidence intervals. All estimates are based on a sample of men displaced due to establishment closures between 1980 and 2007 and a control group selected by coarsened exact matching. Sample selection and matching procedures are detailed in Subsection 2.3.

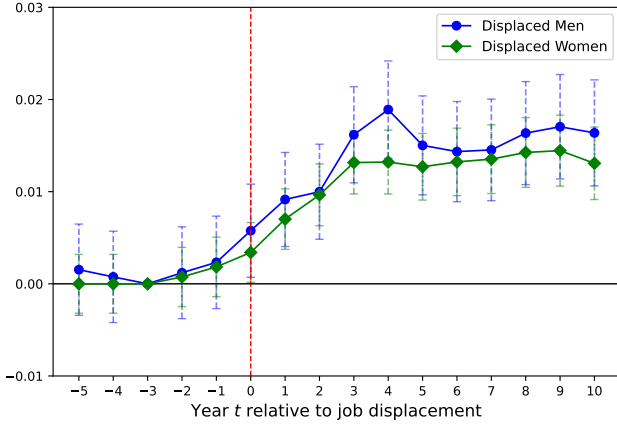
Table B.2: Rematching Patterns of Displaced Men and Women:
New Partners' Age, Education, and No. of Children

Partners' Outcomes	Age	Education	No. of Children
<i>Panel (a): Displaced men</i>			
Treated $\times$ Post-Displacement, γ	-0.26*** (0.09)	-0.04 (0.03)	0.04*** (0.01)
No. of observations	108,982	108,982	108,982
<i>Panel (b): Displaced women</i>			
Treated $\times$ Post-Displacement, γ	-0.01 (0.12)	0.03 (0.04)	-0.004 (0.02)
No. of observations	68,908	68,908	68,908

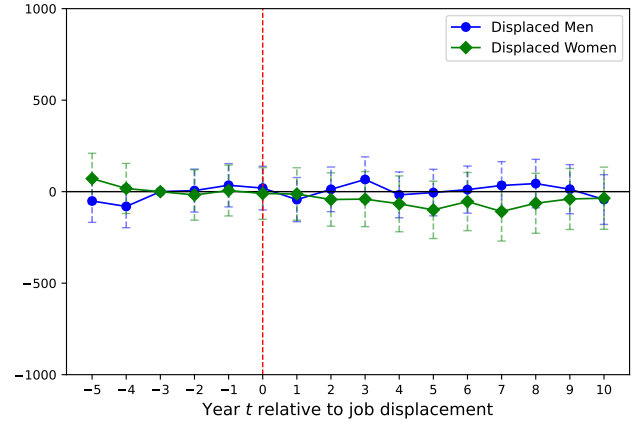
Notes: The table shows the effect of job displacement on the types of partners displaced individuals transition to, measured by partners' age, education (years of schooling), and number of children. Panel (a) reports the effect of men's job displacement on the characteristics of new female partners. Panel (b) reports the effect of women's job displacement on the characteristics of new male partners. The table reports coefficient estimates of γ from Equation (2). Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.5: Robustness Checks

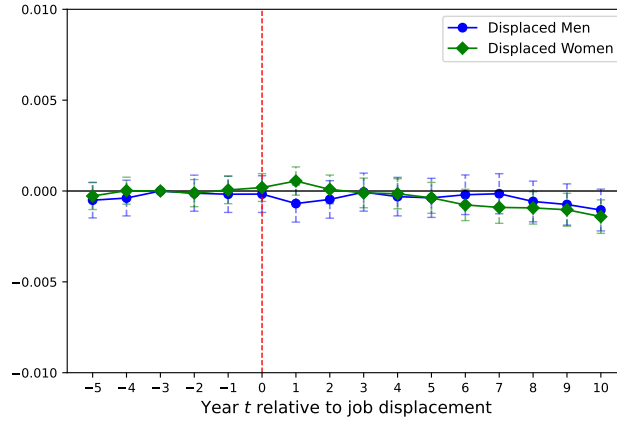
(A) Relocations to a New Municipality



(B) Earnings of Single Women in Location of Residence



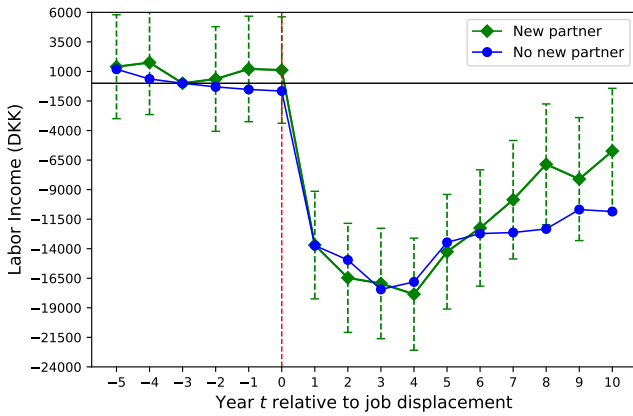
(C) Sex Ratio in Location of Residence



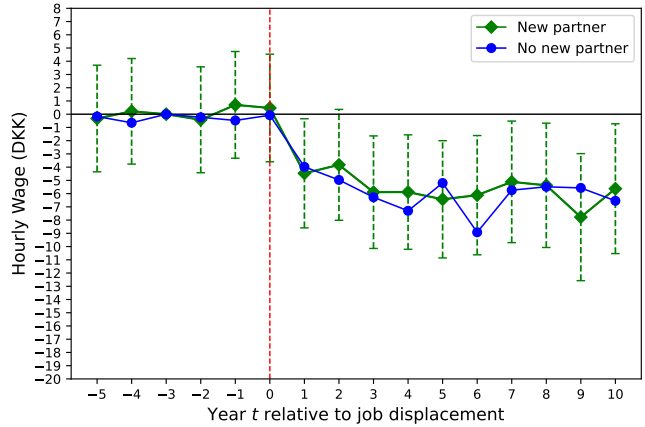
Notes: Panel A shows the effect of job displacement on the probability of residing in a municipality different from that in $t = -3$ for at least three consecutive years. Panel B shows the effect of displacement on the difference in the average annual labor income of single women and of single men, and Panel C on the difference in the local sex ratio $\left(\frac{\# \text{ single women}}{\# \text{ single men}} \right)$, between the individual's municipality of residence in year t and in $t = -3$. Estimates correspond to ϕ_τ from Equation (1) and are shown separately for displaced men and women. All estimates are based on individuals who experienced an establishment closure between 1980 and 2007 and an equally sized control group selected by exact matching. Sample selection and matching procedures are described in Subsection 2.3.

Figure B.6: Labor Market Outcomes of Displaced Individuals:
Labor Income and Hourly Wages, by Rematching Status

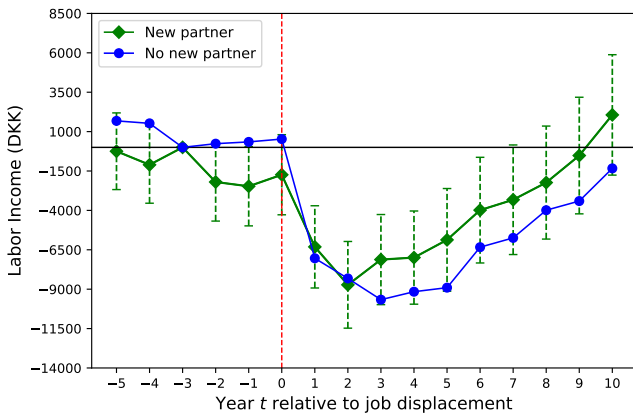
(A) Men's Labor Income (in DKK)



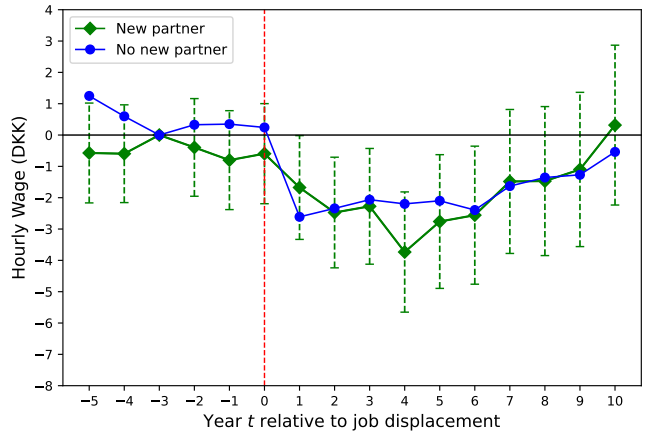
(B) Men's Hourly Wage (in DKK)



(C) Women's Labor Income (in DKK)



(D) Women's Hourly Wage (in DKK)



Notes: The figure shows the impact of job displacement on annual labor income and hourly wages, separately for displaced individuals who do and do not find a new partner in the event-time window $t = 1, \dots, 10$. Panels A and B show results for displaced men; Panels C and D show results for displaced women. Labor income includes zeros for nonemployed individuals, while hourly wages are conditional on employment. Both outcomes are measured in DKK (CPI 2004). The estimates are obtained by fully interacting empirical specification (1) with an indicator for finding a new partner in $t = 1, \dots, 10$. Dashed vertical lines represent 95% confidence intervals for the interaction terms, i.e. for the difference between individuals who do and do not find a new partner in $t = 0, \dots, 10$.

Table B.3: Robustness to Excluding Displaced Men with High Post-Displacement Income Growth

	Baseline	Excl. Top 2%	Excl. Top 10%
Partner Labor Income	3278.61** (1614.72)	4044.12** (1657.42)	3433.61** (1695.73)
No. of observations	108,982	102,086	97,332
Partner Wage	3.98*** (1.46)	4.46*** (1.52)	4.16*** (1.58)
No. of observations	60,362	56,388	53,514

Notes: The table re-estimates the rematching specifications from Table B.2, excluding displaced men in the top 2% and top 10% of the post-displacement income-growth distribution. Post-displacement income growth is measured as the difference between average labor income in $t = 6, \dots, 10$ and labor income in the pre-displacement period $t = -3$. Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Robustness to Excluding Displaced Women with High Post-Displacement Income Growth

	Baseline	Excl. Top 2%	Excl. Top 10%
Partner Labor Income	2282.65 (2342.12)	6879.96** (2745.99)	5241.70* (2824.55)
No. of observations	68,908	53,108	51,378
Partner Wage	0.72 (1.52)	5.16*** (1.79)	4.55** (1.84)
No. of observations	46,628	35,968	34,682

Notes: The table re-estimates the rematching specifications from Table B.2, excluding displaced women in the top 2% and top 10% of the post-displacement income-growth distribution. Post-displacement income growth is measured as the difference between average labor income in $t = 6, \dots, 10$ and labor income in the pre-displacement period $t = -3$. Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Effects of Men's Displacement on Separation by Women's Pre-Displacement Employment and Income

	(1)	(2)	(3)	(4)
Treatment	0.008*** (0.002)	0.014*** (0.004)	0.010*** (0.002)	0.016*** (0.004)
Employed		-0.026*** (0.003)		
Treatment $\times$ Employed		-0.009* (0.004)		
Above Median Income			-0.003 (0.003)	
Treatment $\times$ Above Med. Inc.			-0.009** (0.004)	
Income				-0.004*** (0.001)
Treatment $\times$ Income				-0.005*** (0.002)
Male Partner's Income	-0.009*** (0.001)	-0.009*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
Observations	144, 446	144, 446	142, 268	142, 268

Notes: The table reports heterogeneity in the effect of male job displacement on separation. All employment and income variables are measured prior to the male partner's displacement. The outcome variable is an indicator equal to one if an individual separates at any point in the ten years following displacement (i.e., in $t = 1, \dots, 10$), and zero otherwise. Column (1) reports a baseline specification including only treatment status. Column (2) allows the treatment effect to vary with female partner's employment status prior to the male partner's displacement. Column (3) allows the treatment effect to depend on whether the woman's labor income in period $t = -3$ is above or below the sample median in $t = -3$. Column (4) allows the treatment effect to vary continuously with the female partner's pre-displacement labor income. Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.7: Labor Income of Displaced Individuals' Partners



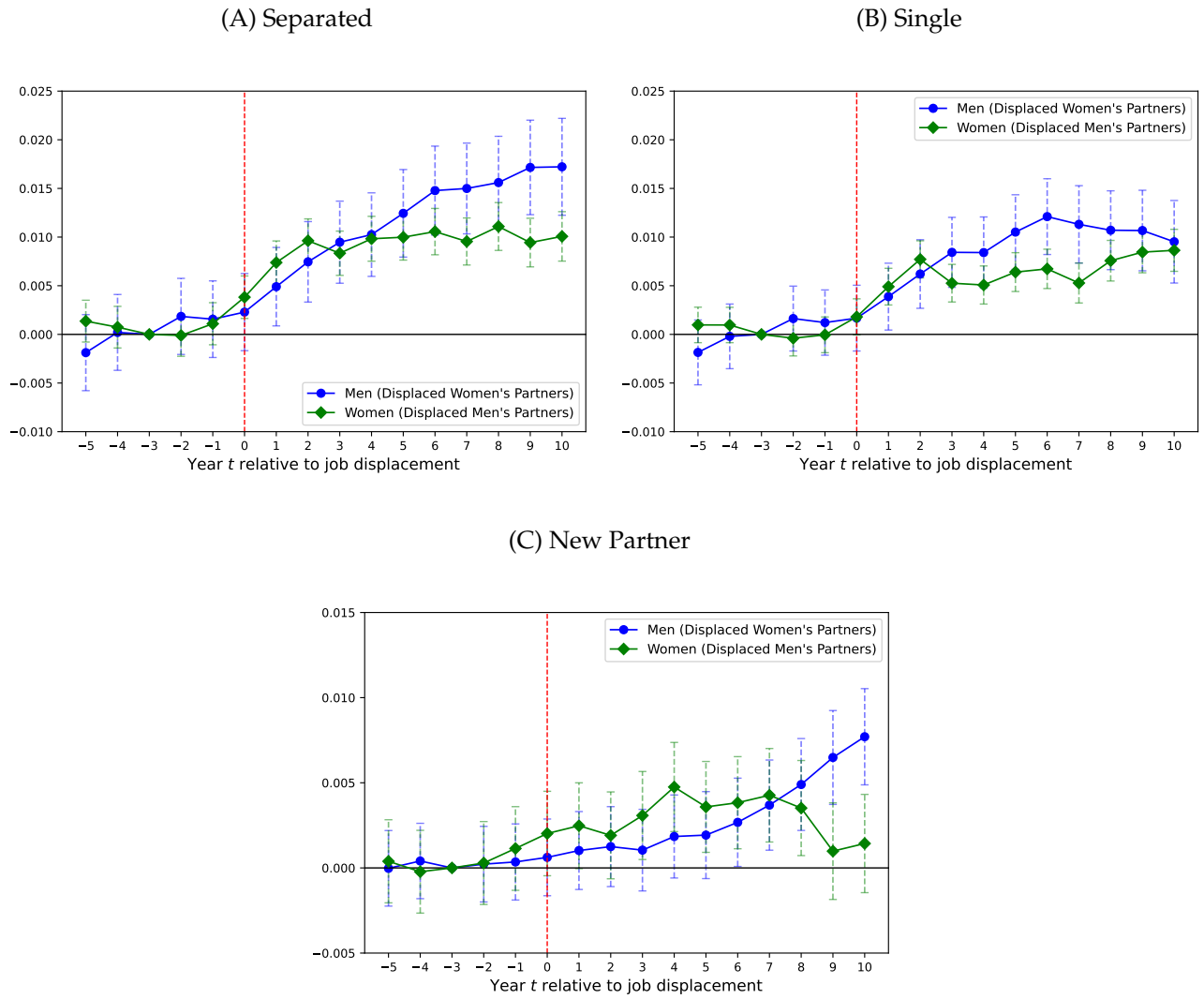
Notes: The figure shows the effect of job displacement on the annual labor income of displaced individuals' partners, including zeros for non-employed individuals. Labor income is measured in DKK (CPI 2004). The values correspond to coefficient estimates of ϕ_τ from empirical specification (1), estimated separately for displaced men's and women's partners. Dashed vertical lines indicate 95% confidence intervals. The underlying sample consists of men and women displaced as part of establishment closures between 1980–2007, and matched control groups selected by coarsened exact matching. The specific sample selection criteria and matching procedure are described in Subsection 2.3.

Table B.6: Rematching Patterns of Displaced Individuals' Partners:
New Partners' Income, Wage, and Work Hours

New Partners' Outcomes	Labor Income	Wage	Work Hours
<i>Panel (a): Partners of displaced men</i>			
Treated $\times$ Post-Displacement, γ	−1657.57 (2806.91)	1.62 (2.75)	−0.16 (0.27)
No. of observations	18,140	15,402	13,884
<i>Panel (b): Partners of displaced women</i>			
Treated $\times$ Post-Displacement, γ	−79.89 (3092.00)	2.84 (2.65)	0.22 (0.21)
No. of observations	22,638	20,576	18,588

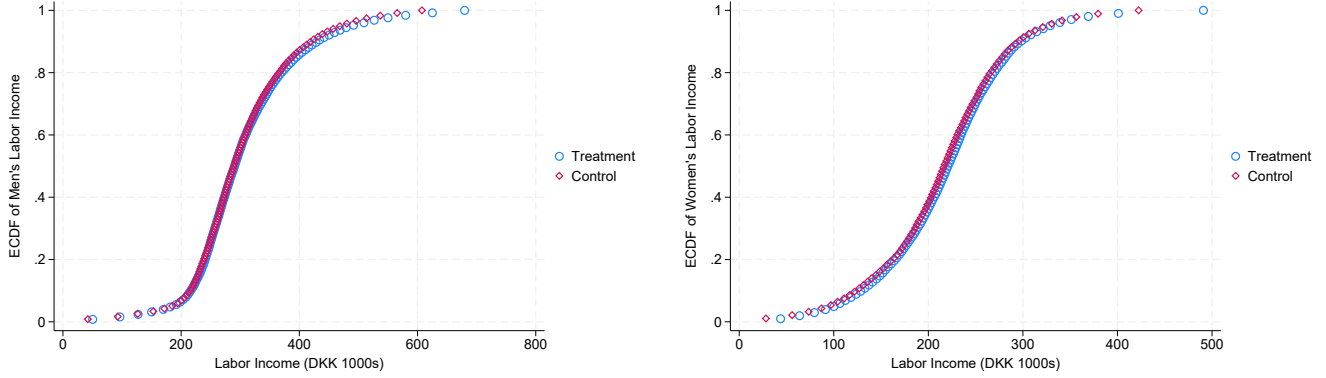
Notes: The table shows the effect of job displacement on the types of new partners matched with the original partners of displaced individuals, measured by new partners' labor income, hourly wage (conditional on employment), and weekly work hours (including zeros for non-employed individuals). Panel (a) reports rematching patterns for the original partners of displaced women. Panel (b) reports rematching patterns for the original partners of displaced men. The table reports coefficient estimates of γ . Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.8: Relationship Status of Displaced Individuals' Partners



Notes: Panels A–C show the effects of job displacement on displaced individuals' partners' relationship status. Panel A displays the effect on the probability of being separated from the displaced individual. Panel B shows the effect on the probability of being single (i.e., unmarried and not cohabiting), and Panel C shows the effect on the probability of being matched (married or cohabiting) with a new partner who is distinct from the displaced individual. Dashed vertical lines around point estimates indicate 95% confidence intervals. The values in Panels A–C correspond to coefficient estimates of ϕ_τ from Equation (1).

Figure B.9: Pre-Displacement Income of Couples Who Break Up:
Empirical CDFs of Men's Labor Income



Notes: The figure displays the pre-displacement labor income distribution of men who experience a breakup. The plotted empirical CDFs are based on pre-displacement labor income in $t \in \{-5, \dots, -3\}$ (before the male partner's actual or placebo displacement) for individuals who experience a breakup between $t = 0$ and $t = 10$ (after the male partner's actual or placebo displacement). Each dot in the plot represents an average across 100 individuals, aggregated to comply with Statistics Denmark's data confidentiality policy. The underlying sample includes men who were displaced as part of an establishment closure between 1980 and 2007, and an equally sized control group selected by coarsened exact matching. The specific sample selection criteria and matched sampling algorithm are described in Subsection 2.3.

Table B.7: Calibrated Parameter Values

Parameter	Symbol	1D Model	2D Model	Comment
Discount rate	r	0.05	0.05	fixed
Risk aversion	η	1.5	1.5	fixed
Bargaining power	μ_f	0.50	0.50	fixed
Separation rate	δ	0.06	0.06	data estimate
Meeting rate scaling factor	$\bar{\lambda}$	1.00	1.00	fixed
Love shock mean	μ_z	7963.97	91768.12	calibrated
Love shock standard deviation	σ_z	3034.07	44.07	calibrated
Income NAM utility parameter 1D	κ_1	0.79	–	calibrated
Income PAM utility parameter 1D	κ_2	0.34	–	calibrated
Income NAM utility parameter 2D	ω_1	–	13844.73	calibrated
Unobserved PAM utility parameter 2D	ω_2	–	7724811.39	calibrated
Correlation Income/Unobserved dimension	ρ	–	0.71	calibrated

Notes: This table reports the calibrated parameter values used in the quantitative versions of our framework in the 1D and the 2D version, see Section 5.2.

C Decomposing the Cross-Sectional Correlation in Partners' Incomes

We decompose the empirical association between spouses' incomes into the shares driven by observed characteristics and a residual attributable to unobserved characteristics. Consider the regression

$$y_f = \beta_0 + \beta_1 y_m + \beta_2' X_m + \beta_3' X_f + \epsilon \quad (\text{C.1})$$

where y_f and y_m denote the labor income of the female and male partner in a couple, and X_f, X_m are vectors of characteristics other than income. Interpreted through the lens of the multidimensional matching framework, our evidence from job displacements suggests that sorting on income, conditional on all other characteristics, is negative. The coefficient β_1 reflects this direct negative effect, but also captures the indirect effect arising from sorting on unobserved characteristics omitted from the regression. β_1 thus provides a lower bound on the contribution of this indirect effect to the empirical association between partners' incomes.⁶⁴

In Table C.1, we report estimates from regression (C.1), varying the set of variables included in X_m and X_f . The “raw” coefficient estimate of β_1 obtained by regressing the female partner's income on the male partner's income without controls is 0.16. Controlling for age fixed effects for both partners reduces the estimate by 0.019 (12%), while controlling for education fixed effects reduces it by 0.077 (48%). Including both sets of fixed effects reduces the estimate by 0.099 (56%) to 0.071. Given that β_1 is a lower bound on the indirect effect of sorting on unobserved characteristics, it follows that at least 44% ($= 0.07/0.16 \cdot 100\%$) of the raw coefficient estimate reflects sorting on unobserved characteristics omitted from the regression (that is, characteristics other than income, age, and education). This may include characteristics such as anthropometrics (Oreffice and Quintana-Domeque, 2010), physical attractiveness (Fisman, Iyengar, Kamenica, and Simonson, 2006), or personality traits (Dupuy and Galichon, 2014), which are unavailable in the Danish register data. While we are not aware of studies that directly estimate how controlling for attractiveness or personality affects the correlation between partners' incomes, previous work documents positive assortative matching on these characteristics (e.g., Hitsch, Hortaçsu, and Ariely, 2010; Oreffice and Quintana-Domeque, 2010; Chiappori et al., 2012; Dupuy and Galichon, 2014). Importantly, these

⁶⁴Formally, suppose the multidimensional agent types are $q_f = (y_f, X_f, U_f)$ for women and $q_m = (y_m, X_m, U_m)$ for men, where U_f, U_m are characteristics omitted from the regression. Regression (C.1) estimates the conditional mean $\mathbb{E}[y_f|y_m, X_m, X_f]$, whose slope can be decomposed into a direct and an indirect effect, analogous to Proposition 3:

$$\begin{aligned} \mathbb{E}[y_f|y_m'', X_m, X_f] - \mathbb{E}[y_f|y_m', X_m, X_f] &= \text{DE} + \text{IE}, \\ \text{where} \\ \text{and} \quad \text{DE} &= \int \mathbb{E}[y_f|y_m'', X_m, U_m, X_f, U_f] - \mathbb{E}[y_f|y_m', X_m, U_m, X_f, U_f] dG(U_f, U_m|y_m'', X_m, X_f) \\ \text{IE} &= \int \mathbb{E}[y_f|y_m', X_m, U_m, X_f, U_f] dG(U_f, U_m|y_m'', X_m, X_f) \\ &\quad - \int \mathbb{E}[y_f|y_m', X_m, U_m, X_f, U_f] dG(U_f, U_m|y_m', X_m, X_f). \end{aligned}$$

Given that sorting on income conditional on other characteristics is negative (i.e., under NAM(1)) it follows that $\text{DE} \leq 0$ and therefore $\mathbb{E}[y_f|y_m'', X_m, X_f] - \mathbb{E}[y_f|y_m', X_m, X_f] \leq \text{IE}$, which implies in the limit that $\beta_1 \leq \text{IE}$.

same characteristics have also been shown to be positively associated with earnings (e.g., Hamermesh and Biddle, 1994; Oreffice and Quintana-Domeque, 2016; Fletcher, 2013; Flinn, Todd, and Zhang, 2018; Alderotti, Rapallini, and Traverso, 2023). Together, these findings in the previous literature suggest that omitted characteristics of this kind may contribute to the observed cross-sectional correlation between partners' incomes.

Table C.1: Regressing Wives' on Husbands' Income, Controlling for Age and Education

	(1)	(2)	(3)	(4)
Husband's labor income ($\hat{\beta}_1$)	0.160*** (0.000558)	0.141*** (0.000523)	0.083*** (0.000539)	0.071*** (0.000571)
<i>Covariates</i>				
Male education FE	No	No	Yes	Yes
Female education FE	No	No	Yes	Yes
Male age FE	No	Yes	No	Yes
Female age FE	No	Yes	No	Yes
Observations	3,034,432	3,034,432	3,034,432	3,034,432

Notes: This table reports coefficient estimates of β_1 from equation (C.1) for varying sets of control variables X_f and X_m . All specifications are estimated on our full sample of married or cohabiting couples, observed between 1980 and 2007. Standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

D Semiparametric Estimation of Local Sorting Patterns

Sample and Setup We begin with the universe of Danish couples and retain the first year in which each couple is observed as married or cohabiting. We then restrict the sample to the same calendar period (1980–2007) and age range (25–58) as in our job-displacement analysis, and apply the same restrictions to singles. Closely following [Chiappori et al. \(2026\)](#), we discretize male and female log-labor incomes using cutoffs from the 1995 gender-specific log-income distributions.⁶⁵ From the lowest income category up to the 99th percentile, log-labor incomes are partitioned into 24 equally spaced bins. Above the 99th percentile, additional cutoffs are placed at the 99.2, 99.4, 99.6, 99.8, and 99.95 percentiles.

Semiparametric Estimation Following [Chiappori et al. \(2026\)](#), let

$$S_{IJ} = 2 \ln(\mu_{IJ}) - \ln(\mu_{I0}) - \ln(\mu_{0J}), \quad (\text{D.1})$$

where μ_{IJ} denotes the number of couples in which the woman is in income category I and the man in category J (where I and J index the income bins defined above), and μ_{I0} and μ_{0J} denote the numbers of single women in category I and single men in category J , respectively. Note that S_{IJ} corresponds to the marital surplus in the [Choo and Siow \(2006\)](#) model. However, its second cross-difference has a model-free interpretation, as it is twice the local log-odds ratio, a measure of local assortative matching (see, e.g., [Chiappori et al., 2025](#)). Following [Chiappori et al. \(2026\)](#), we estimate S_{IJ} semiparametrically and use the resulting estimate of its second cross-difference as our measure of local sorting. The surplus is specified as a polynomial

$$S^\lambda(I_x, J_y) = \sum_{\substack{l,r \geq 0 \\ l+r \leq L}} \lambda_{l,r} I_x^l J_y^r, \quad (\text{D.2})$$

and we estimate the coefficients by the Poisson regression estimator proposed by [Chiappori et al. \(2026\)](#). Following [Chiappori et al. \(2026\)](#), we set $L = 6$ and evaluate the cross-partial derivative of the fitted polynomial

$$\frac{\partial^2 S^\lambda(I_x, J_y)}{\partial I_x \partial J_y} = \sum_{\substack{l+r \leq L \\ l,r > 0}} l r \lambda_{l,r} I_x^{l-1} J_y^{r-1}. \quad (\text{D.3})$$

Figure 4 plots $\frac{\partial^2 S^\lambda(I_x, J_y)}{\partial I_x \partial J_y}$ over an equispaced grid (of 100 bins) covering the bottom 99% of the gender-specific income distributions, omitting the upper tail to focus the plot on the region where our sample used

⁶⁵We select 1995 as our reference year as it is in the middle of our sample period. As in the implementation of [Chiappori et al. \(2026\)](#), we set log-labor income for all individuals with income of less than 1 DKK equal to zero. [Chiappori et al. \(2026\)](#) group all observations below the 50th percentile into the lowest income category and exclude that category from the subsequent analysis. We instead retain below-median observations to characterize local sorting patterns in the lower part of the bivariate income distribution, where a large part of our sample used for the analysis of job displacements is located.

in the analysis of job displacements is located. Regions with a positive value of $\frac{\partial^2 S^\lambda(I_x, J_y)}{\partial I_x \partial J_y}$ are classified as regions exhibiting local positive sorting (shaded in red), while negative values are classified as regions exhibiting local negative sorting (shaded in blue).

E Model Appendix

E.1 Nash Bargaining

We assume that couples split the match surplus by agreeing on transfers, t_f and t_m , via Nash-bargaining. Define the marital surplus of a man of type q_m who is matched with a woman of type q_f by

$$S_m(q_f, q_m) = V_m^1(q_f, q_m) - V_m^0(q_m) = \frac{u_m^1(q_f, q_m) + t_m - rV_m^0(q_m)}{r + \delta}, \quad (\text{E.1})$$

and the marital surplus of a woman of type q_f who is matched with a man of type q_m by

$$S_f(q_f, q_m) = V_f^1(q_f, q_m) - V_f^0(q_f) = \frac{u_f^1(q_f, q_m) + t_f - rV_f^0(q_f)}{r + \delta}. \quad (\text{E.2})$$

Under Nash-bargaining the transfers, t_f and t_m , solve:

$$\begin{aligned} \max_{t_f, t_m} \quad & S_m(q_f, q_m)^{(1-\mu_f)} S_f(q_f, q_m)^{\mu_f} \\ \text{s.t.} \quad & t_m + t_f = 0. \end{aligned}$$

Using (E.1) and (E.2), the Nash bargaining solution is:

$$(1 - \mu_f) \left(\frac{u_m^1(q_f, q_m) + t_f - rV_f^0(q_f)}{r + \delta} \right) = \mu_f \left(\frac{u_m^1(q_f, q_m) + t_m - rV_m^0(q_m)}{r + \delta} \right). \quad (\text{E.3})$$

Equation (E.3) can be solved for the transfers, t_m and t_f :

$$t_m = rV_m^0(q_m) - u_m^1(q_f, q_m) + (1 - \mu_f) (u_m^1(q_f, q_m) + u_f^1(q_f, q_m) - rV_m^0(q_m) - rV_f^0(q_f)) \quad (\text{E.4})$$

$$t_f = rV_f^0(q_f) - u_f^1(q_f, q_m) + \mu_f (u_m^1(q_f, q_m) + u_f^1(q_f, q_m) - rV_m^0(q_m) - rV_f^0(q_f)). \quad (\text{E.5})$$

Together with (E.1) and (E.2) it follows that

$$S_m(q_f, q_m) = V_m^1(q_f, q_m) - V_m^0(q_m) = (1 - \mu_f) S(q_f, q_m), \quad (\text{E.6})$$

$$S_f(q_f, q_m) = V_f^1(q_f, q_m) - V_f^0(q_f) = \mu_f S(q_f, q_m). \quad (\text{E.7})$$

E.2 Quantitative Model Specification

This section provides a detailed description of the quantitative specification of our framework that we calibrate and use to generate simulation results in Section 5.2.

Type spaces and distributions Women and men are fully characterized by their types, $q_f \in Q_f$ and $q_m \in Q_m$, respectively. As described in the main text in Section 4.1, we allow for multidimensional

types: $Q_f = Q_m = \prod_{k=1}^K [q_k, \bar{q}_k]$, where each dimension, k , of the Cartesian product represents a distinct attribute. In the one-dimensional version of our framework we impose $K = 1$, and consider $K = 2$ for the bidimensional case.

We denote the PDFs (and CDFs) of the male and female type distributions in the population by $l_m(q_m)$ and $l_f(q_f)$ ($L_m(q_m)$ and $L_f(q_f)$). The masses of men and women are normalized to one, $\int L_m(q_m) dq_m = 1$ and $\int L_f(q_f) dq_f = 1$. At any given point in time, each individual is either single or married. Let $g_f(q_f)$ and $g_m(q_m)$ ($G_m(q_m)$ and $G_f(q_f)$) denote the endogenous PDFs (CDFs) of female and male singles. The masses of singles are endogenous, and denoted by $\mathcal{G}_m = \int g_m(q_m) dq_m$ and $\mathcal{G}_f = \int g_f(q_f) dq_f$. We denote the endogenous bivariate PDF of married individuals by $c(q_m, q_f)$, and the mass of married couples by $\mathcal{C} = \iint c(q_m, q_f) dq_m dq_f$. These definitions imply that $l_m(q_m) = \int c(q_m, q_f) dq_f + g_m(q_m)$ and $l_f(q_f) = \int c(q_m, q_f) dq_m + g_f(q_f)$.

Matching technology As described in the main text in Section 4.1, we assume a quadratic matching function $\Lambda(\mathcal{G}_m, \mathcal{G}_f) = \bar{\lambda} \mathcal{G}_m \mathcal{G}_f$ (see, e.g., [Mortensen, 2011](#)). The meeting rates for women and men thus equal $\lambda_f = \frac{\Lambda(\mathcal{G}_m, \mathcal{G}_f)}{\mathcal{G}_f} = \bar{\lambda} \mathcal{G}_m$ and $\lambda_m = \frac{\Lambda(\mathcal{G}_m, \mathcal{G}_f)}{\mathcal{G}_m} = \bar{\lambda} \mathcal{G}_f$.

Matching probabilities As described in the main text in Section 5.2, we assume that model agents experience a match-specific shock, z , which is experienced by both partners and fixed for the duration of the match. We denote flow utilities net of the match-specific shock by $\tilde{u}_g^1(q_f, q_m)$, i.e., $u_g^1(q_f, q_m) = \tilde{u}_g^1(q_f, q_m) + z$ for $g \in \{f, m\}$. Under these assumptions, the probability that a man of type q_m and a woman of type q_f is given by $\alpha(q_m, q_f) = 1 - F_z\left(-\frac{S(q_f, q_m)}{2}\right)$, where F_z denotes the CDF of the match specific shock, $z \sim N(\mu_z, \sigma_z)$.

Equilibrium Characterization and Solution We derive four equations that characterize a steady state equilibrium in the described setup. We start from the steady-state-condition, which requires that match creation equals match destruction for any given combination of men's and women's types, q_f and q_m :

$$\delta c(q_m, q_f) = g_m(q_m) \lambda_m \frac{g_f(q_f)}{\mathcal{G}_f} \alpha(q_m, q_f) = \bar{\lambda} g_m(q_m) g_f(q_f) \alpha(q_m, q_f), \quad \forall (q_m, q_f). \quad (\text{E.8})$$

Integrating (E.8) over women's type, q_f , yields the steady state flow condition for men of type q_m :

$$\delta \int c(q_m, q_f) dq_f = \bar{\lambda} g_m(q_m) \int g_f(q_f) \alpha(q_m, q_f) dq_f. \quad (\text{E.9})$$

Substituting $l_m(q_m) - g_m(q_m) = \int c(q_m, q_f) dq_f$ yields:

$$\delta (l_m(q_m) - g_m(q_m)) = \bar{\lambda} g_m(q_m) \int g_f(q_f) \alpha(q_m, q_f) dq_f, \quad (\text{E.10})$$

which can be solved for $g_m(q_m)$:

$$g_m(q_m) = \frac{\delta l_m(q_m)}{\delta + \bar{\lambda} \int g_f(q_f) \alpha(q_m, q_f) dq_f}. \quad (\text{E.11})$$

Similarly, for women of type q_f :

$$g_f(q_f) = \frac{\delta l_f(q_f)}{\delta + \bar{\lambda} \int g_m(q_m) \alpha(q_m, q_f) dq_m}. \quad (\text{E.12})$$

Equations (E.11) and (E.12) jointly determine the equilibrium distributions of single women and men.

Next, we use the value of being single, given by equation (4) together with (E.6) to obtain the following extended option-value equation for single men (the option-value for single women is derived by analogous steps):

$$rV_m^0(q_m) = u_m^0(q_m) + \lambda_m \iint \max\{S_m(q_f, q_m), 0\} dF_z(z) \frac{g_f(q_f)}{\mathcal{G}_f} dq_f, \quad (\text{E.13})$$

where $S_m(q_f, q_m) = V^1(q_f, q_m) - V^0(q_m)$ and the integral captures the option value of meeting single women, sampled from $\frac{g_f(q_f)}{\mathcal{G}_f}$, and drawing a match specific shock from F_z .

Using (E.13) together with (5) yields

$$rV_m^1(q_f, q_m) = \tilde{u}_m^1(q_f, q_m) + z + t_m + \delta(V_m^0(q_m) - V_m^1(q_f, q_m)), \quad (\text{E.14})$$

implying for $S_m(q_f, q_m)$:

$$S_m(q_f, q_m) = \frac{\tilde{u}_m^1(q_f, q_m) + z + t_m - rV_m^0(q_m)}{r + \delta}. \quad (\text{E.15})$$

Next, we use $\bar{\lambda} = \frac{\lambda_m}{\mathcal{G}_f}$ and the updated definition of male surplus S_m in (E.15) to substitute the match-specific shock z and the transfer t_m into the value of being a single man of type q_m :

$$rV_m^0(q_m) = u_m^0(q_m) + \frac{\bar{\lambda}}{r + \delta} \iint \max\{\tilde{u}_m^1(q_f, q_m) + z + t_m - rV_m^0(q_m), 0\} dF_z(z) g_f(q_f) dq_f, \quad (\text{E.16})$$

where the transfers, t_m , are given by

$$\begin{aligned} t_m &= rV_m^0(q_m) - \tilde{u}_m^1(q_f, q_m) + (1 - \mu_f) (\tilde{u}_m^1(q_f, q_m) + \tilde{u}_f^1(q_f, q_m) - rV_m^0(q_m) - rV_f^0(q_f)) \\ &\quad - (2\mu_f - 1) z. \end{aligned} \quad (\text{E.17})$$

Using (E.17) together with (E.16) yields:

$$rV_m^0(q_m) = u_m^0(q_m) \quad (\text{E.18})$$

$$\begin{aligned}
& + \frac{\bar{\lambda}(1 - \mu_f)}{r + \delta} \iint \max\{2z + \tilde{u}_m^1(q_f, q_m) + \tilde{u}_f^1(q_f, q_m) \\
& - rV_m^0(q_m) - rV_f^0(q_f), 0\} dF_z(z) g_f(q_f) dq_f,
\end{aligned}$$

By analogous steps we obtain for the value of being a single woman of type q_f :

$$\begin{aligned}
rV_f^0(q_f) &= u_f^0(q_f) \\
& + \frac{\bar{\lambda}\mu_f}{r + \delta} \iint \max\{2z + \tilde{u}_m^1(q_f, q_m) + \tilde{u}_f^1(q_f, q_m) \\
& - rV_m^0(q_m) - rV_f^0(q_f), 0\} dF_z(z) g_m(q_m) dq_m.
\end{aligned} \tag{E.19}$$

In summary, the equilibrium is characterized by equations (E.11), (E.12), (E.18), and (E.19). We use these equations to solve our framework numerically.

Numerical Solution To numerically solve the 1D as well as the 2D specification of our framework, we discretize the type spaces in the income dimension (1D and 2D model) and for the unobserved characteristic (2D model only). For the income dimension, we estimate unconditional (on marital status) income distributions for men and women based on observed annual labor incomes in our estimation sample (see Section 2). Specifically, we discretize the empirical distributions by measuring the density at 50 equally-spaced income grid points.⁶⁶ For the unobserved characteristic in the 2D model, we set up 10 grid points with values ranging from 1 to 10 and construct distributions of male and female types across grid points using a copula, given a mean, a standard deviation, and the correlation between individual level characteristics in the two dimensions, which we denote ρ . We calibrate ρ by targeting the empirical displacement effect. We set the mean of the male and female type distributions in the unobserved dimension to 5.5 (the mid-point between 1 and 10) and the standard deviation to 1.

⁶⁶We estimate the income density at each gridpoint using kernel density estimation. The highest income grid point is the 99th percentile of the male income distribution. The lowest income gridpoint is a small positive amount that we derive from the Danish social security system.